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ASSIGNMENT 05:
LIBRARY MANAGEMENT SYSTEM
WORKFLOW OPTIONS USE CASE

Objective:

To configure and examine workflow options in Zoho Creator for the Library Management System, including form-triggered notifications, scheduled reminders, conditional approvals, fine-payment actions, issue-lifecycle blueprints, bulk overdue processing, report configuration, and final application testing.

Procedure:

Open the Library Management System application in Zoho Creator, configure each workflow option, save the settings, and review the resulting configuration or execution screen.

1. Form Workflows Configuration

A form workflow was configured for the Book Issue Request form to run after successful submission. The workflow sends a confirmation email to Member_Email and includes dynamic values such as Request ID, Book Title, Issue Date and Due Date.

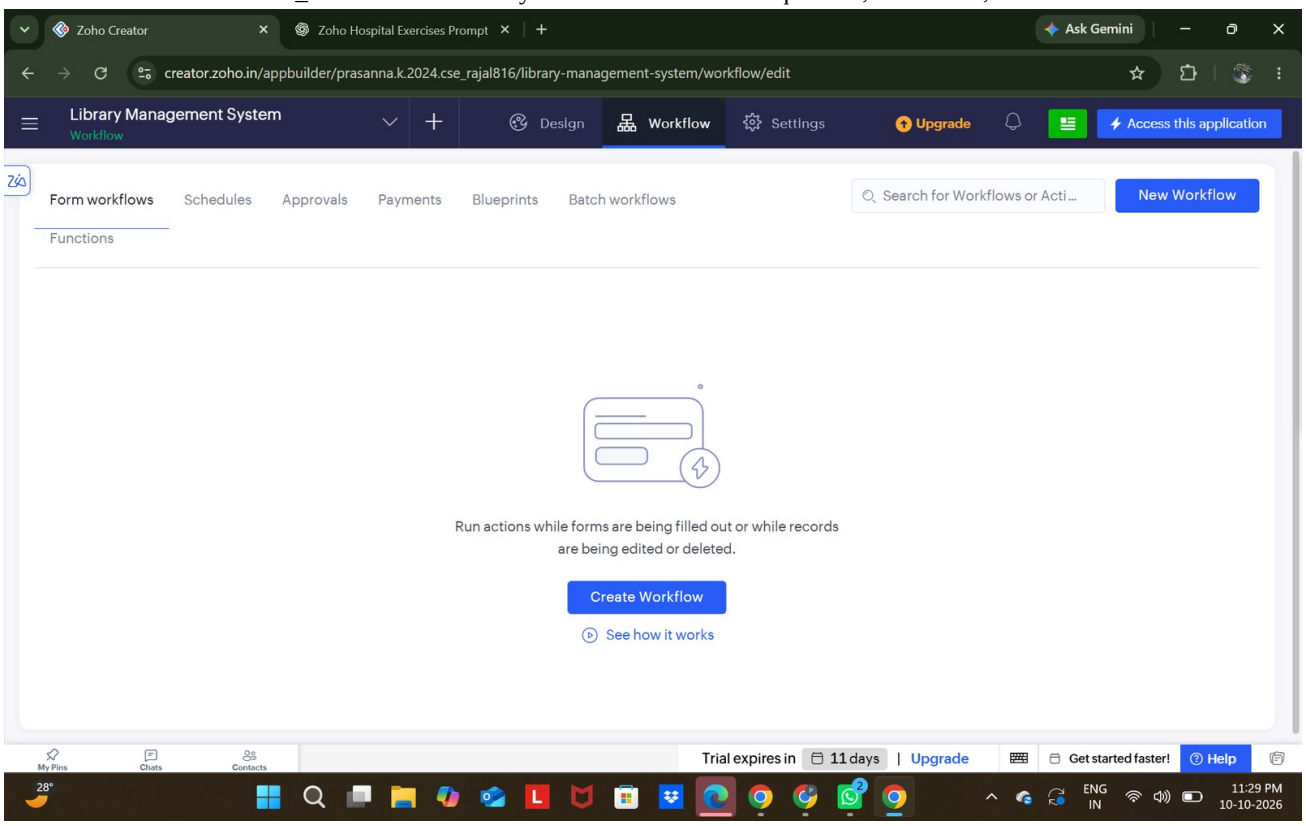


Figure 1: Opening the Workflow module to configure form automation.

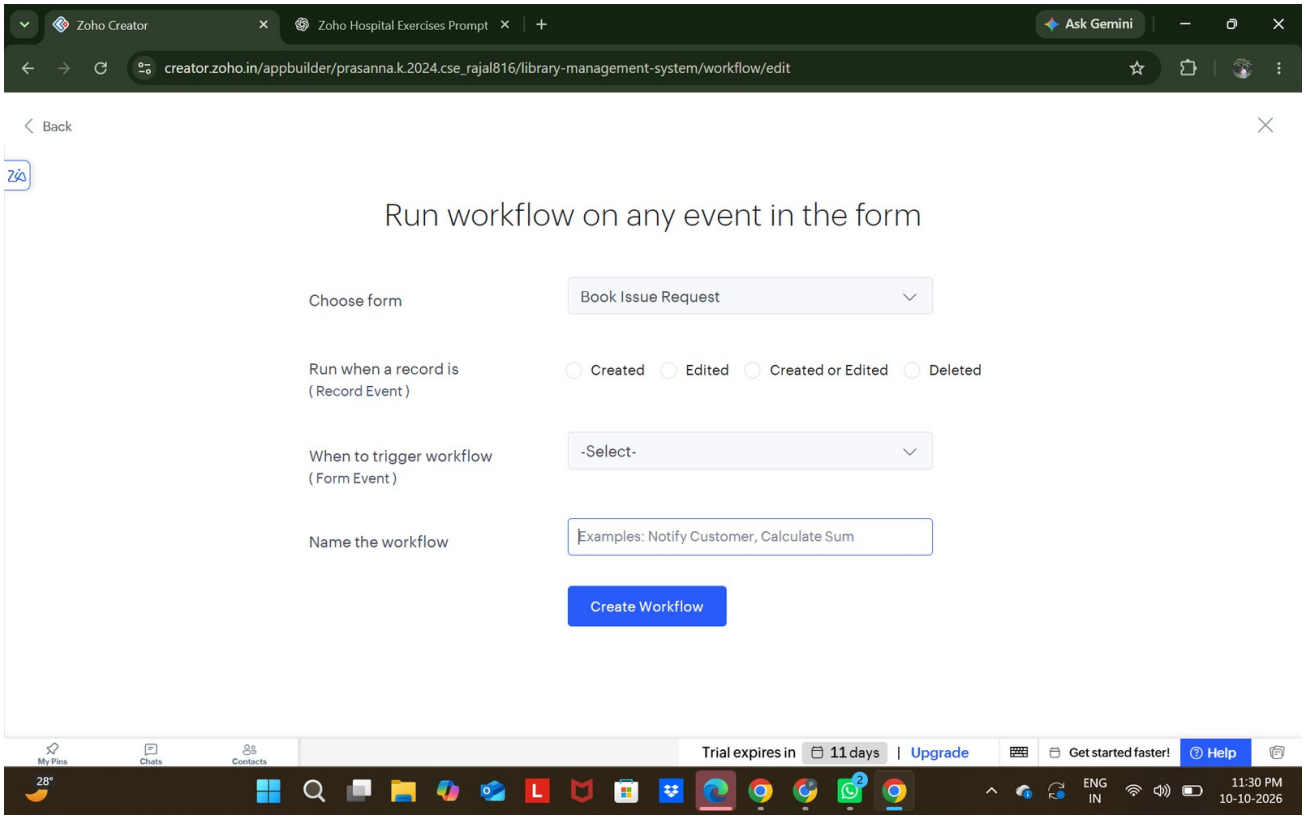


Figure 2: Selecting the form and choosing a record event for the workflow.

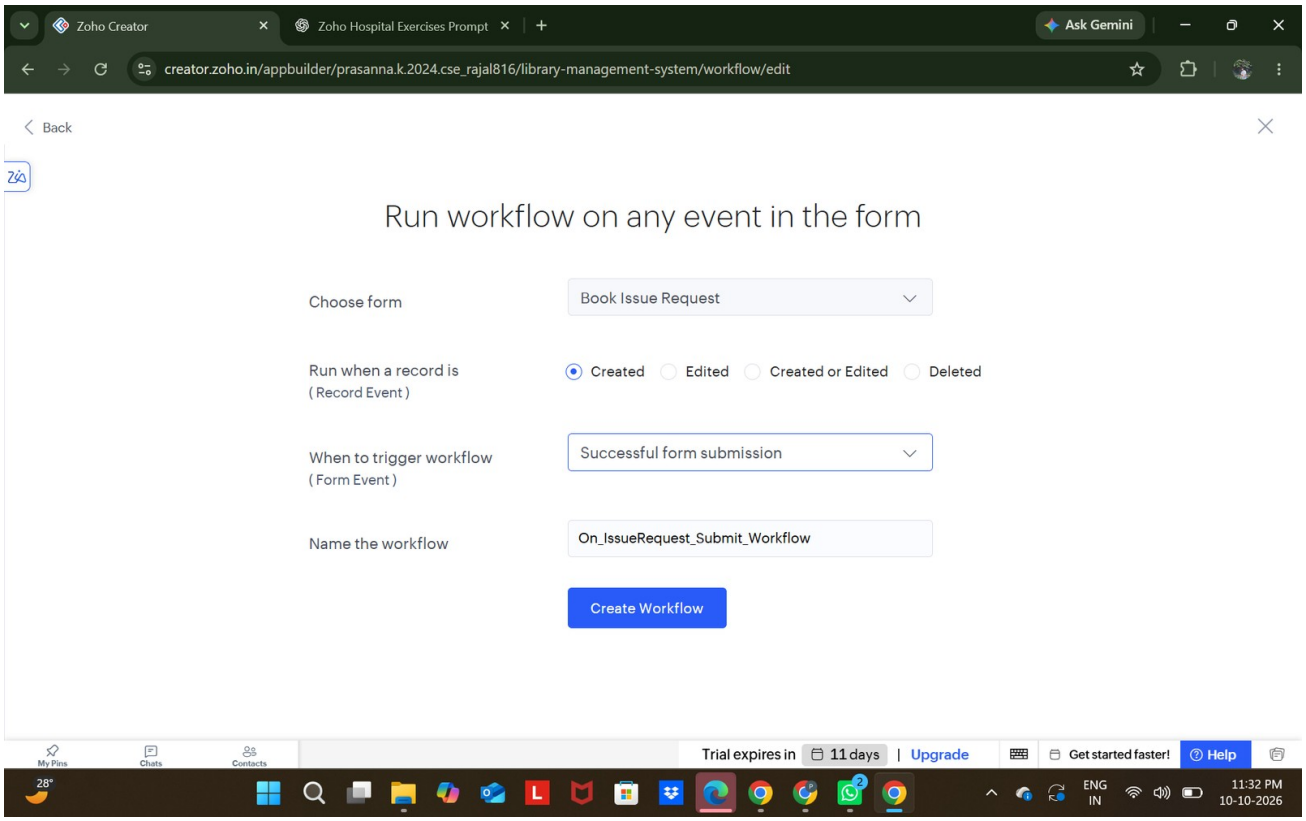


Figure 3: Configuring the successful form-submission trigger and workflow name.

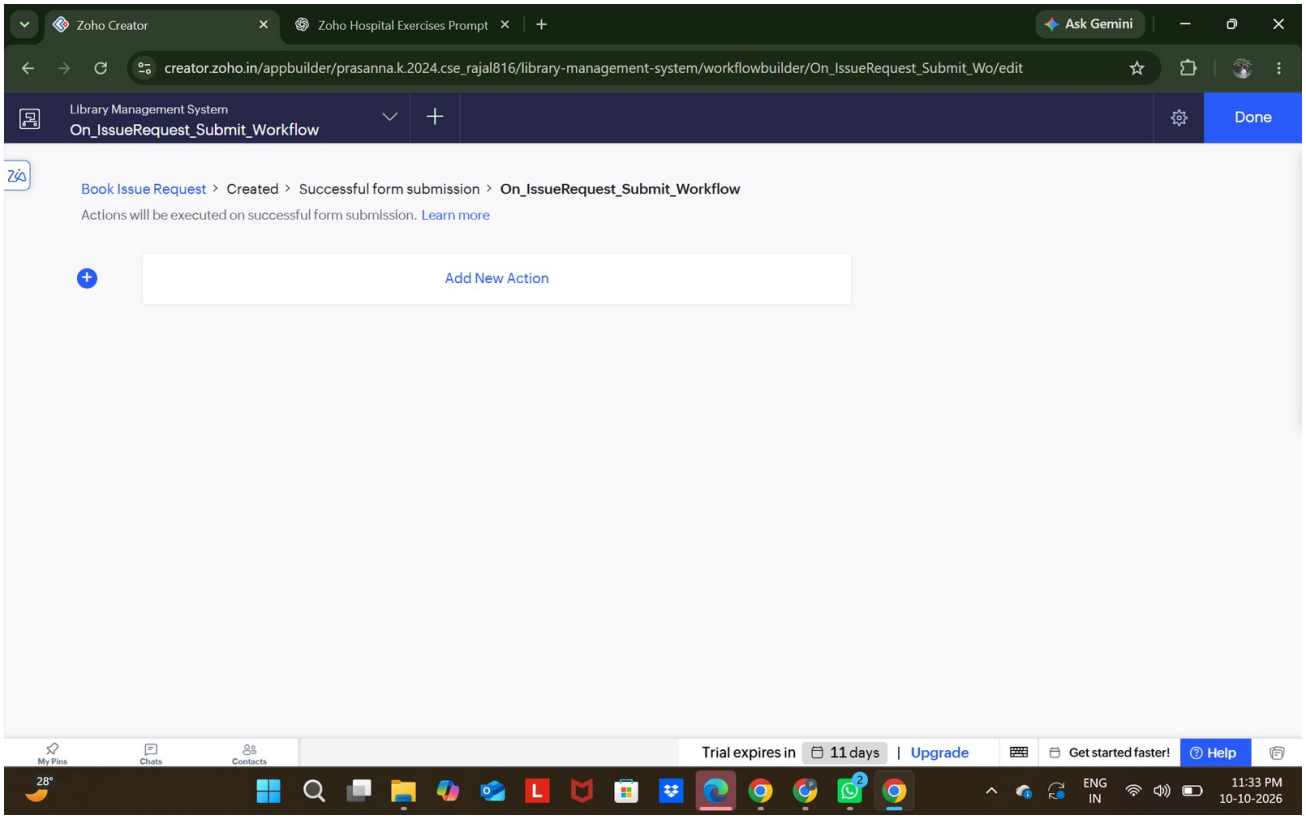


Figure 4: Opening the editor for the new form workflow.

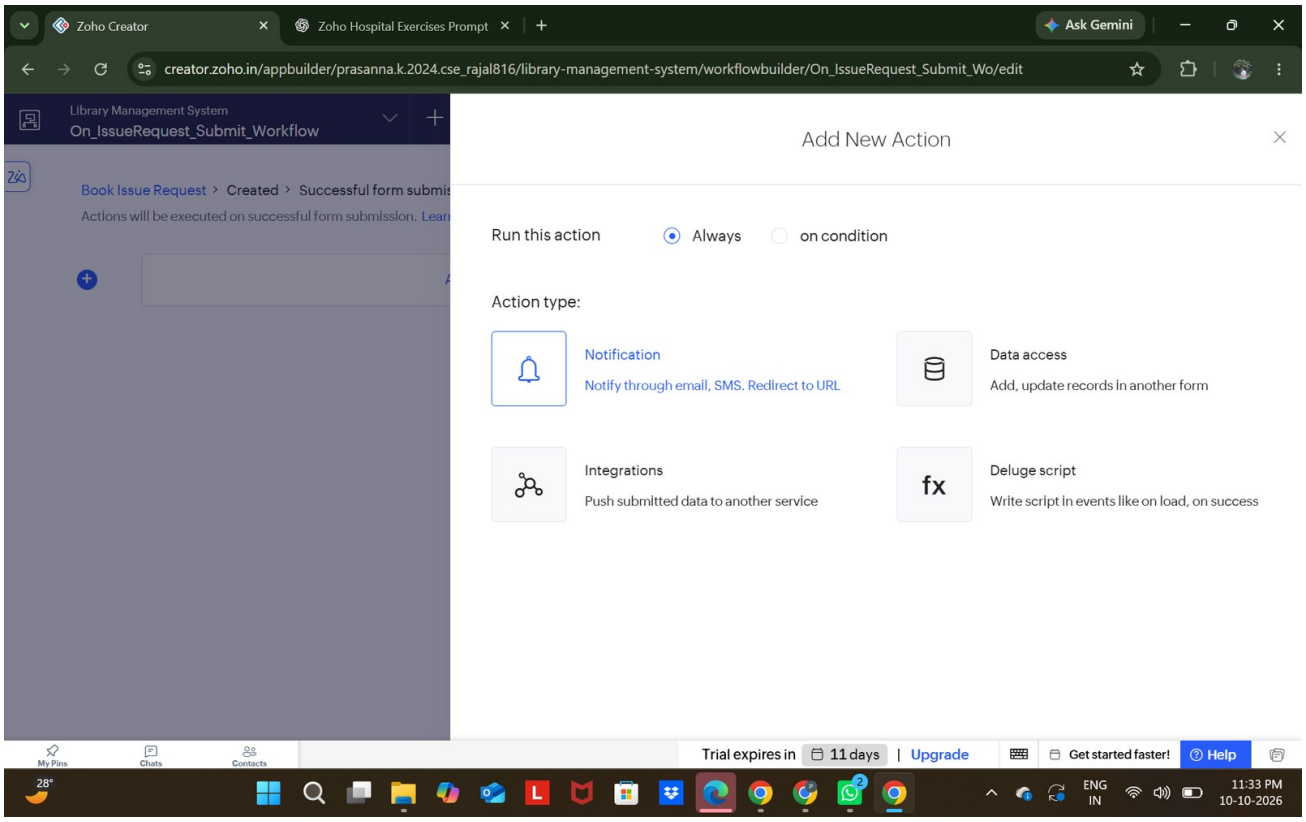


Figure 5: Choosing the type of action to run after submission.

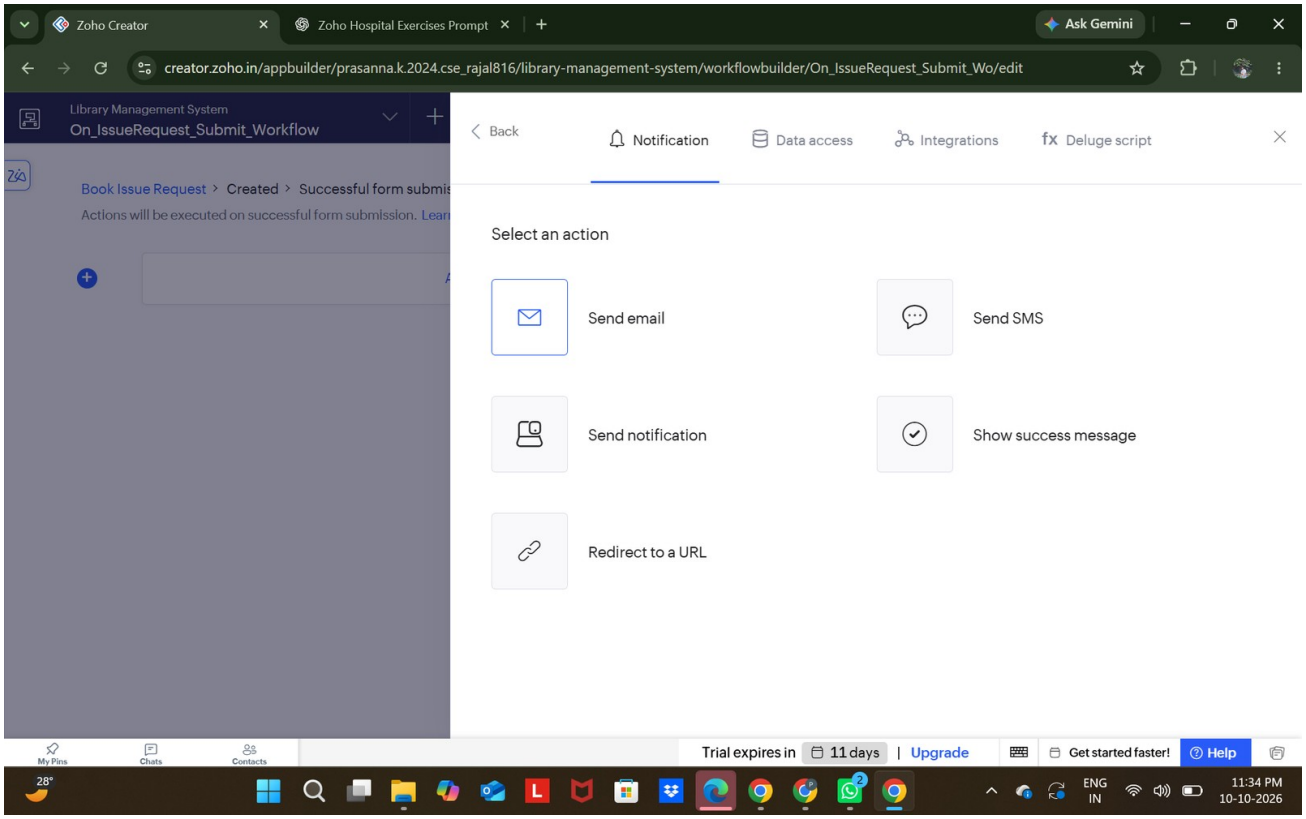


Figure 6: Selecting an email notification action.

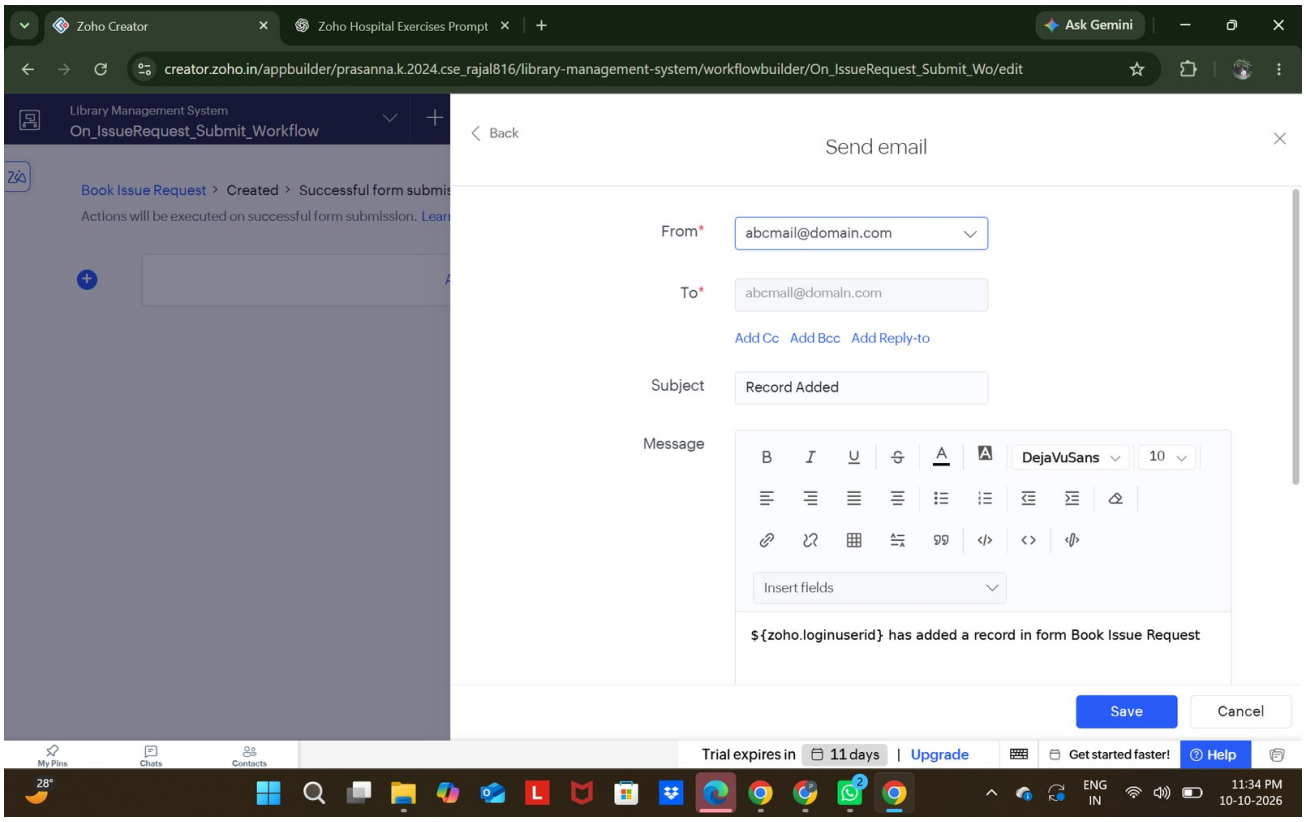


Figure 7: Composing the request-confirmation email with dynamic form fields.

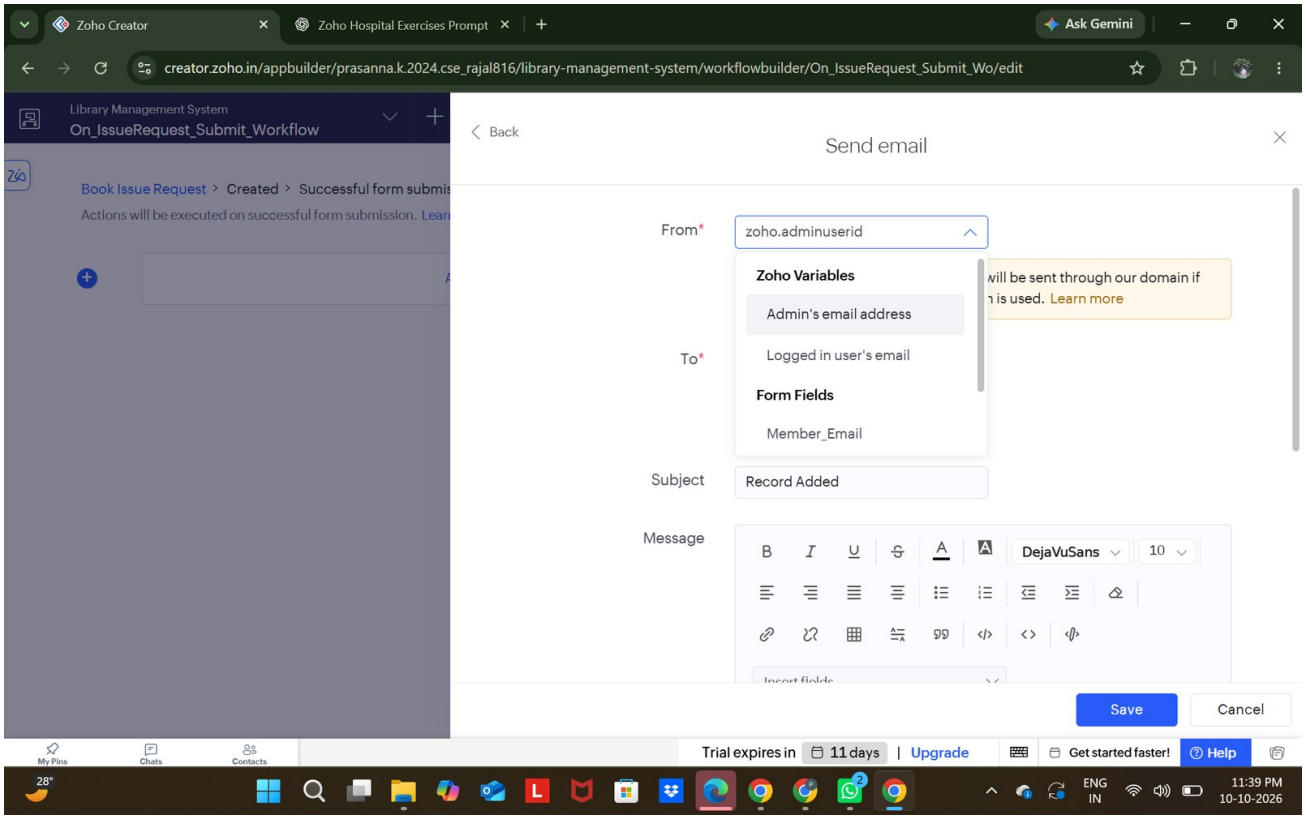


Figure 8: Reviewing the email message and its form-field placeholders.

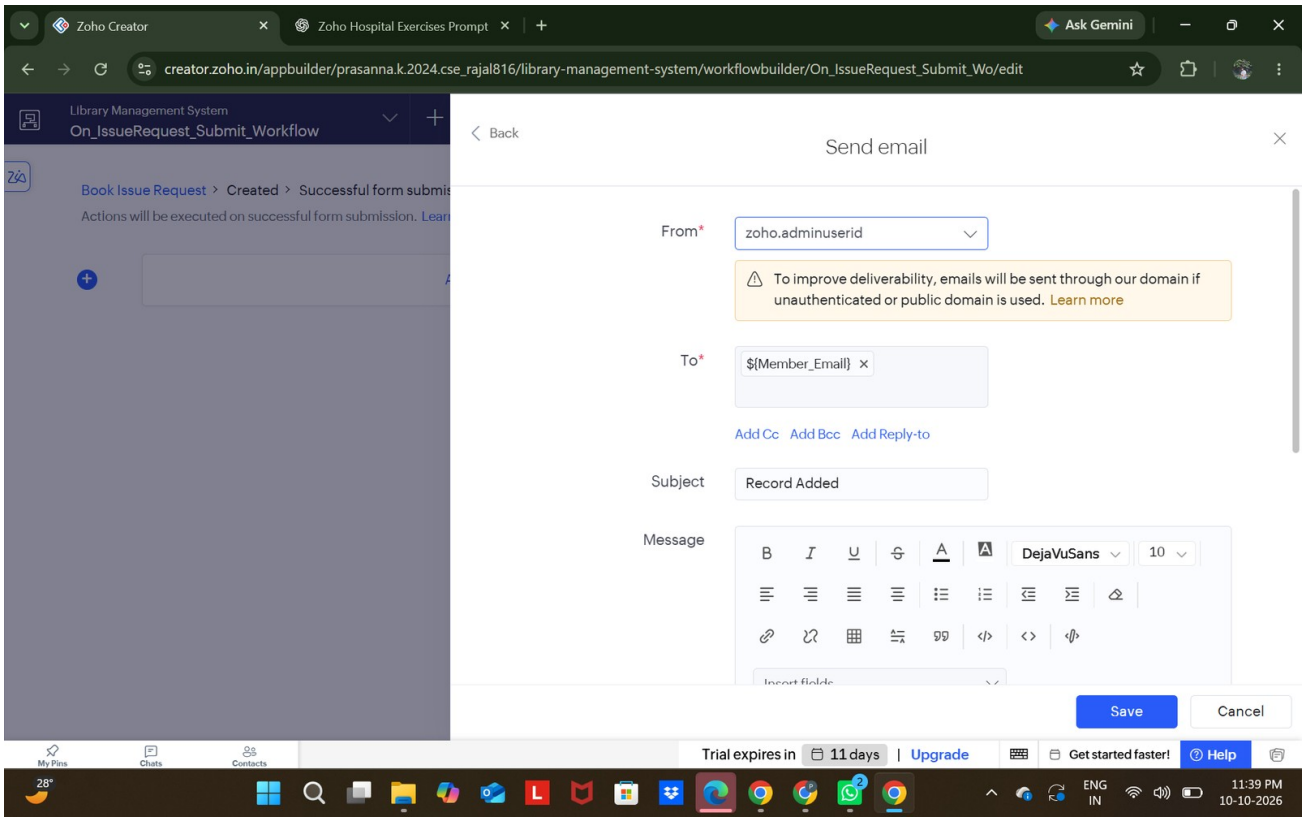


Figure 9: Configuring the sender, recipient and email subject.

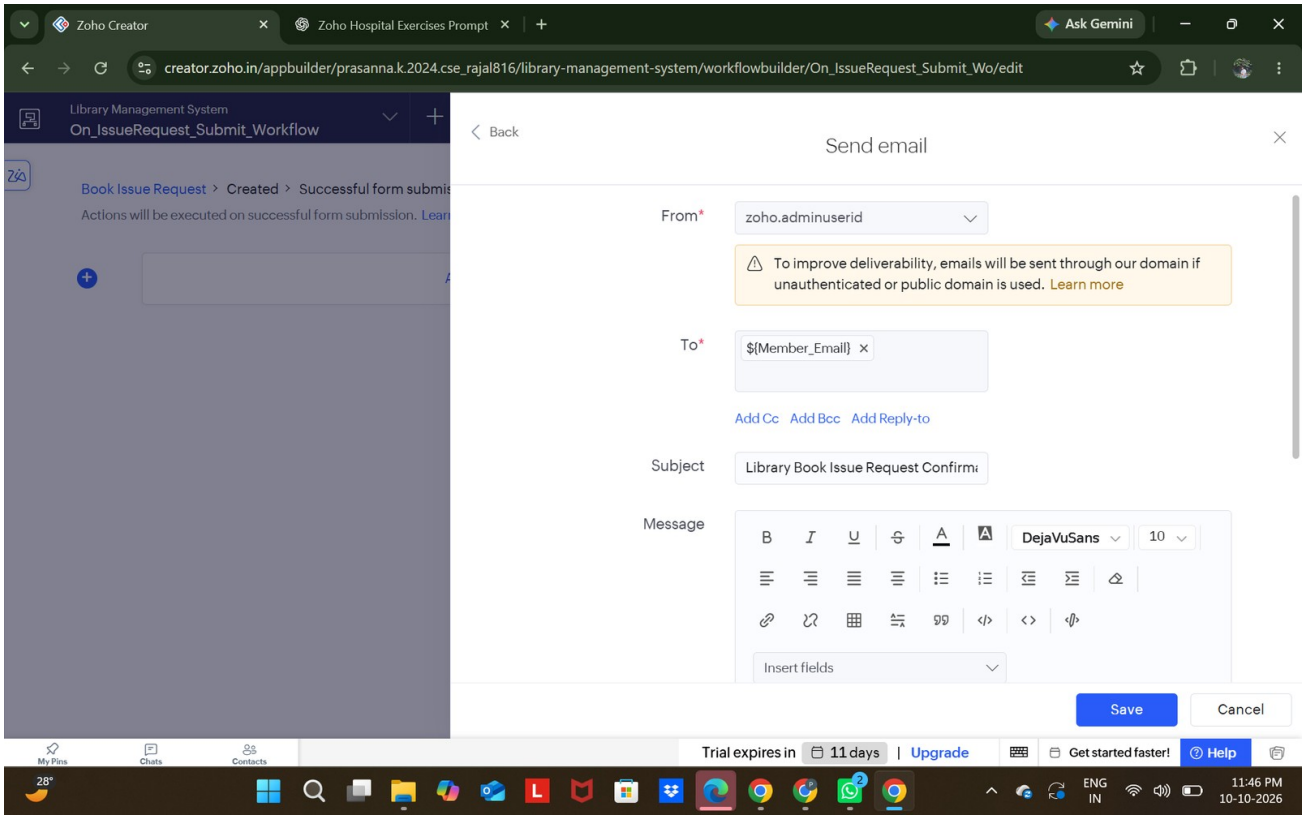


Figure 10: Setting the message details for the submission confirmation.

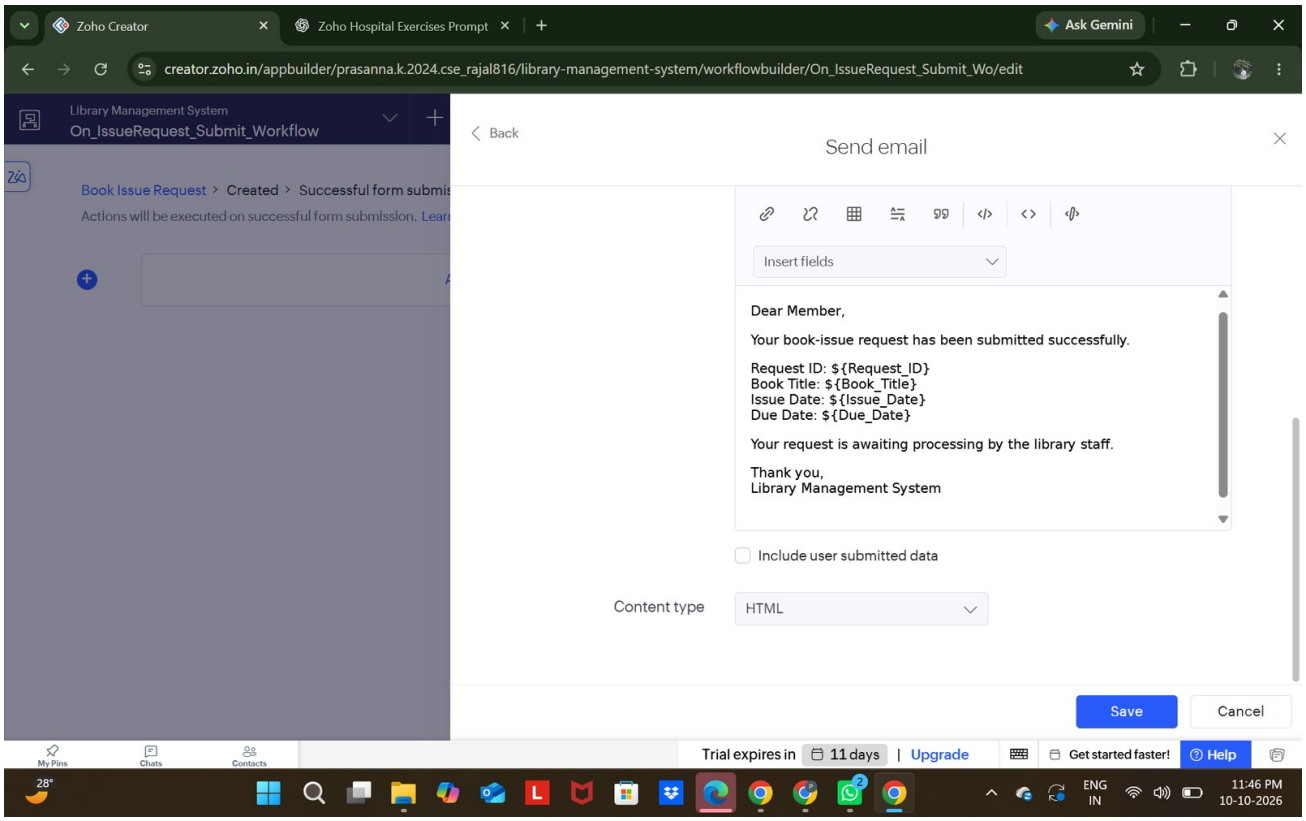


Figure 11: Reviewing the confirmation message with request, book and due-date details.

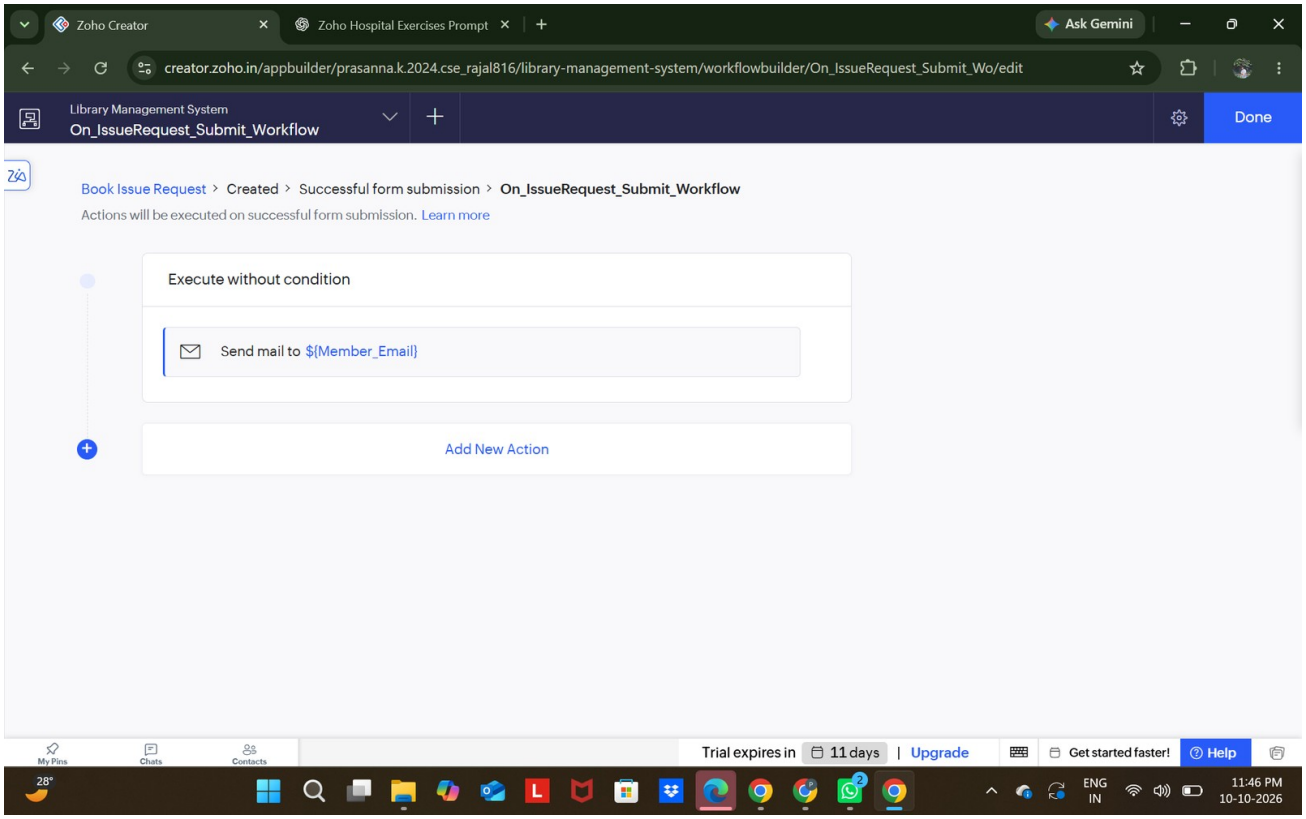


Figure 12: Returning to the workflow editor to review the configured email action.

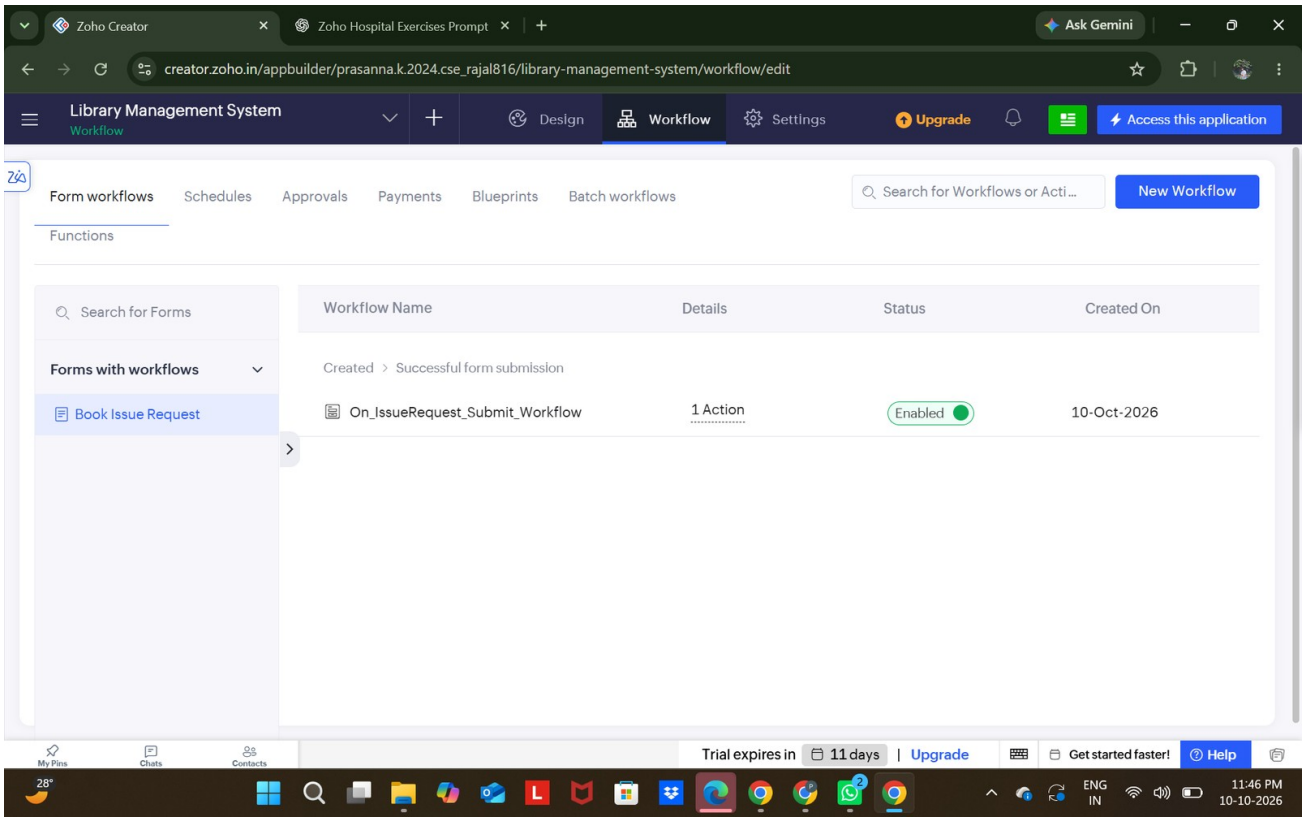


Figure 13: Verifying the form-submission workflow is enabled.

2. Schedules Configuration

The Daily_DueDate_Reminder_Schedule was configured as a daily scheduled workflow. Its Deluge logic checks issued requests with a due date within the next two days, verifies that the reminder has not already been sent, emails the member and updates Reminder_Status to Sent.

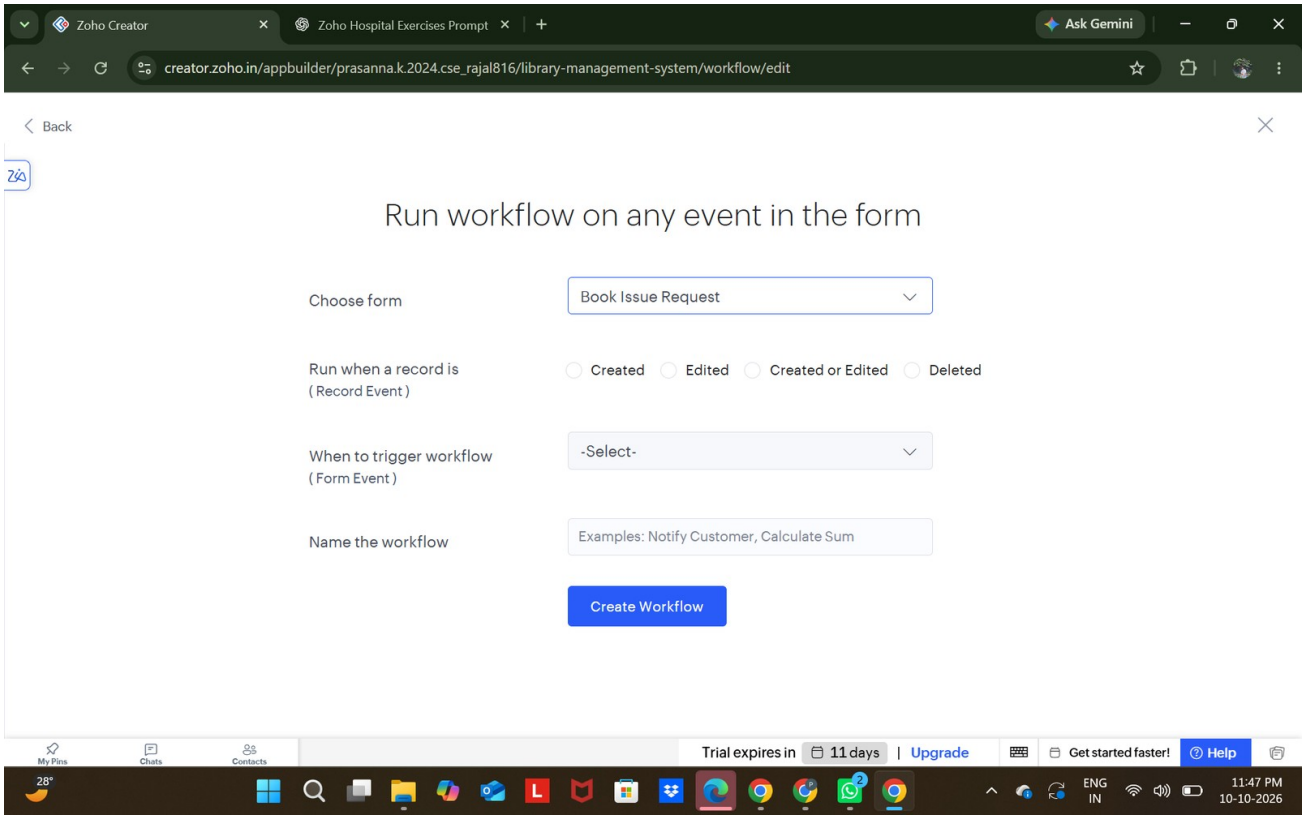


Figure 14: Beginning the next workflow configuration.

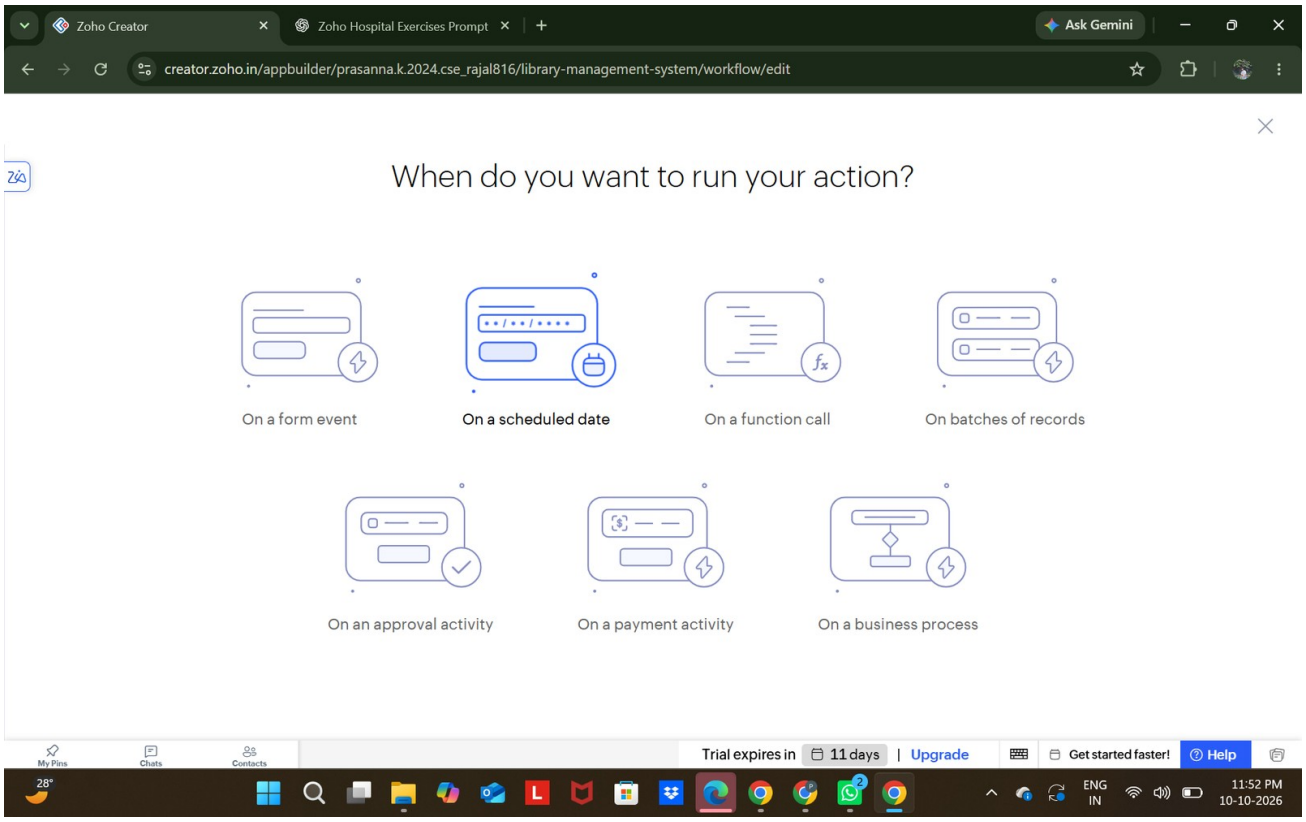


Figure 15: Choosing the scheduled-date trigger for a workflow action.

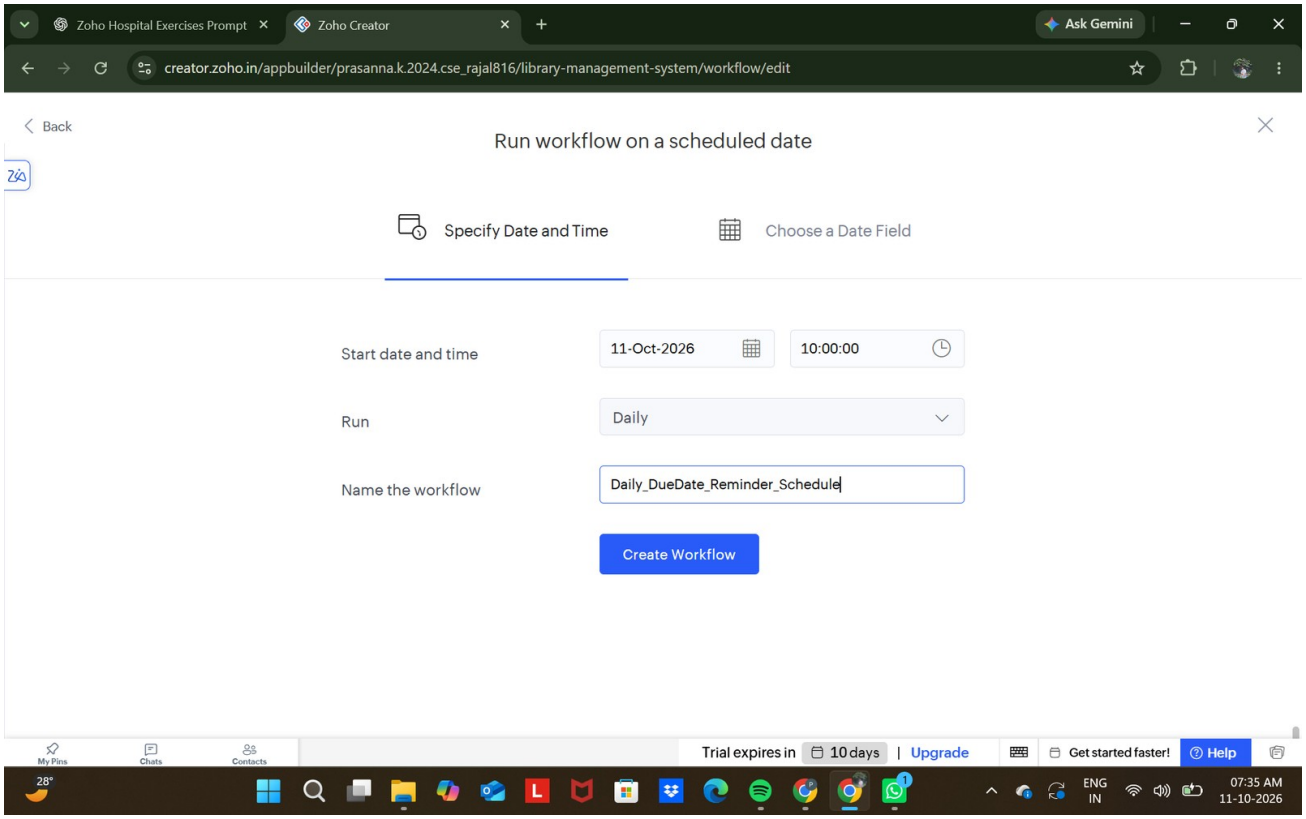


Figure 16: Setting the daily start date, time and schedule workflow name.

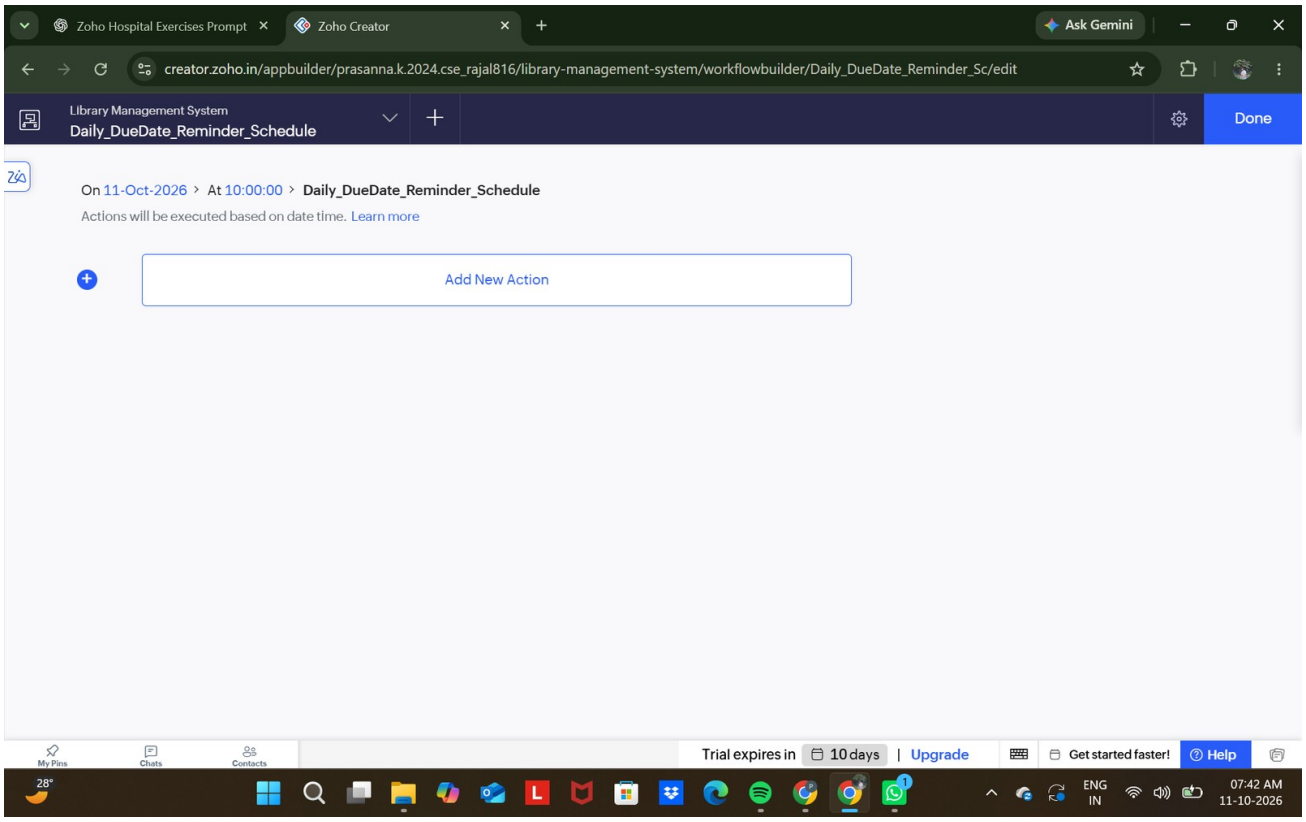


Figure 17: Opening the scheduled workflow editor for the due-date reminder.

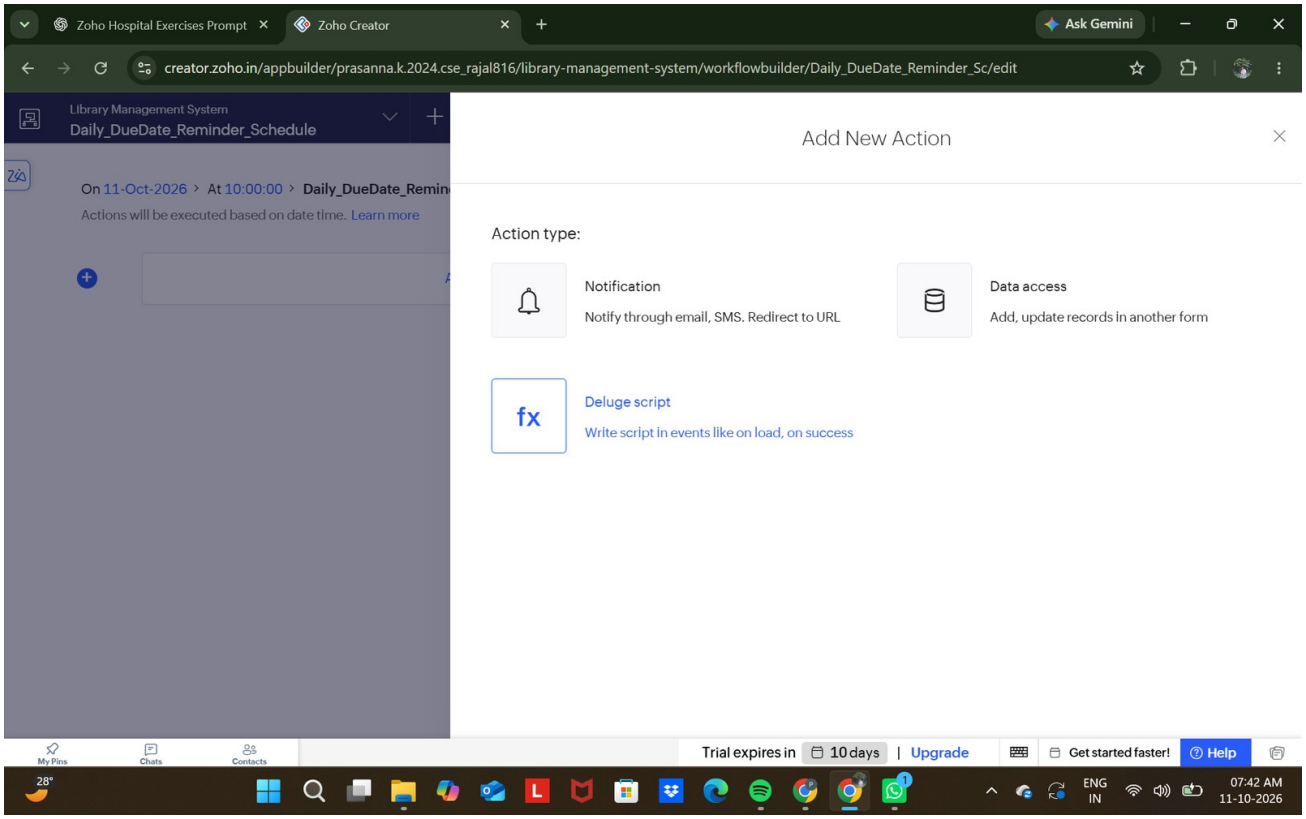


Figure 18: Selecting the Deluge script action for the scheduled workflow.

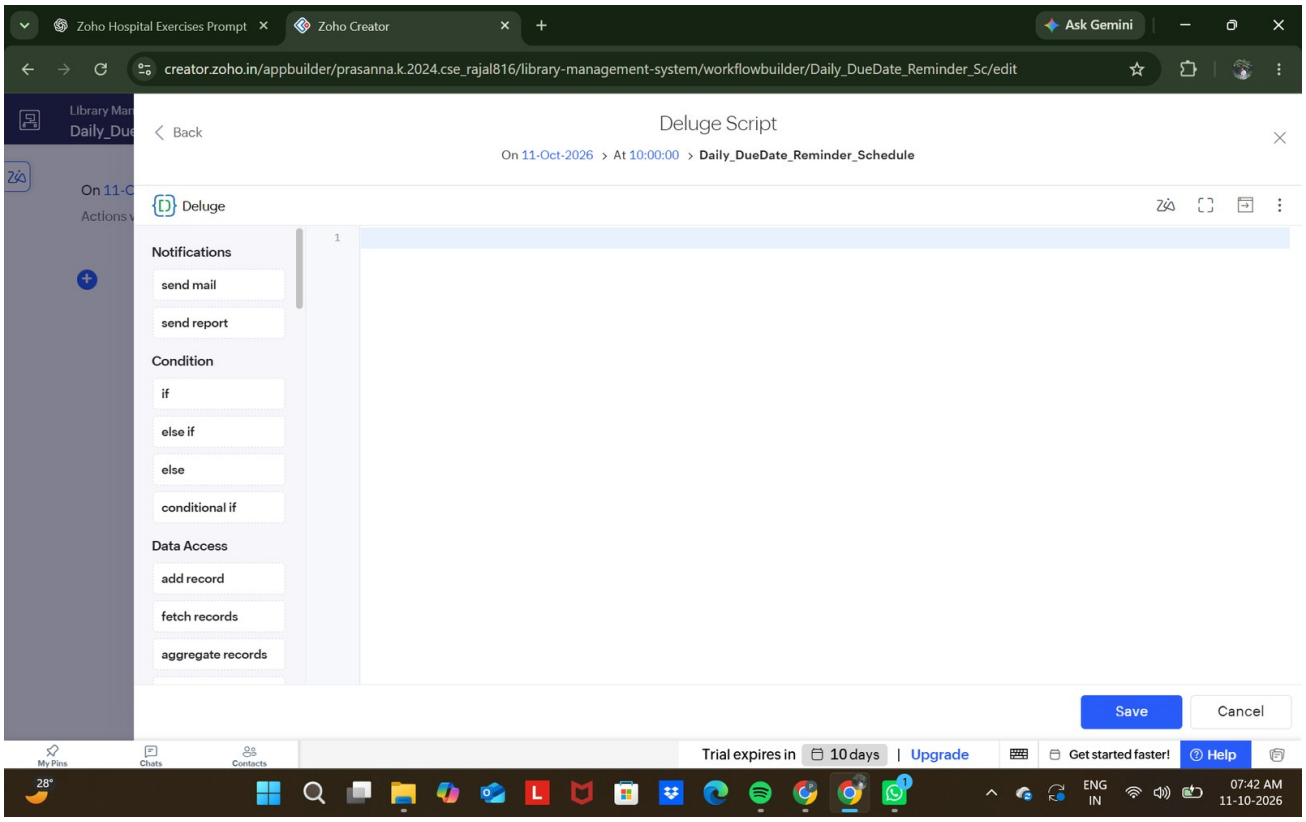


Figure 19: Opening the Deluge editor to implement the reminder logic.

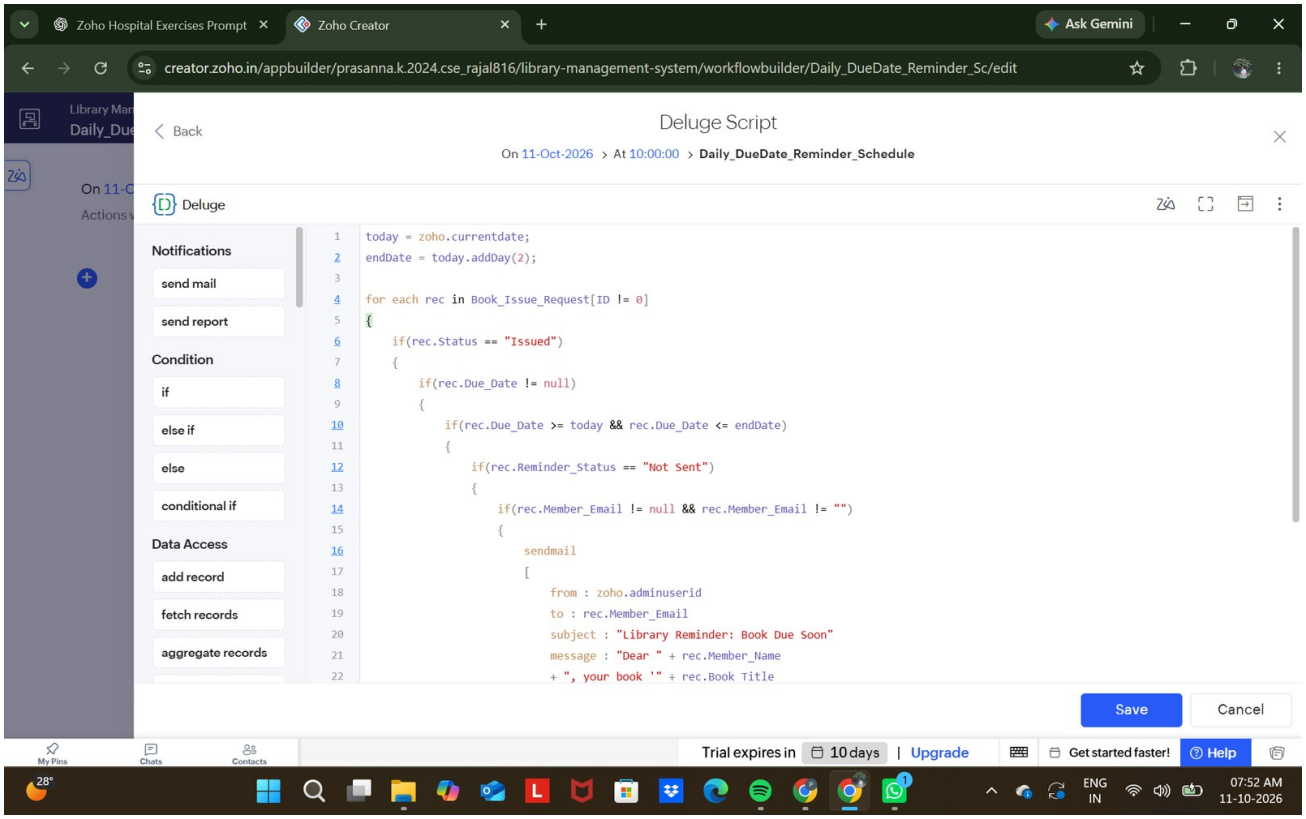


Figure 20: Writing the due-date checks and reminder-status conditions.

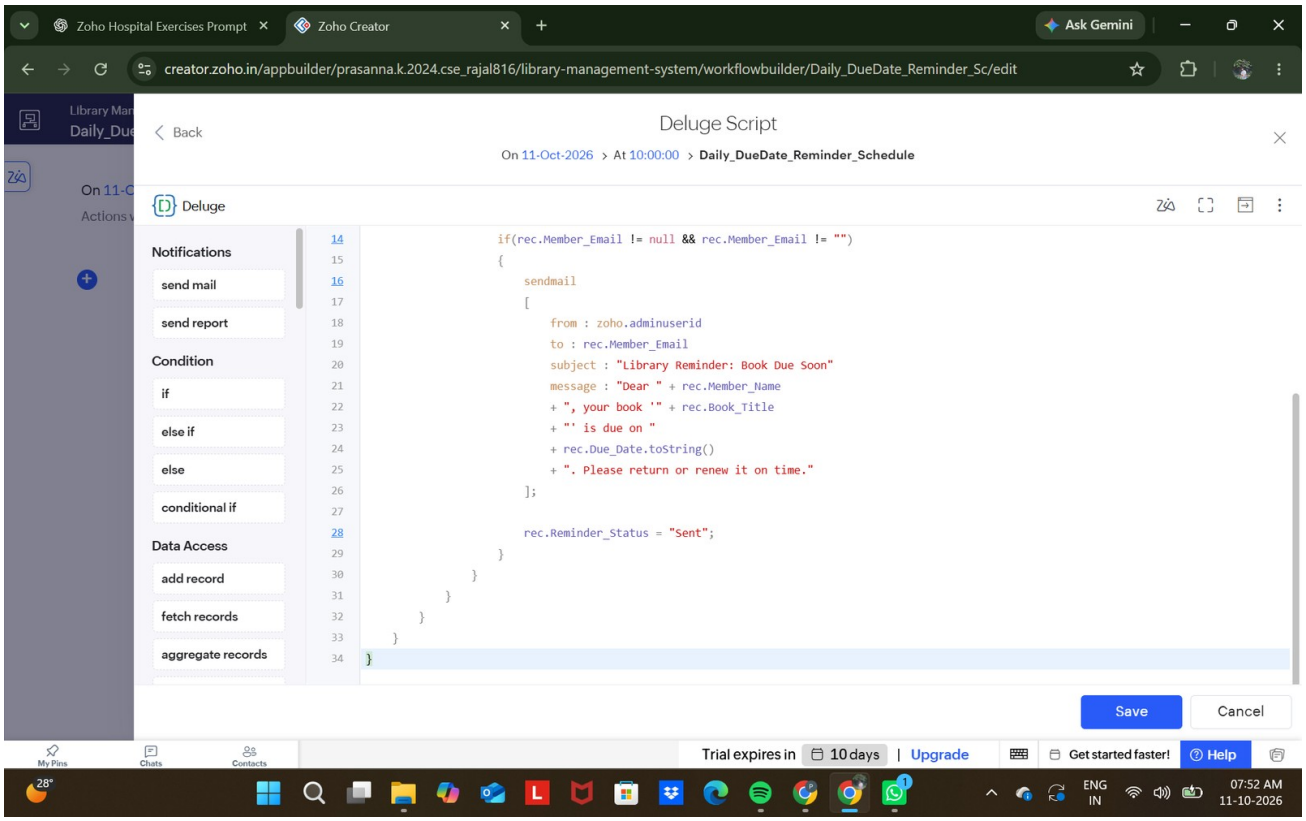


Figure 21: Continuing the script with the member email and reminder message.

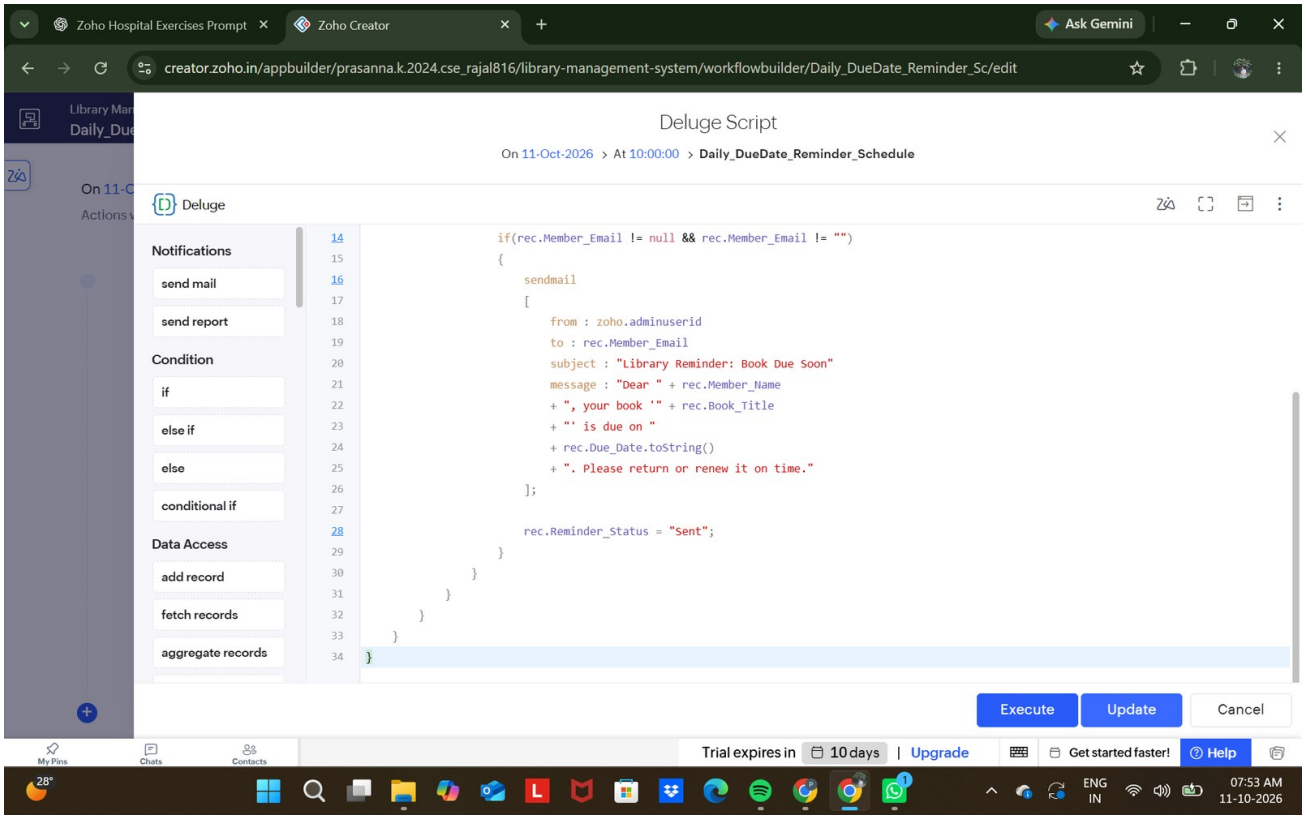


Figure 22: Reviewing the completed reminder email script and status update.

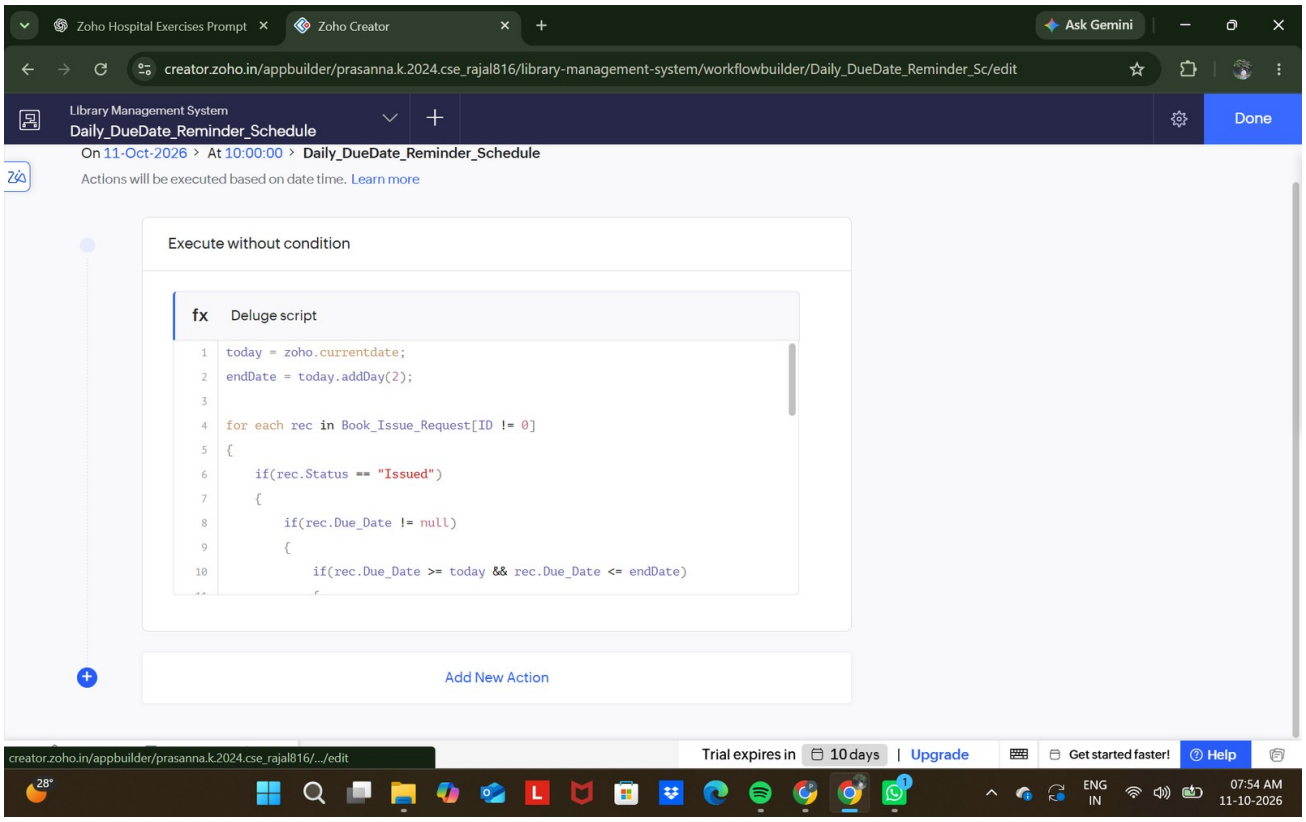


Figure 23: Reviewing the saved scheduled-workflow script.

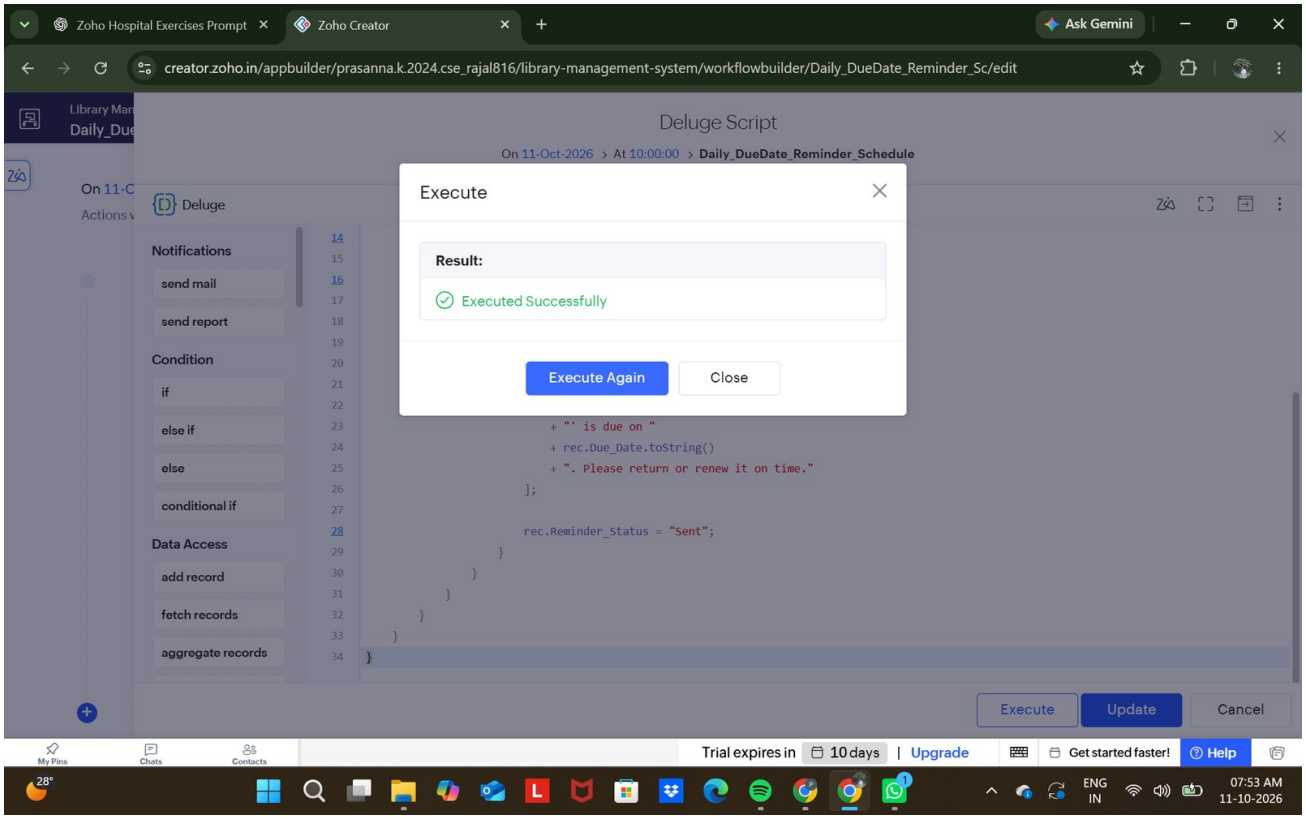


Figure 24: Confirming the scheduled reminder script reports Executed Successfully.

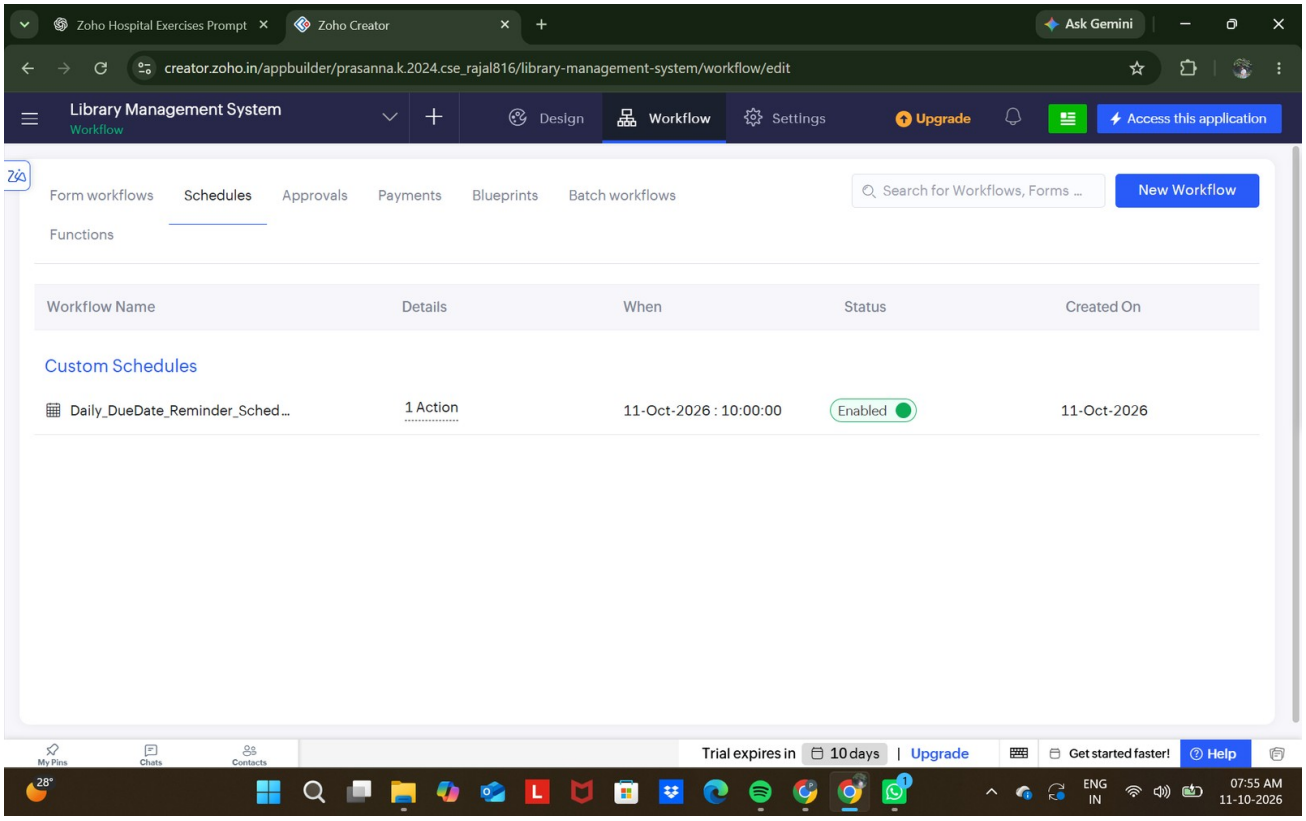


Figure 25: Verifying the Daily_DueDate_Reminder_Schedule is enabled.

3. Approvals Configuration

The RareBook_Issue_Approval process was set up for Book Issue Request records when Book_Category equals Rare/Reference-Only. An approver was added with Approve and Reject decisions, and email notification actions were configured for the approval outcome.

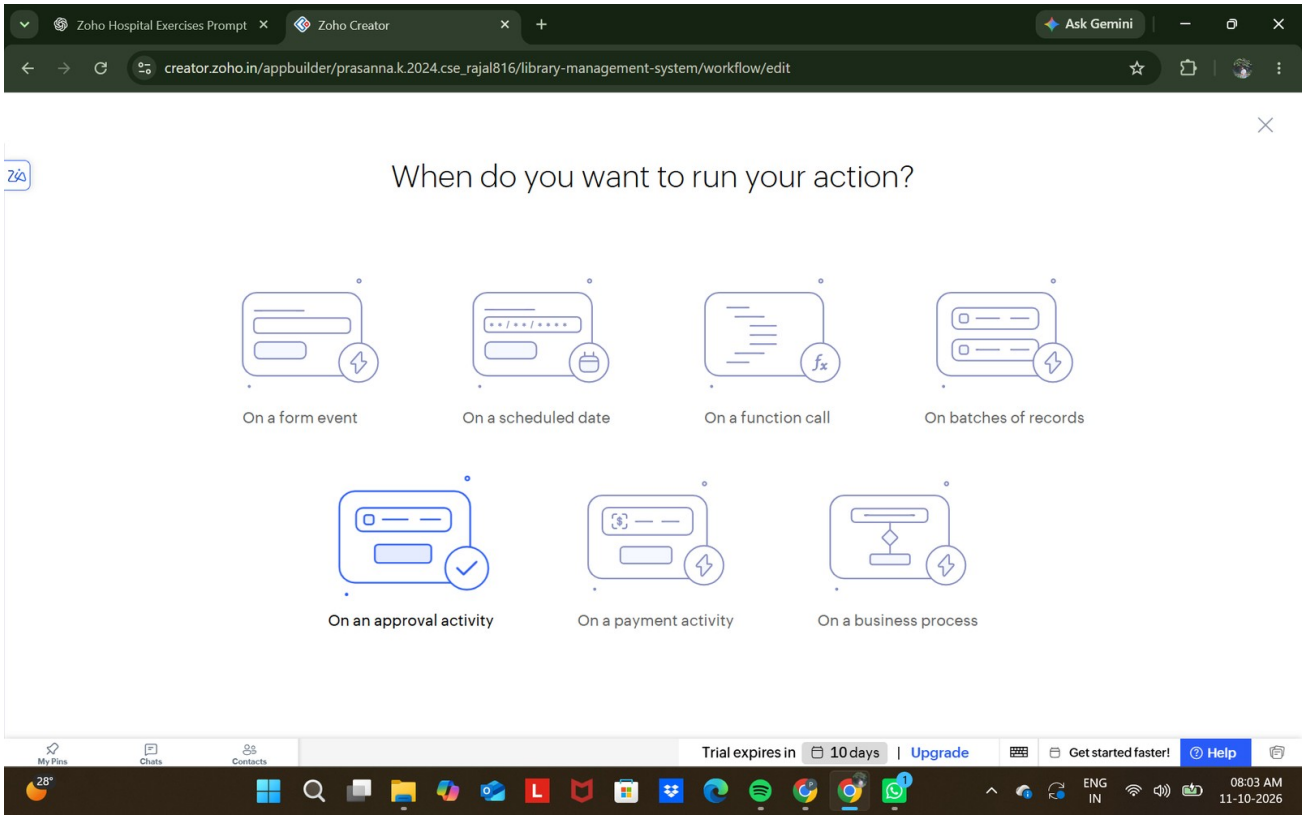


Figure 26: Selecting the approval-activity option for the next workflow.

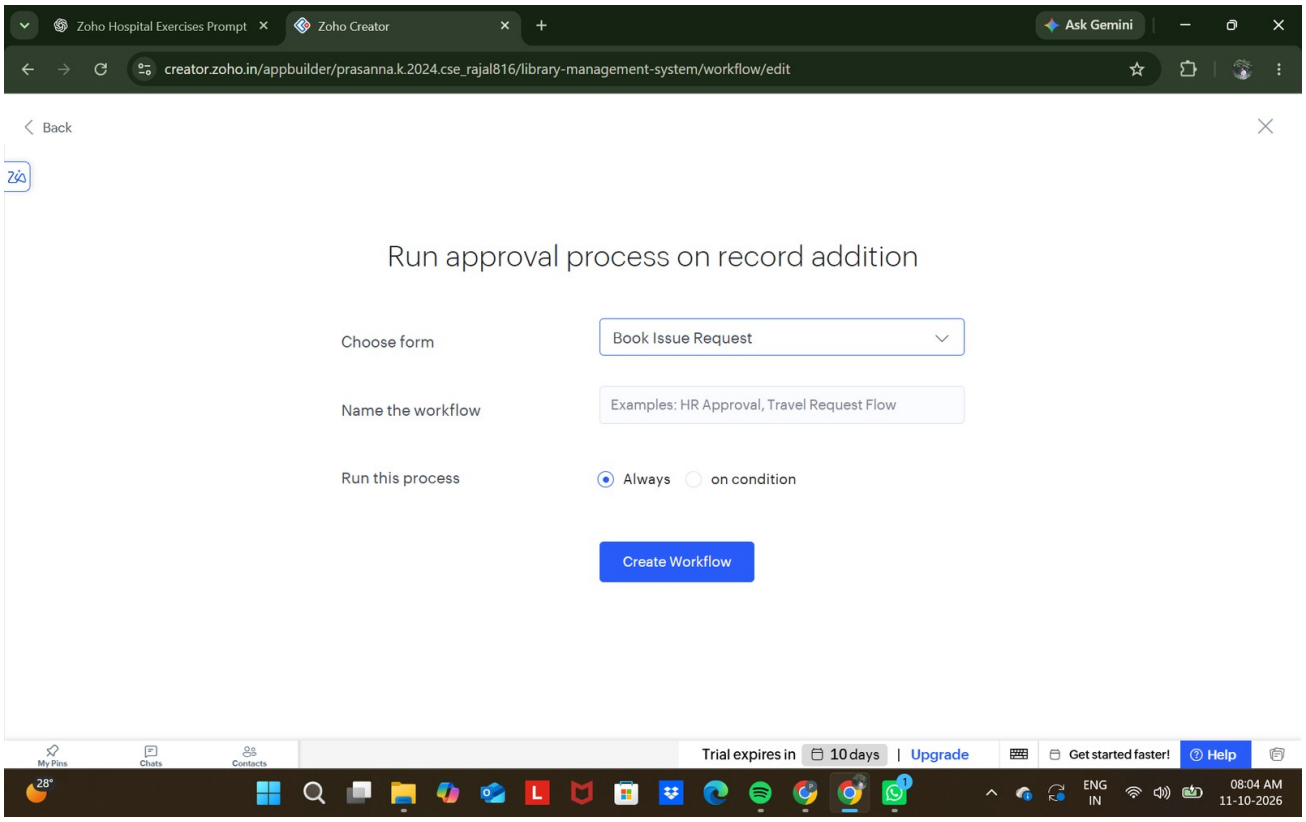


Figure 27: Creating the approval process and specifying its condition.

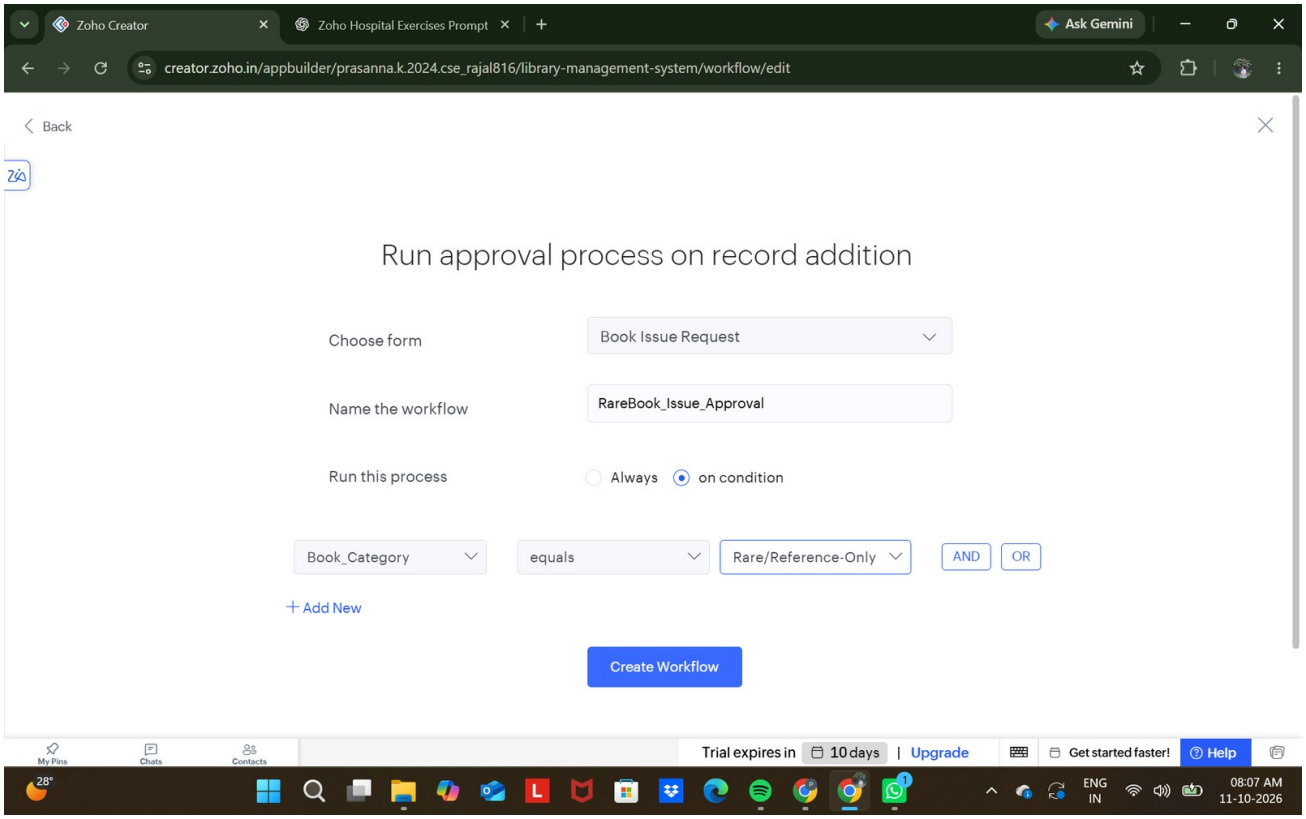


Figure 28: Setting the condition Book_Category equals Rare/Reference-Only.

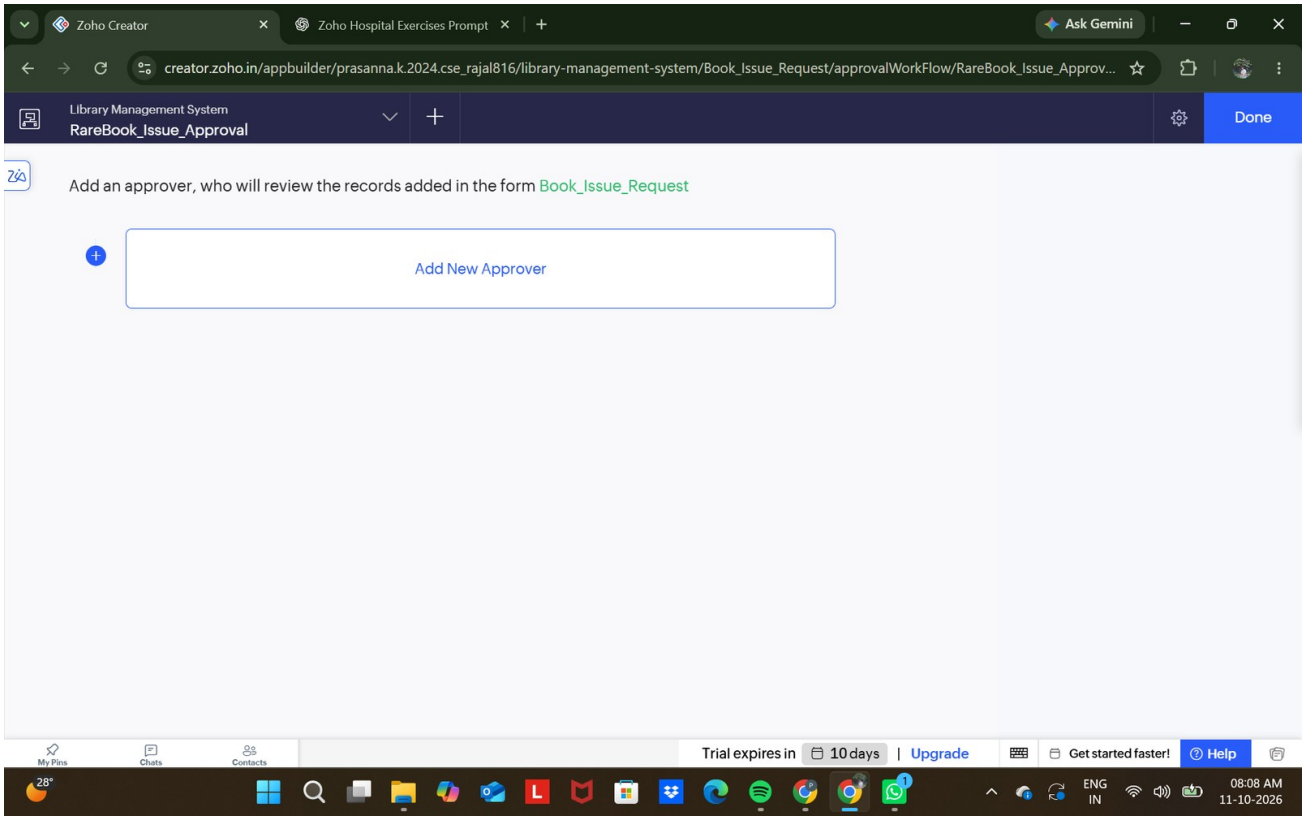


Figure 29: Opening the approval editor to add an approver.

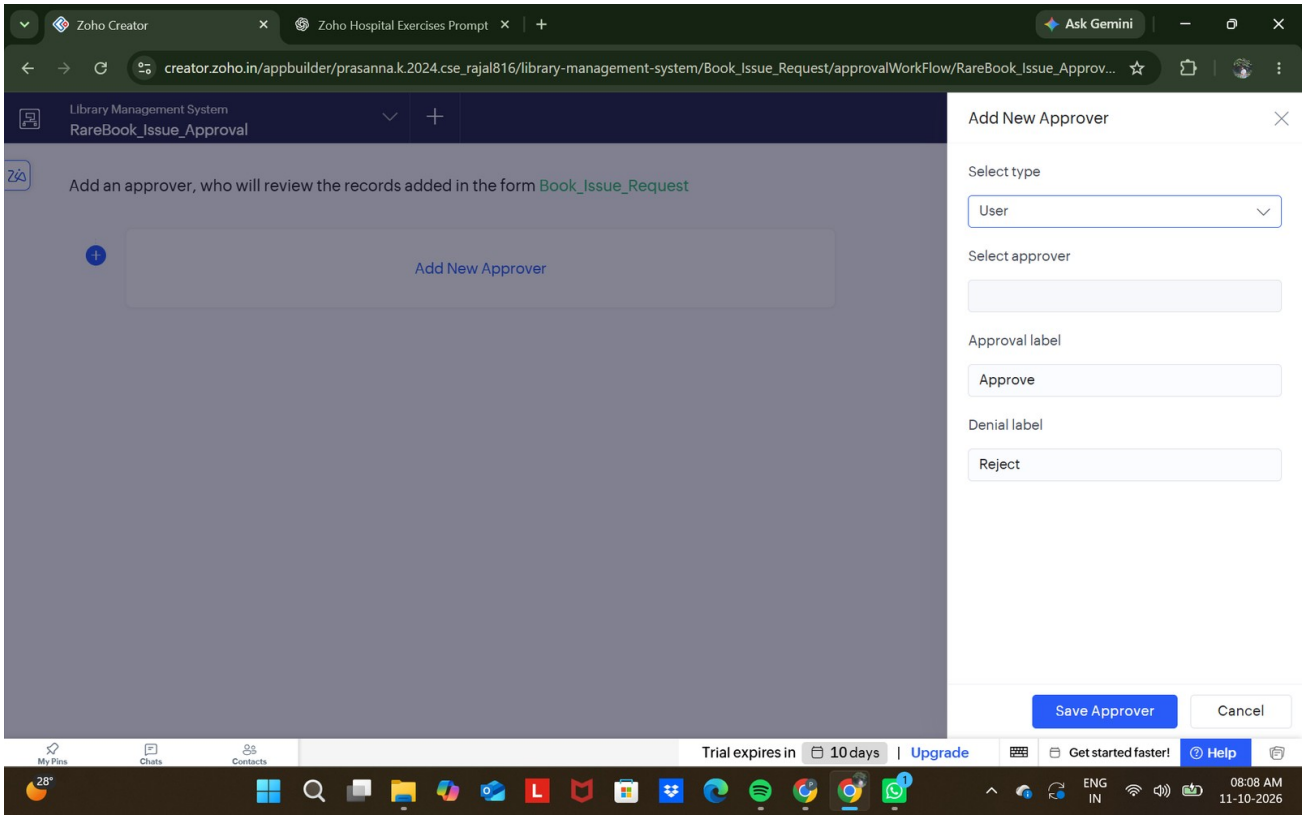


Figure 30: Selecting the approver and setting Approve and Reject labels.

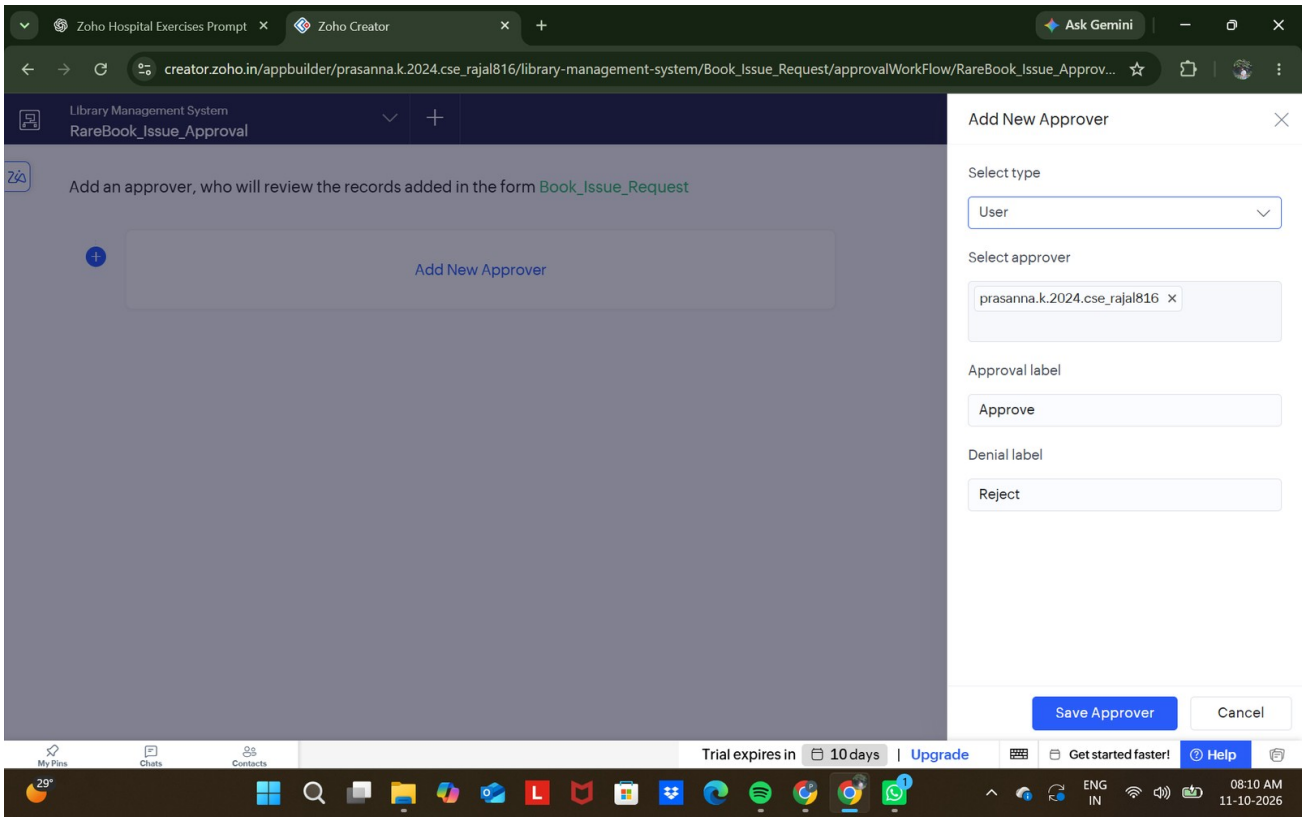


Figure 31: Reviewing the approver configuration.

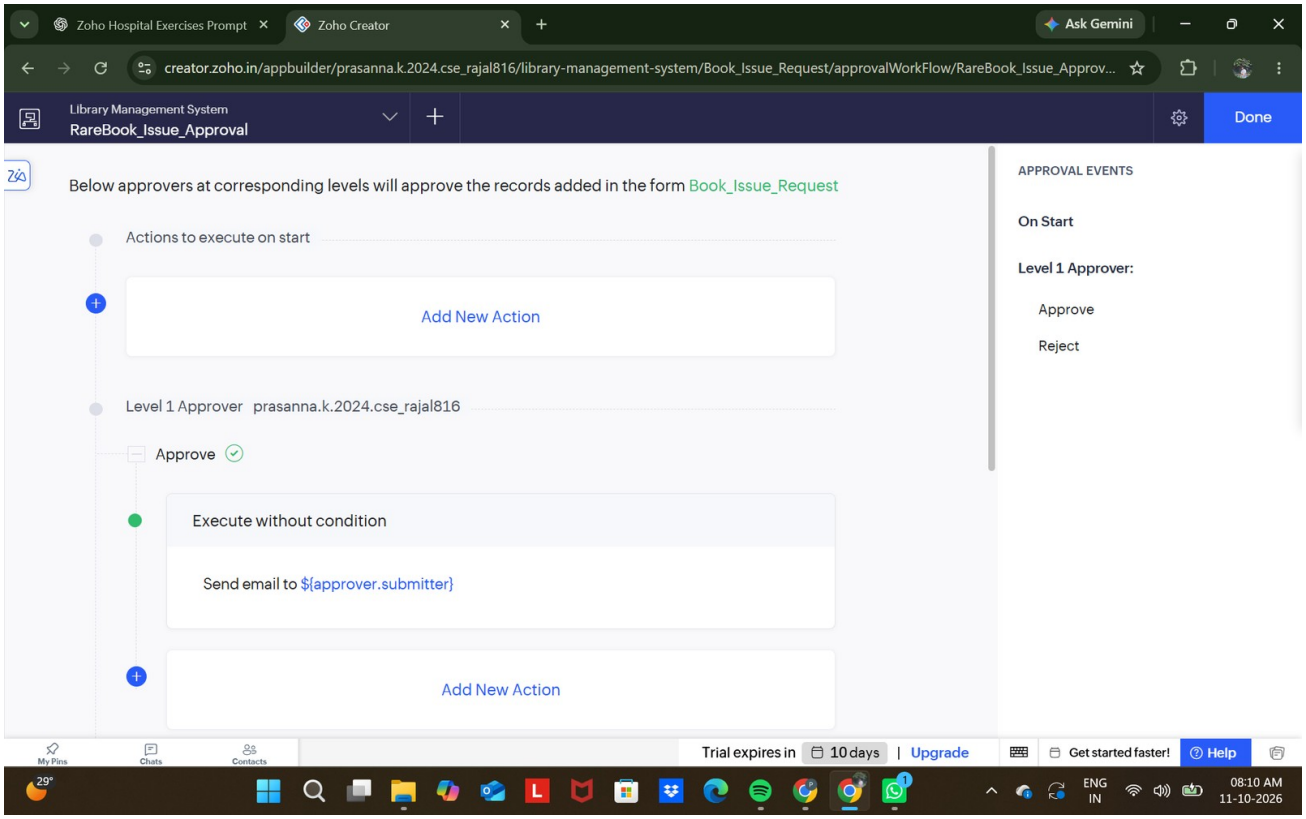


Figure 32: Configuring actions for the approval decision.

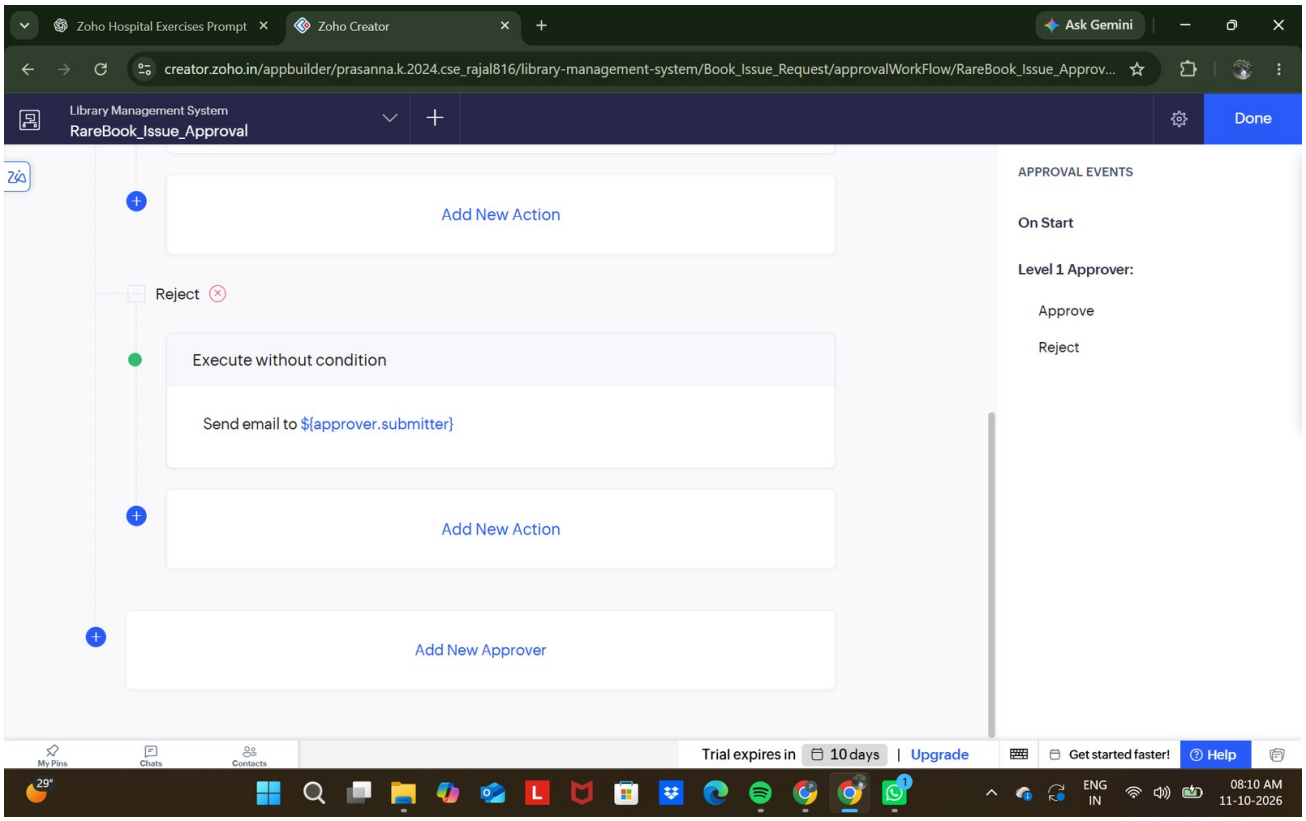


Figure 33: Reviewing email notification actions in the approval process.

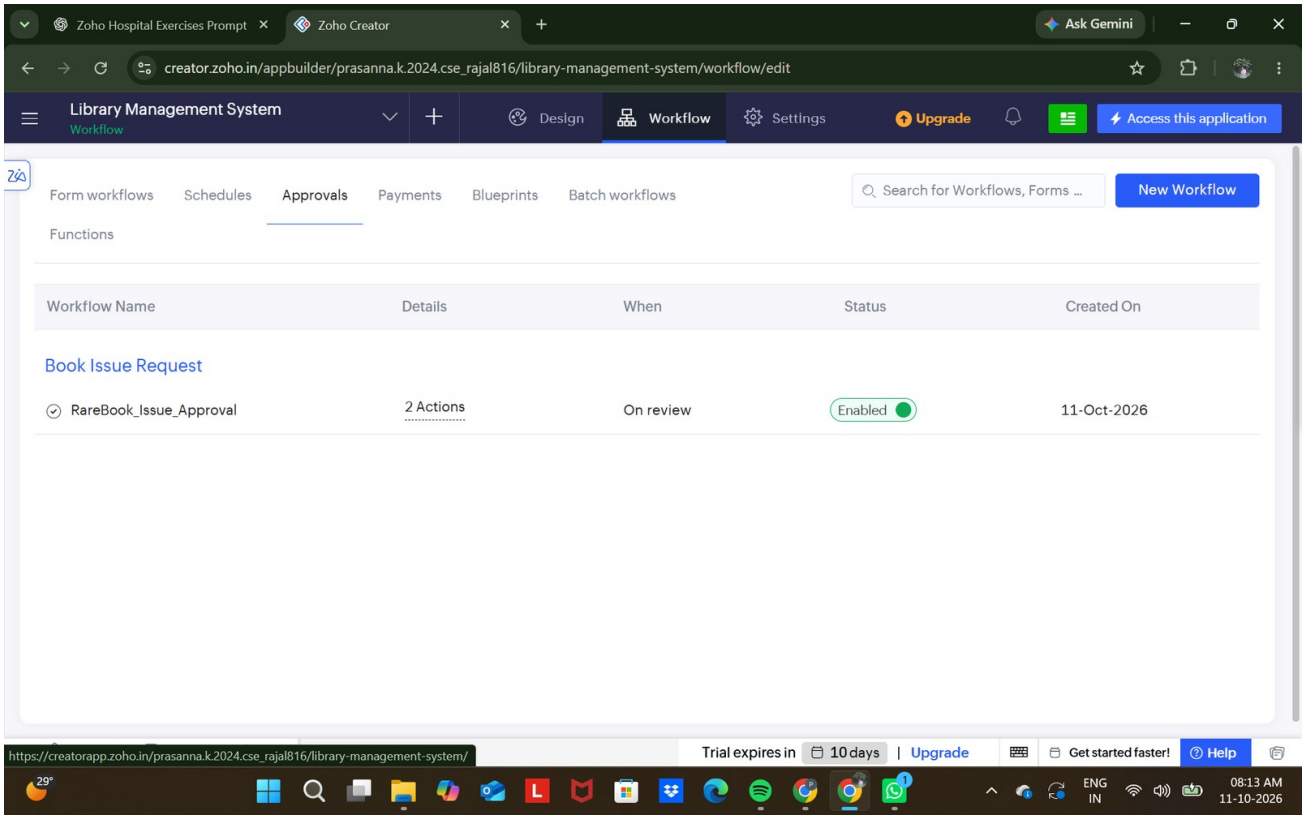


Figure 34: Verifying the RareBook_Issue_Approval process is enabled.

4. Payments Configuration

Fine_Payment_Workflow was configured to use Total_Fine_Payable as the payment amount and to collect payment when the total is greater than zero. Payment Success and Payment Failure branches include notification and record-update actions. During setup, Zoho Creator continued to display a payment-gateway configuration prompt, so live payment collection should be treated as pending gateway confirmation.

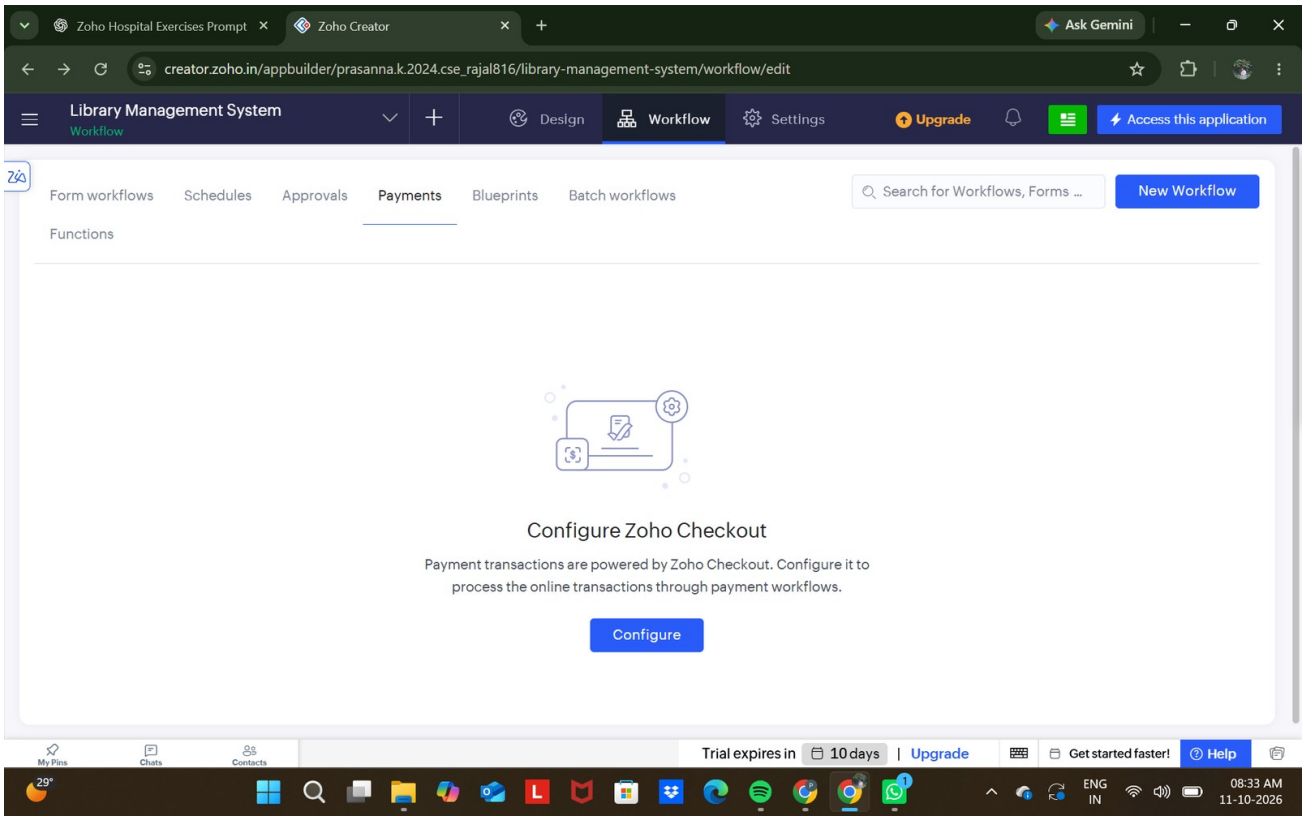


Figure 35: Opening the Payments configuration area and the checkout setup prompt.

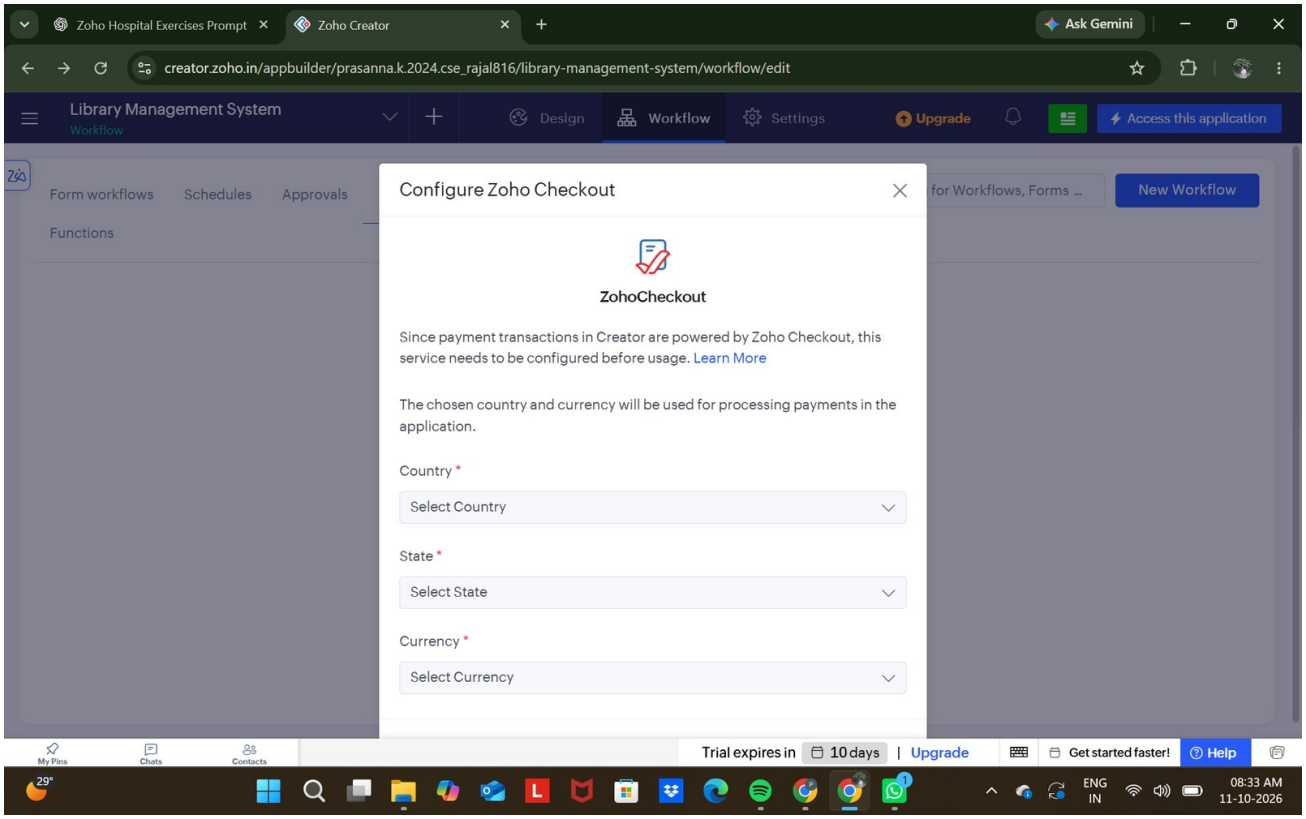


Figure 36: Reviewing the available payment gateway providers.

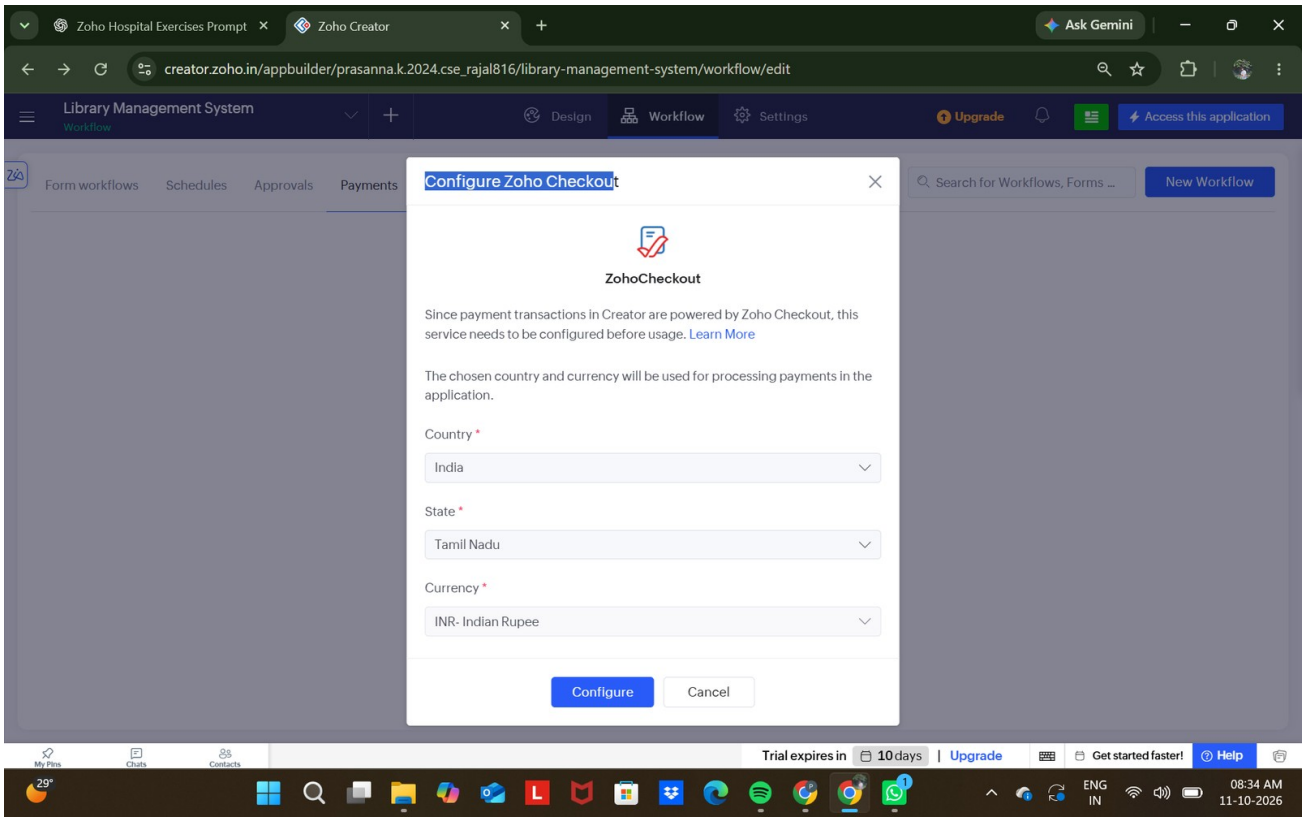


Figure 37: Opening the Zoho Checkout configuration dialog.

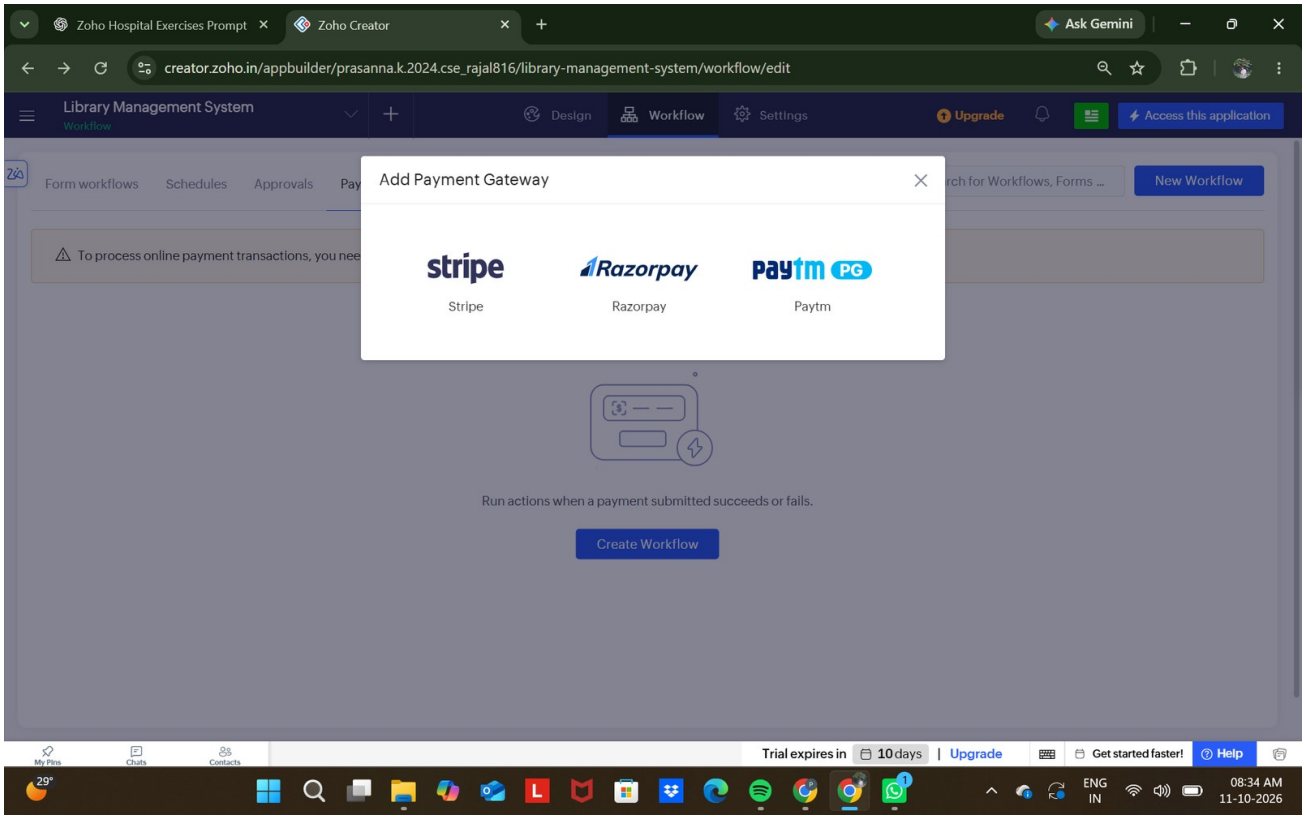


Figure 38: Reviewing country, state and currency settings for checkout.

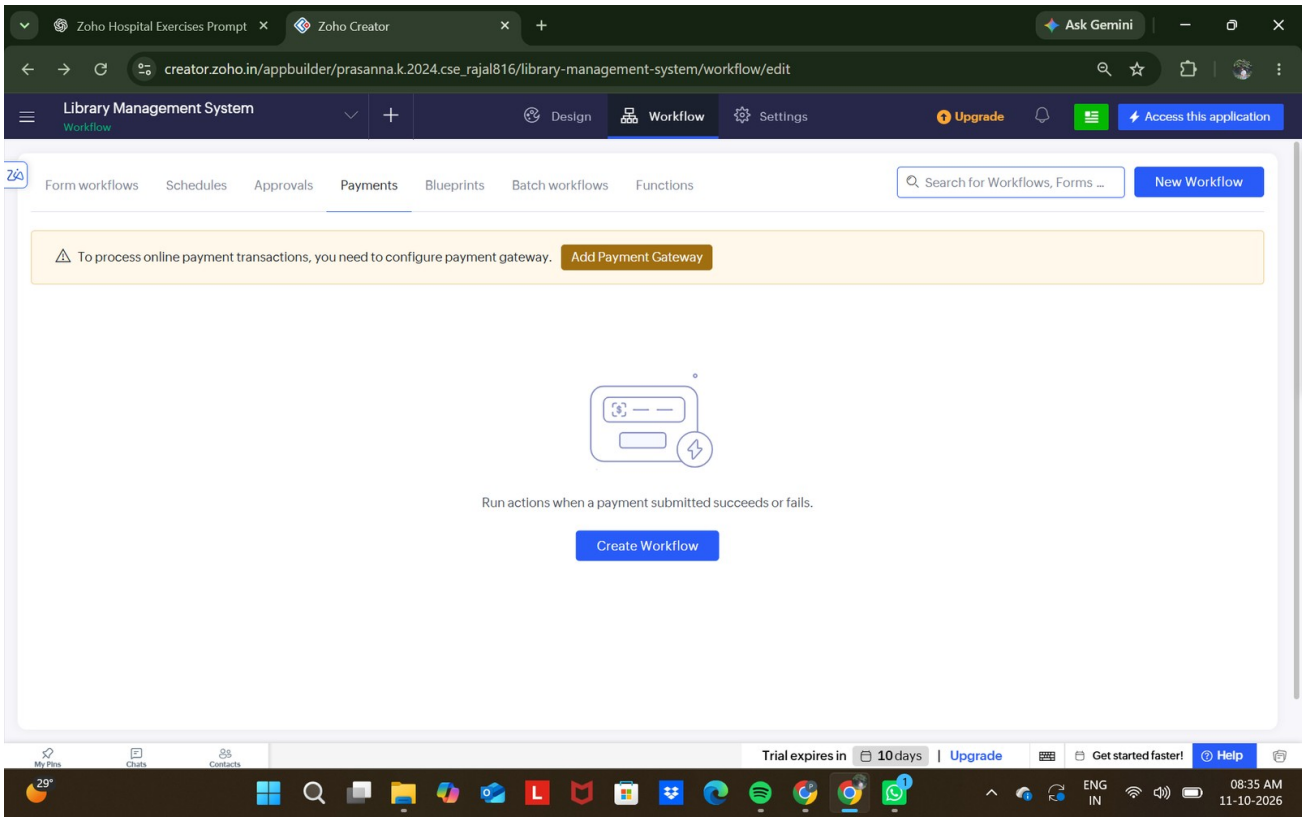


Figure 39: Returning to the payment workflow area and reviewing gateway readiness.

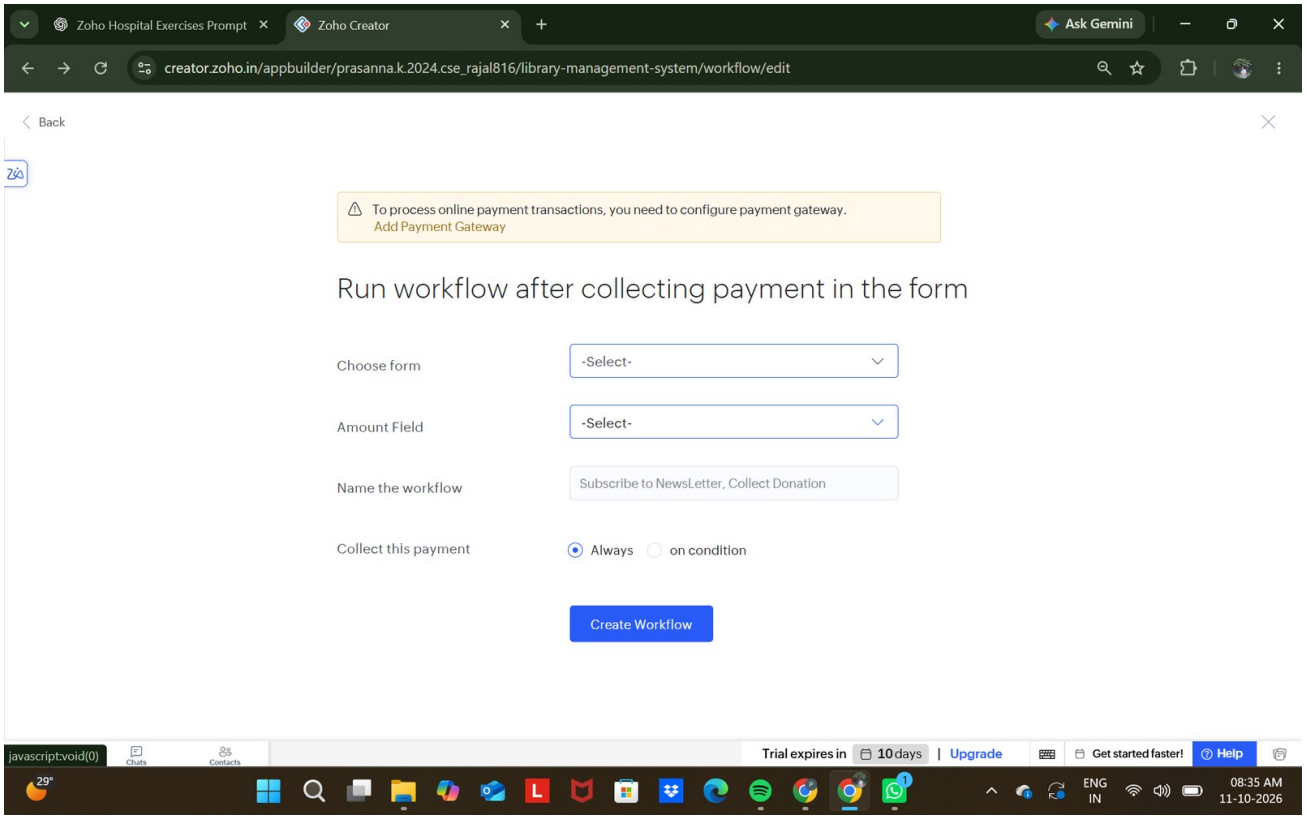


Figure 40: Selecting Book Issue Request and Total_Fine_Payable for the payment workflow.

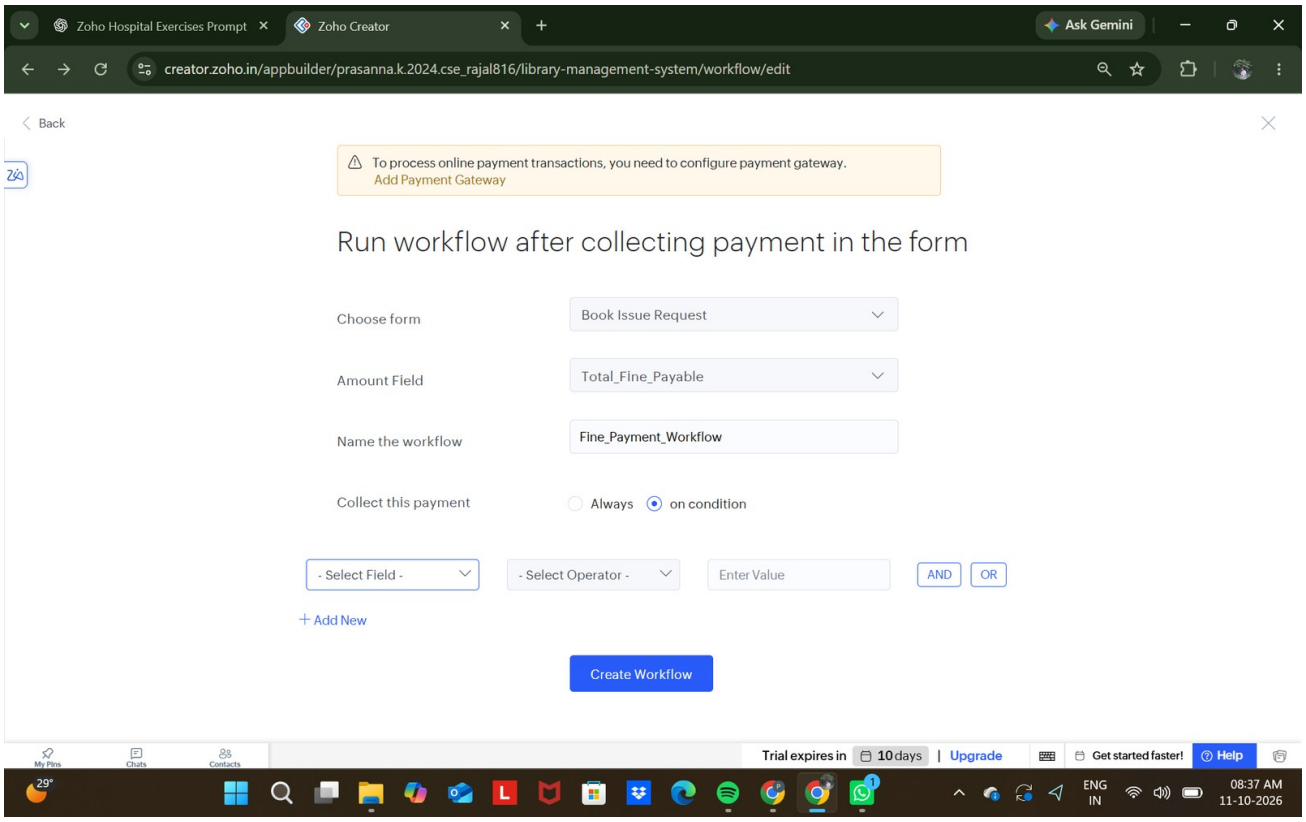


Figure 41: Setting the condition that Total_Fine_Payable must be greater than zero.

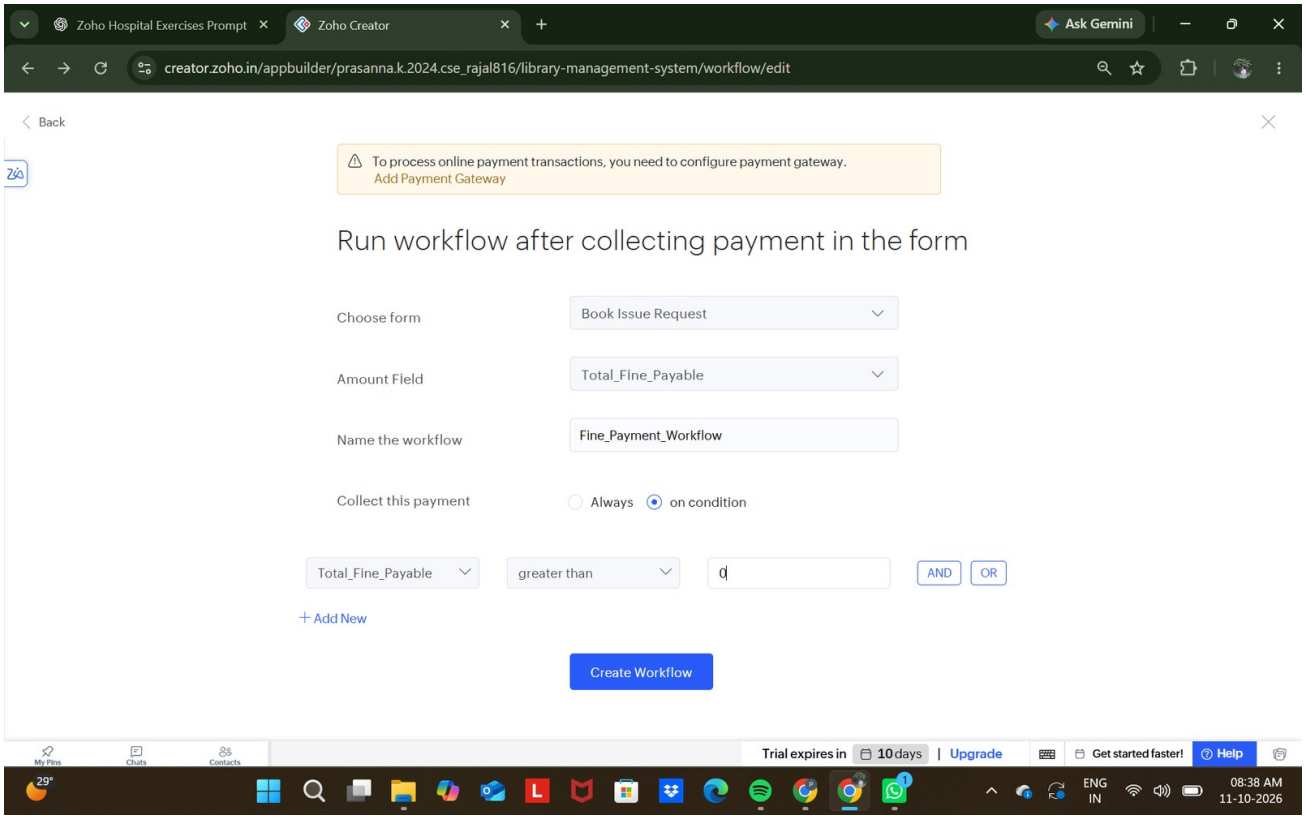


Figure 42: Opening the payment workflow editor after creating the workflow.

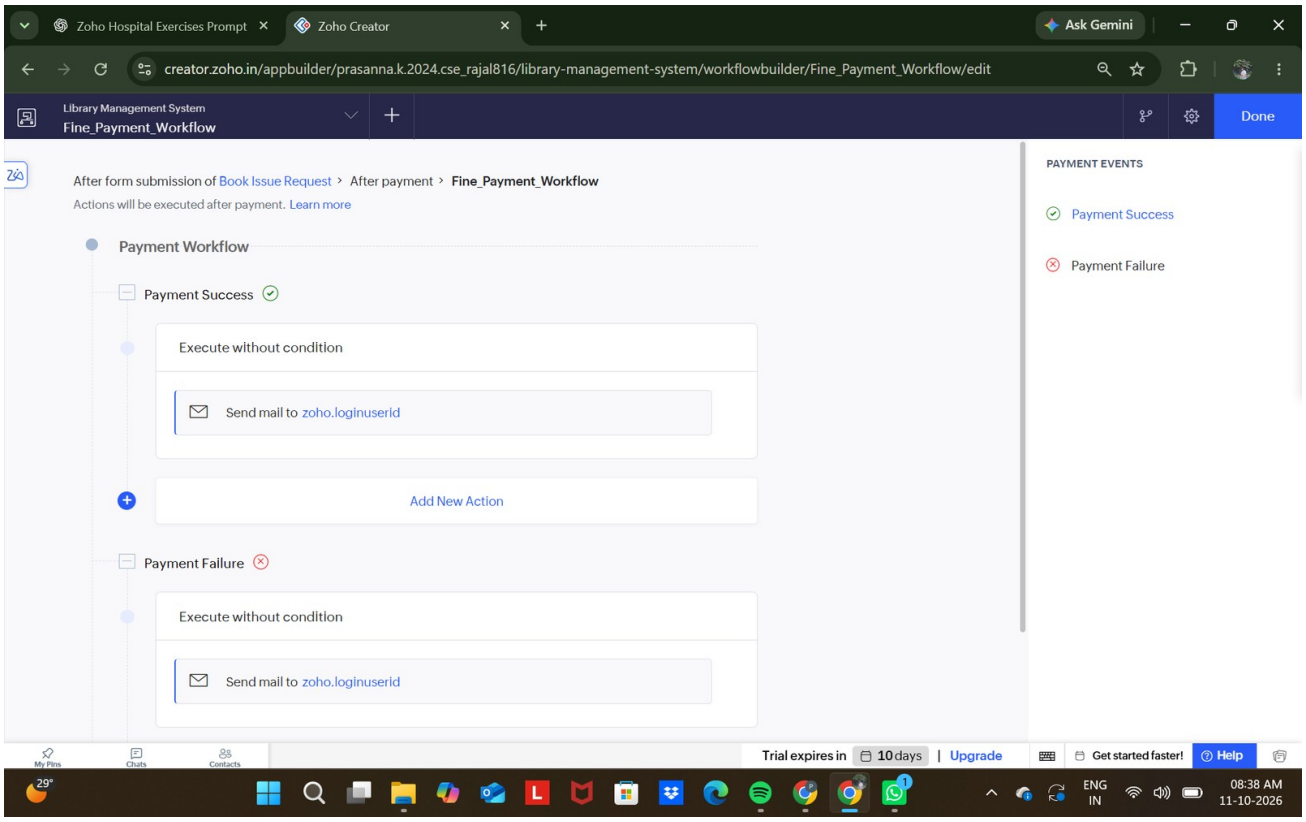


Figure 43: Configuring payment-success actions and recipient notification.

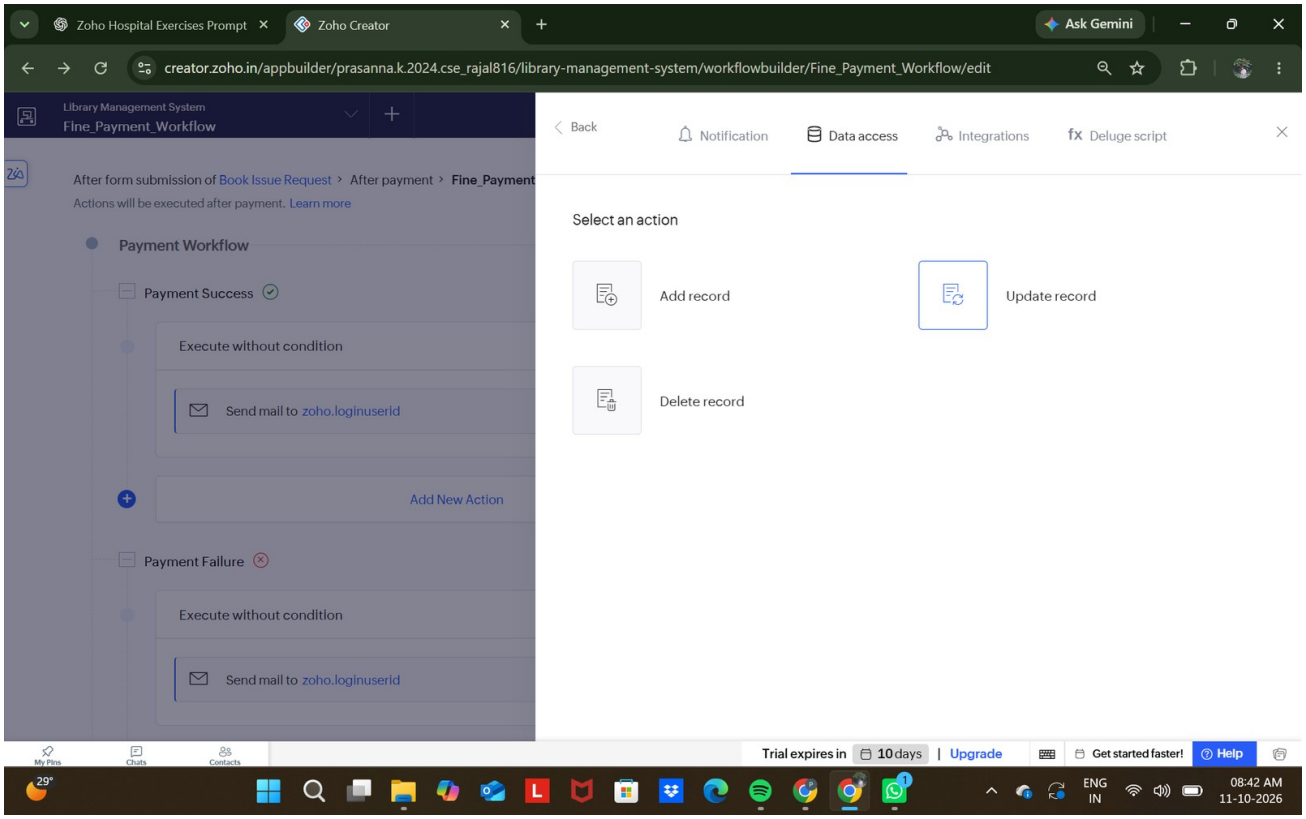


Figure 44: Selecting a data-access action for updating the current record.

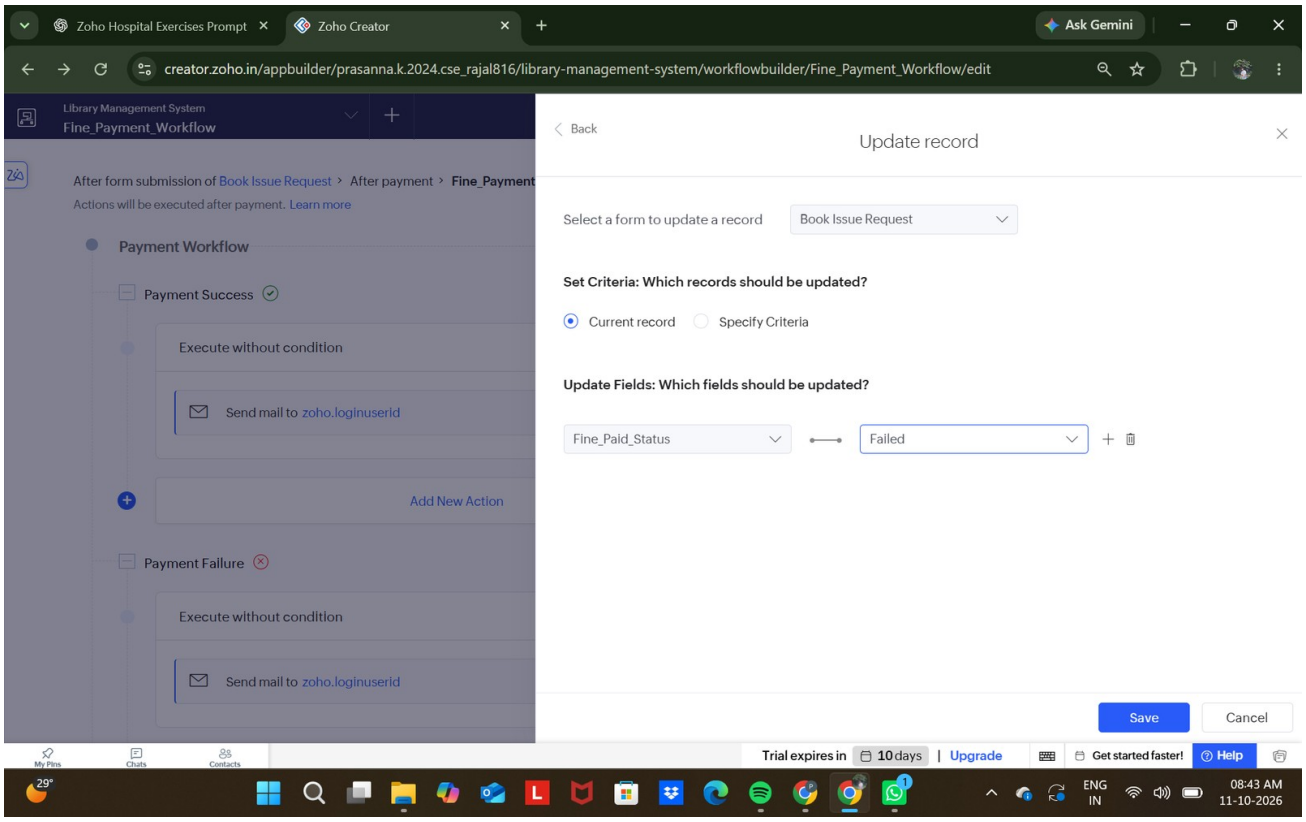


Figure 45: Setting Fine_Paid_Status as a field to update after payment.

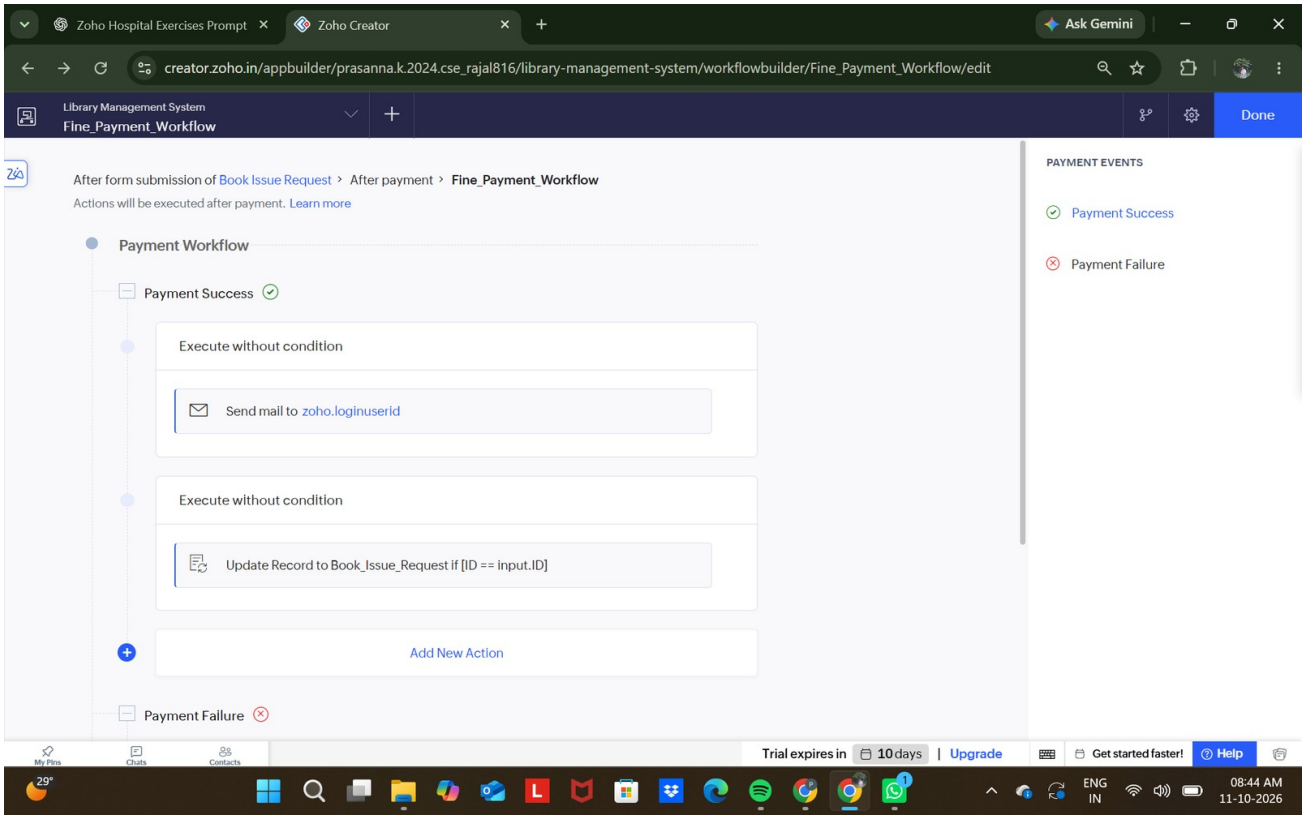


Figure 46: Reviewing the success and failure branches in the payment workflow.

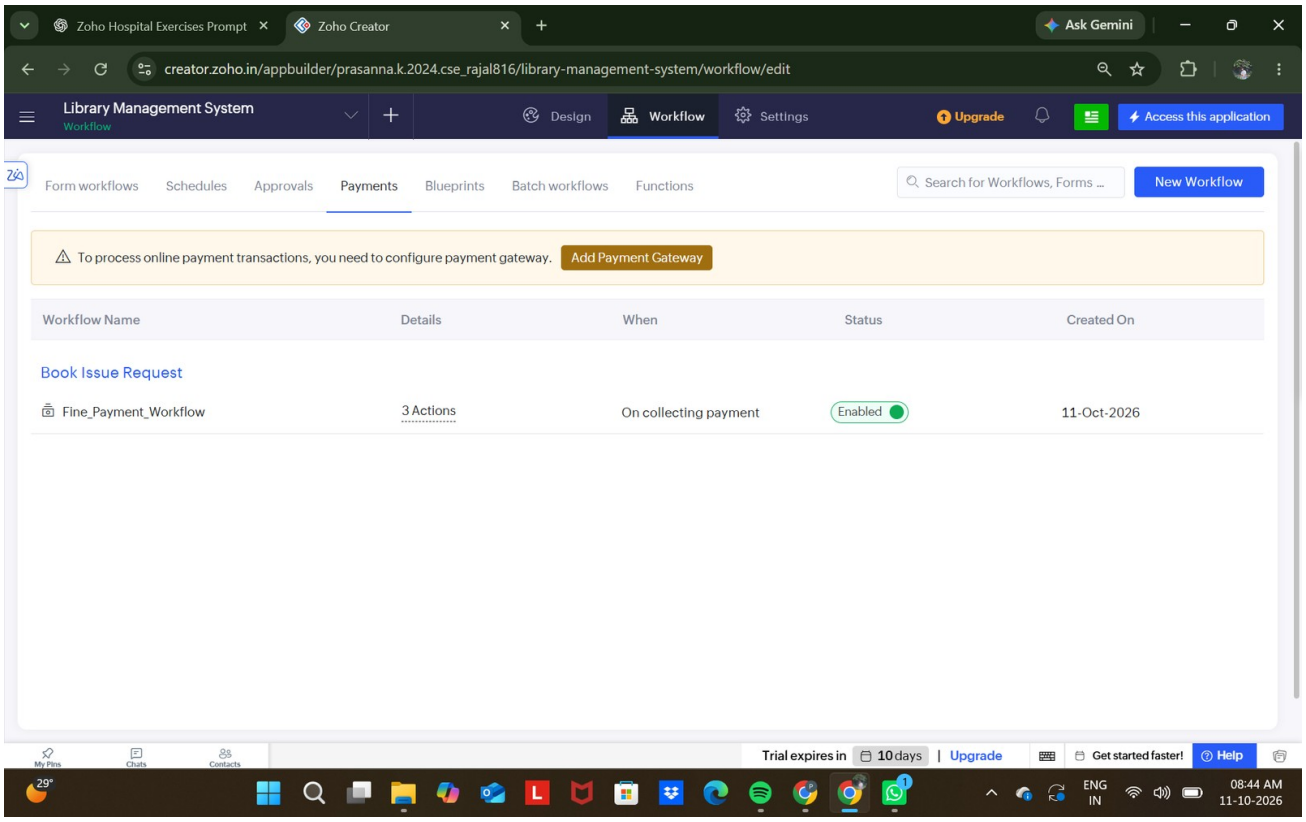


Figure 47: Checking the Fine_Payment_Workflow status in the Payments list.

5. Blueprints Configuration

Issue_Lifecycle_Blueprint was designed for the Book Issue Request form using the Requested, Approved, Issued, Returned and Overdue stages. The main path progresses through request, approval and issue; from Issued, the process can branch to Returned or Overdue.

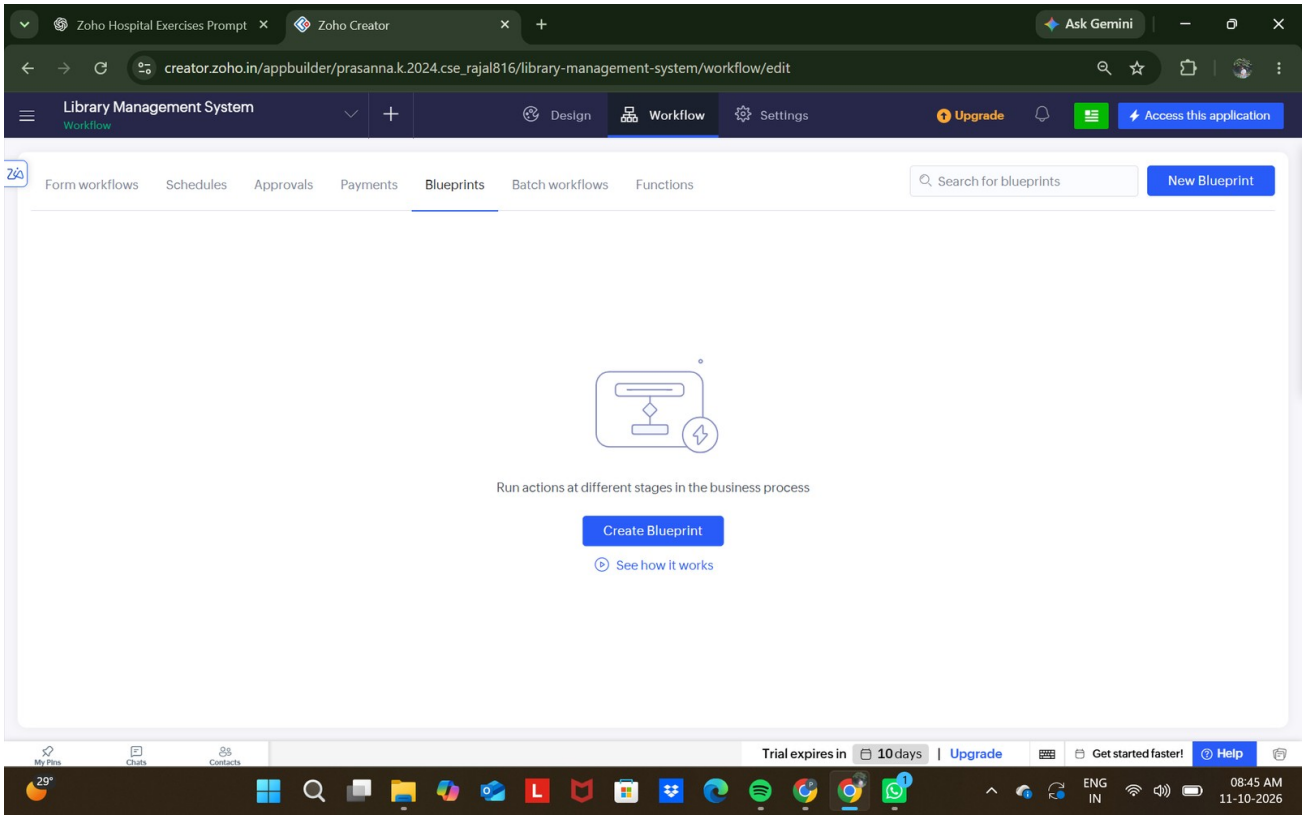


Figure 48: Starting a business-process blueprint for Book Issue Request.

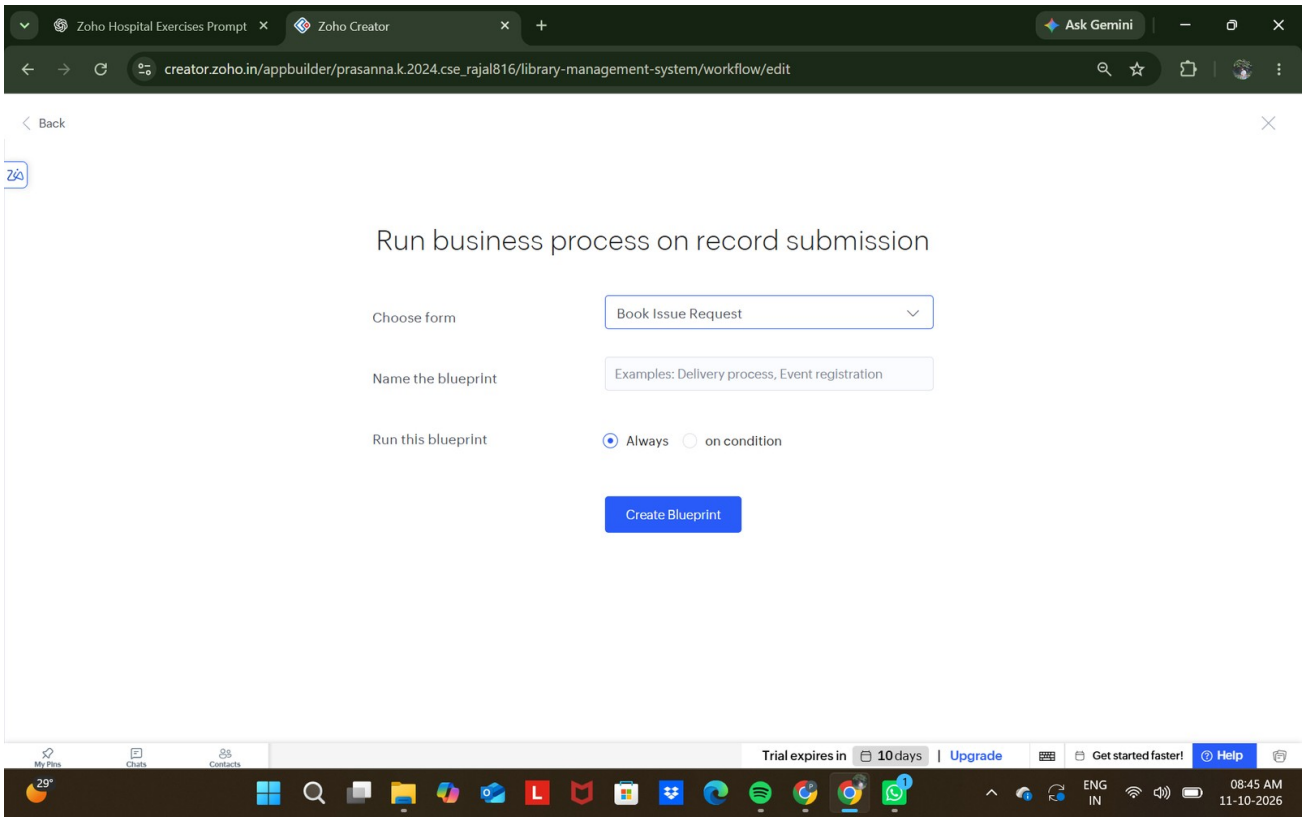


Figure 49: Naming the blueprint for the issue lifecycle.

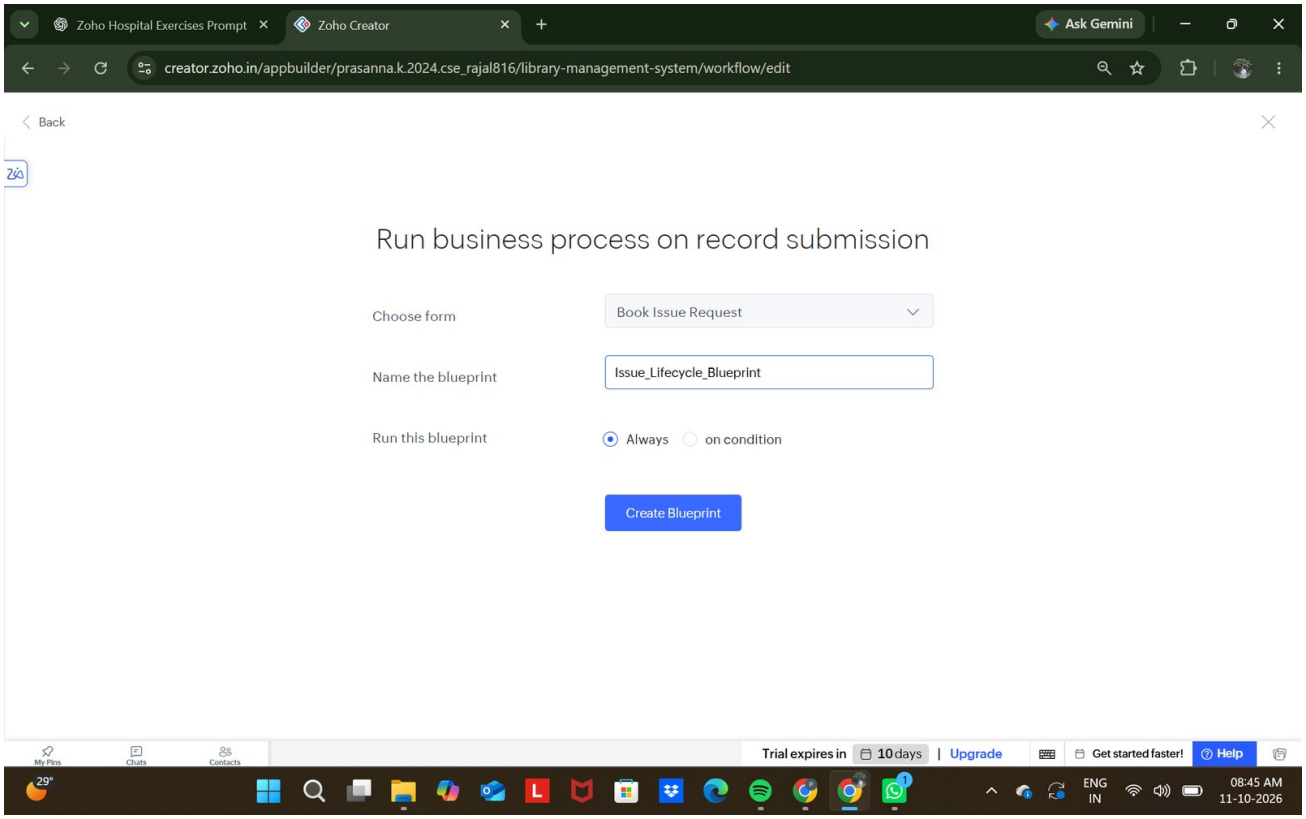


Figure 50: Reviewing the blueprint setup before opening the canvas.

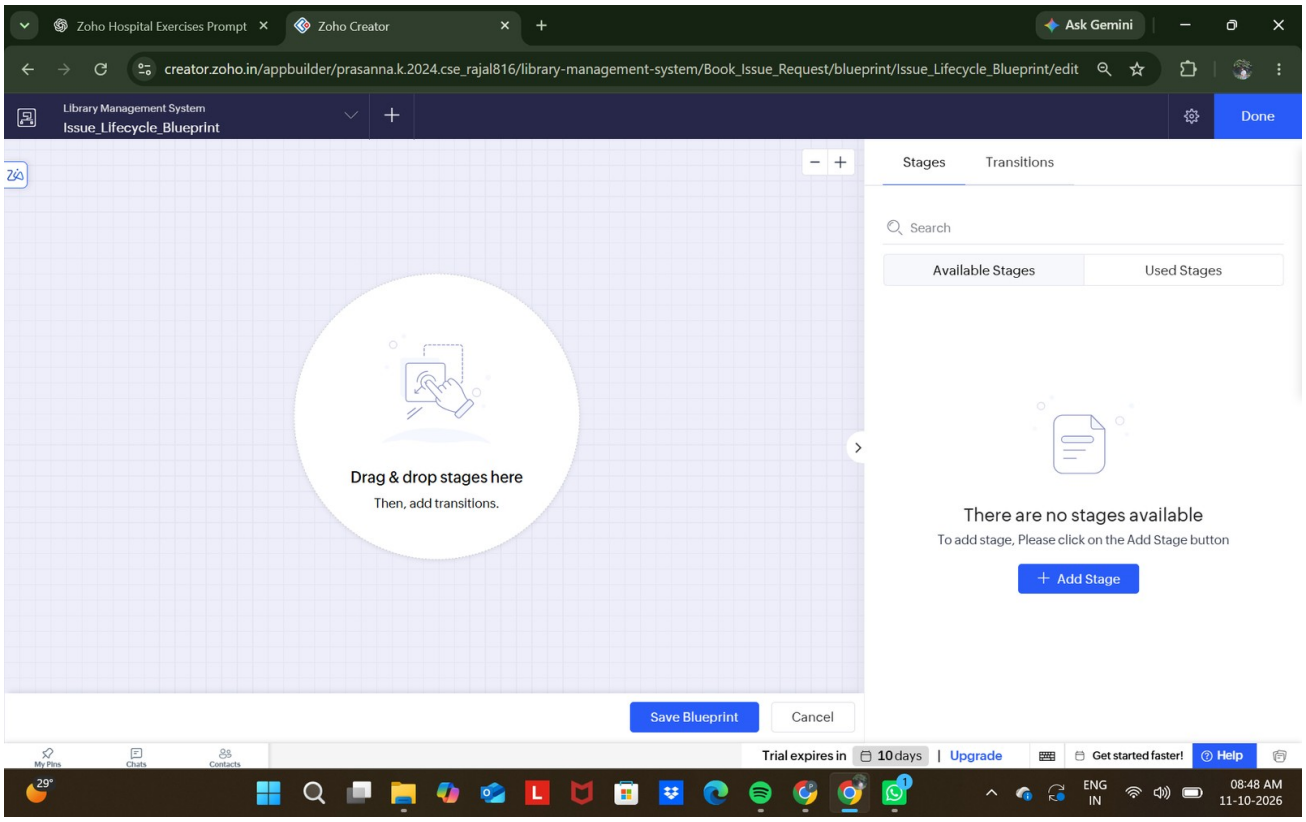


Figure 51: Opening the blueprint canvas and the stage panel.

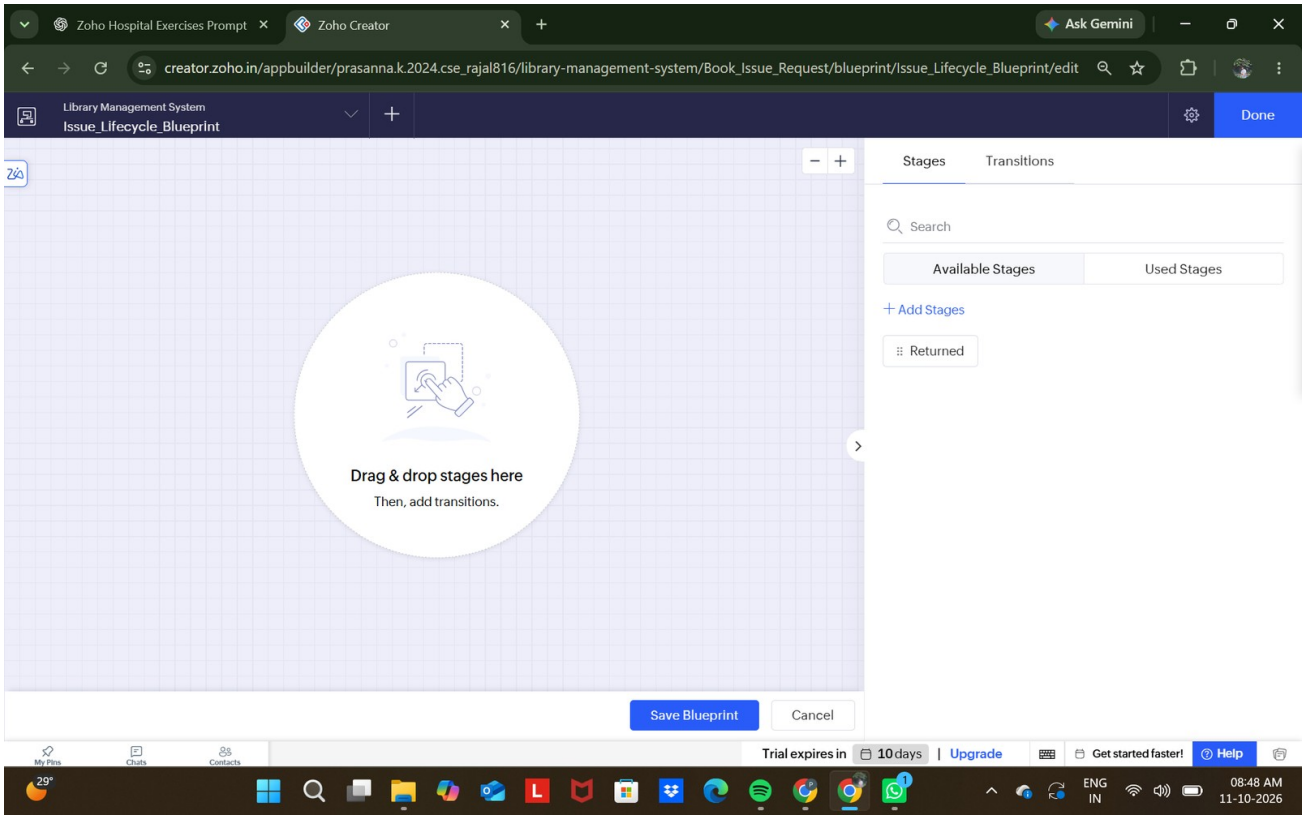


Figure 52: Opening the stage-creation options.

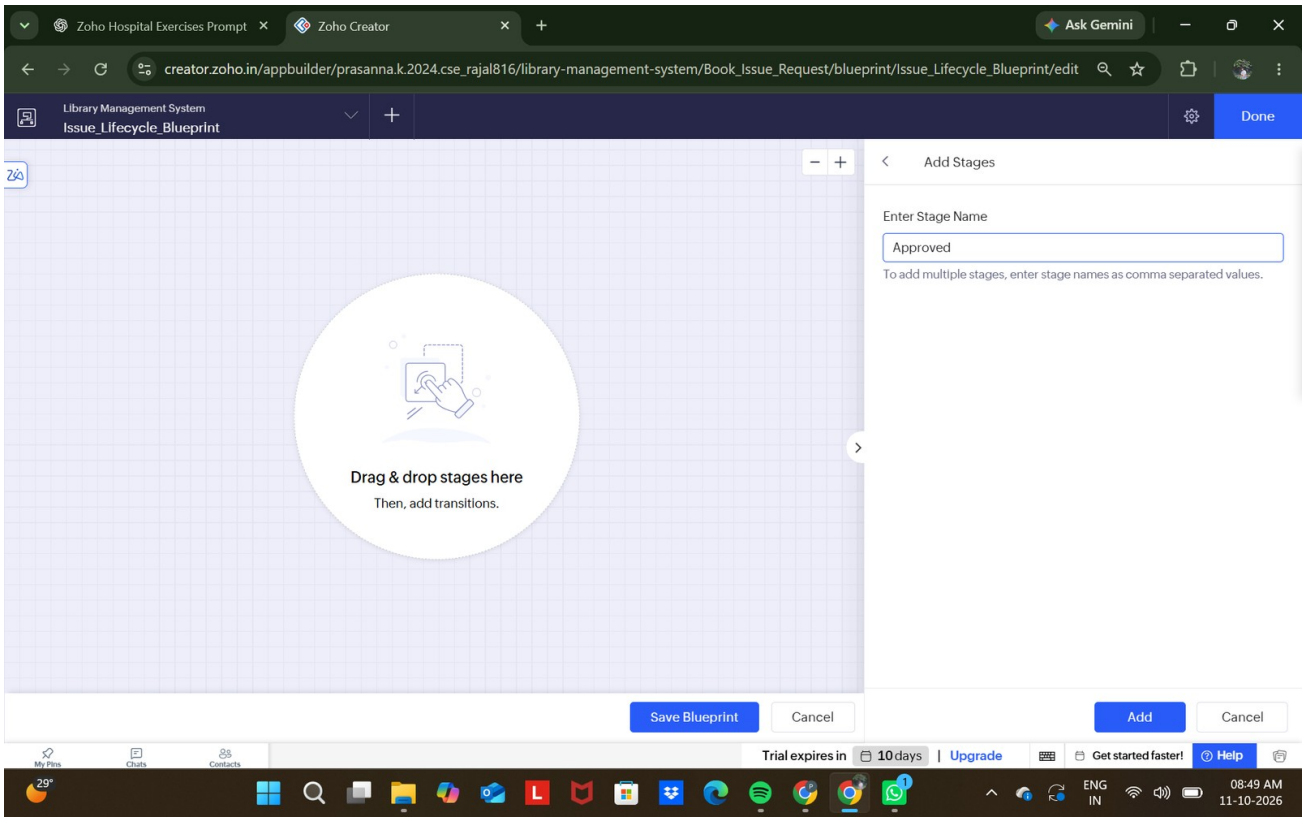


Figure 53: Entering a lifecycle stage in the Add Stages panel.

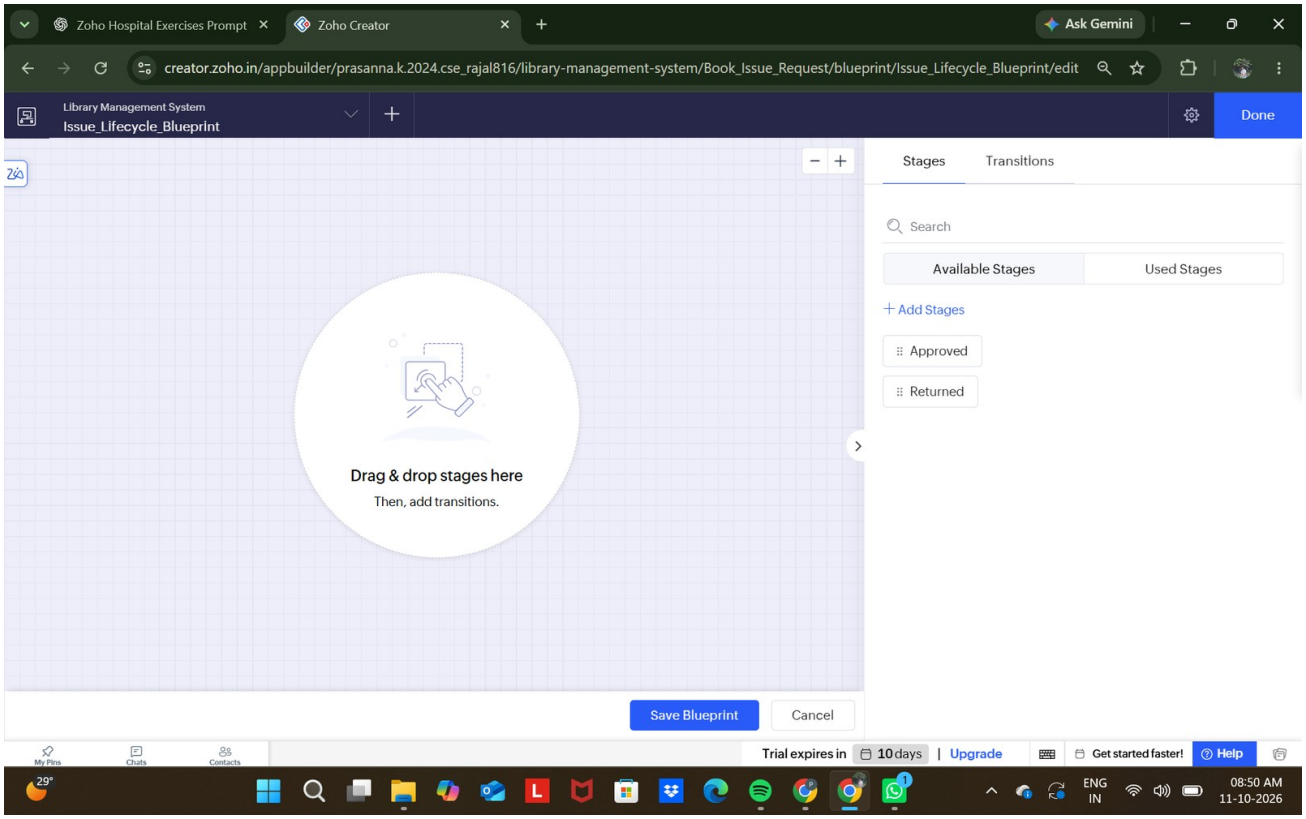


Figure 54: Reviewing stage options available for the blueprint.

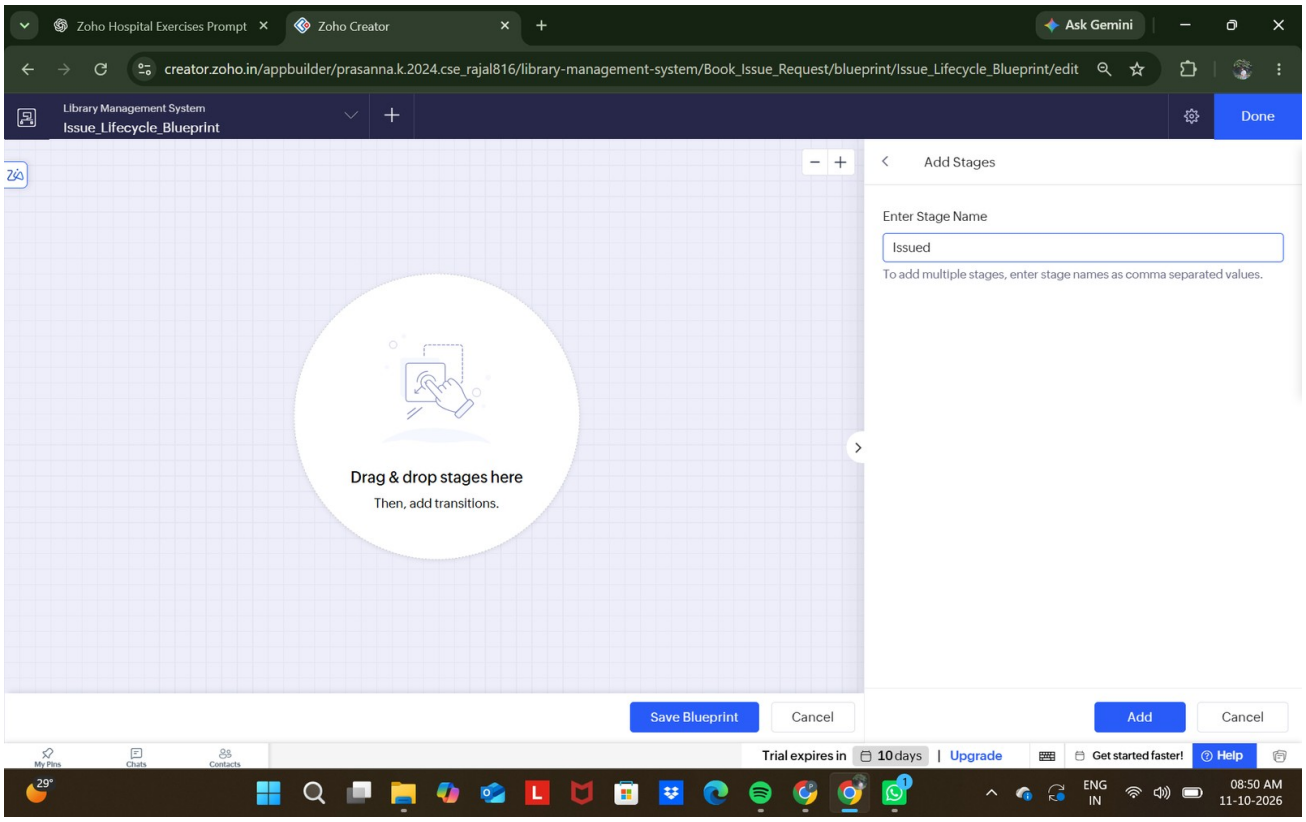


Figure 55: Adding a stage to the blueprint canvas.

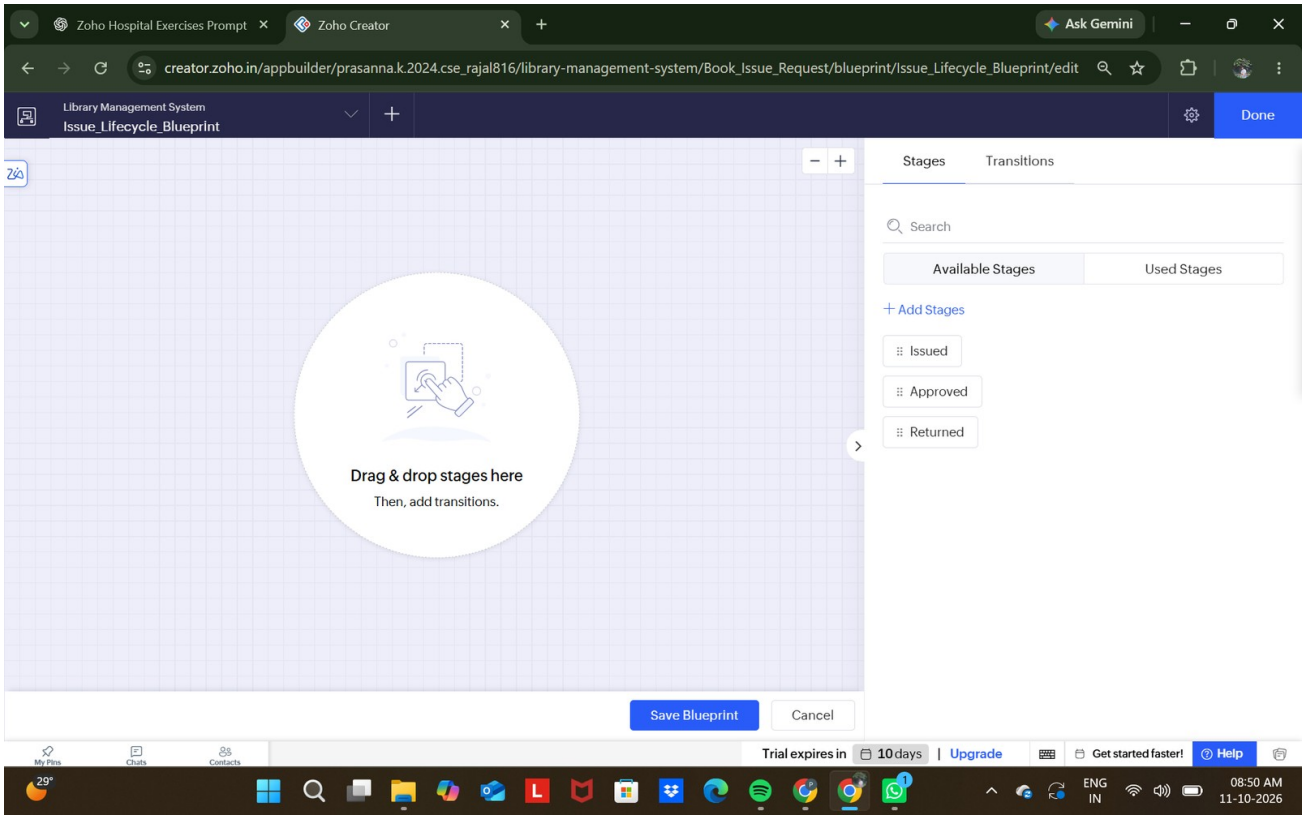


Figure 56: Continuing the creation of issue-lifecycle stages.

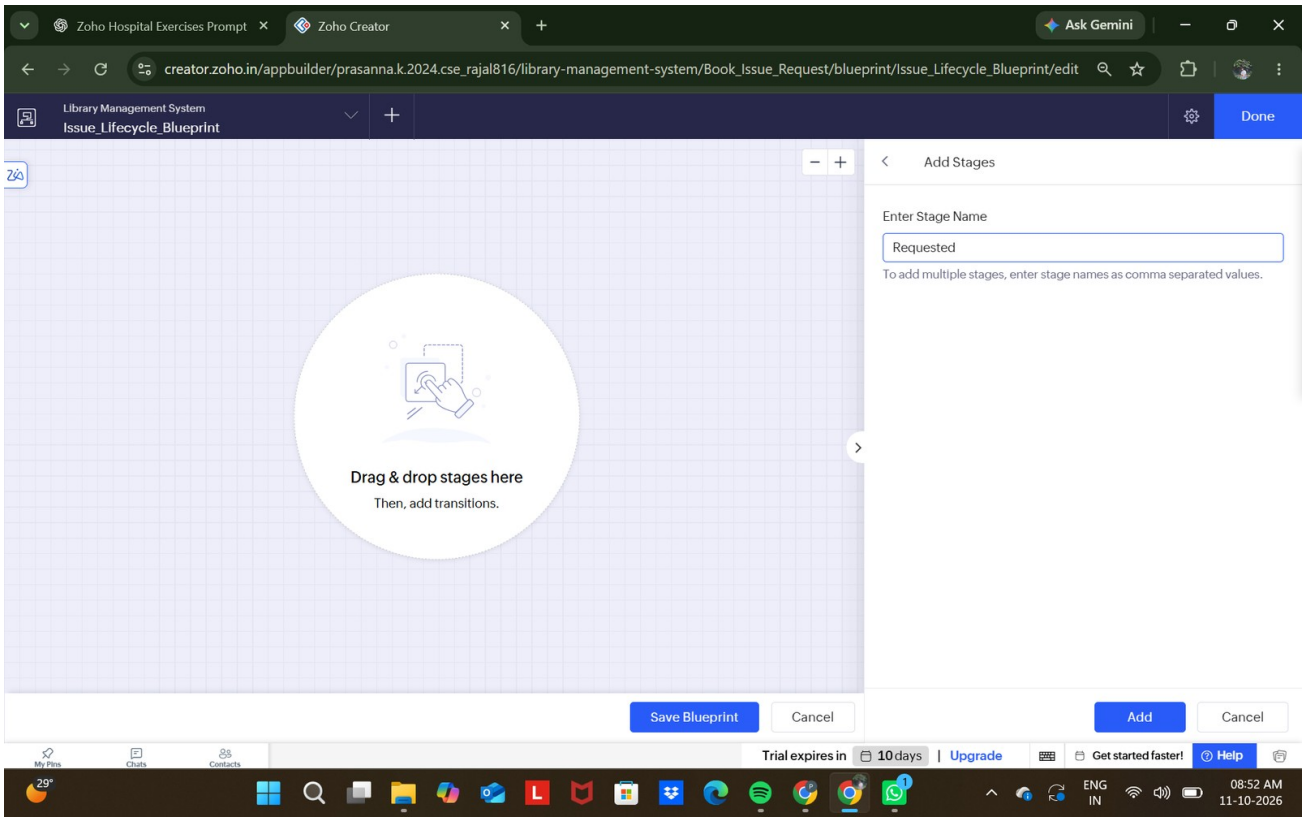


Figure 57: Opening stage options while building the lifecycle path.

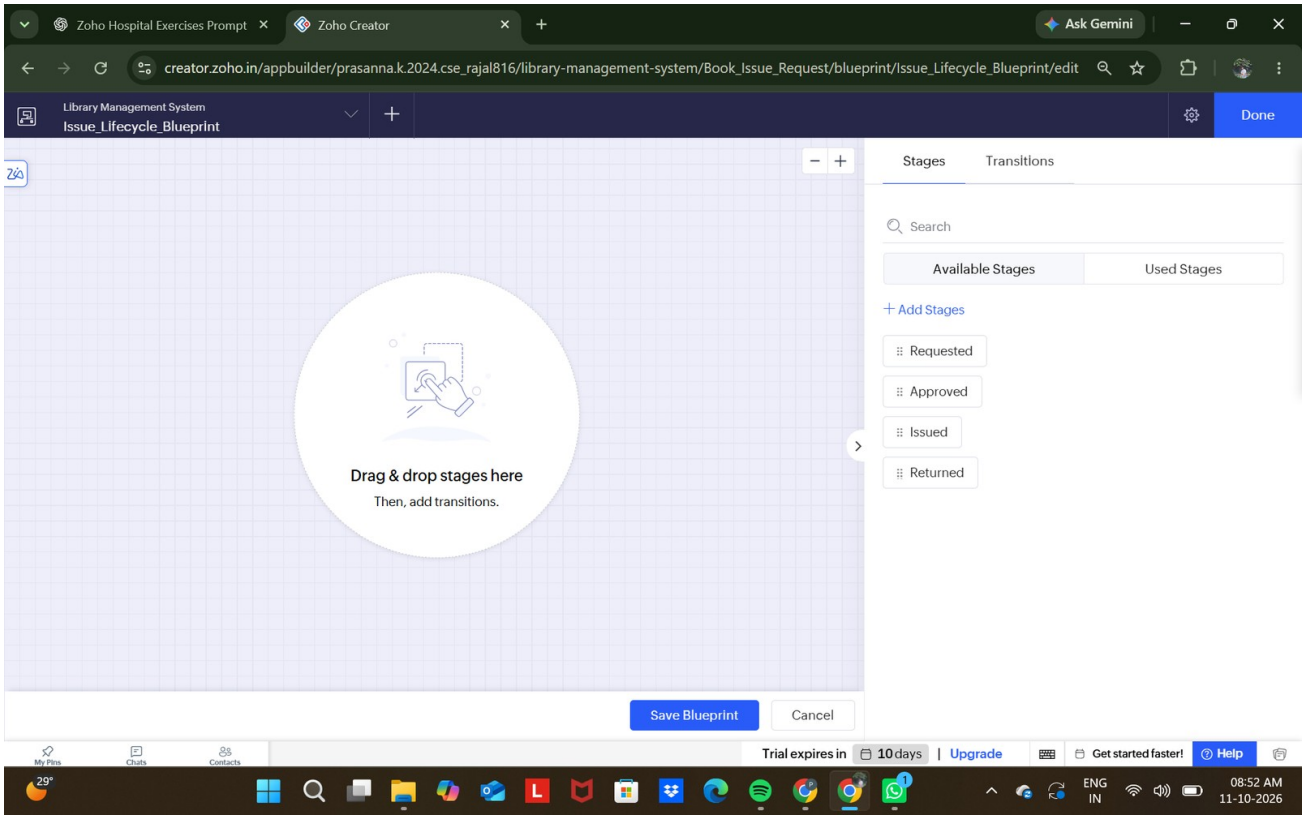


Figure 58: Entering another stage name.

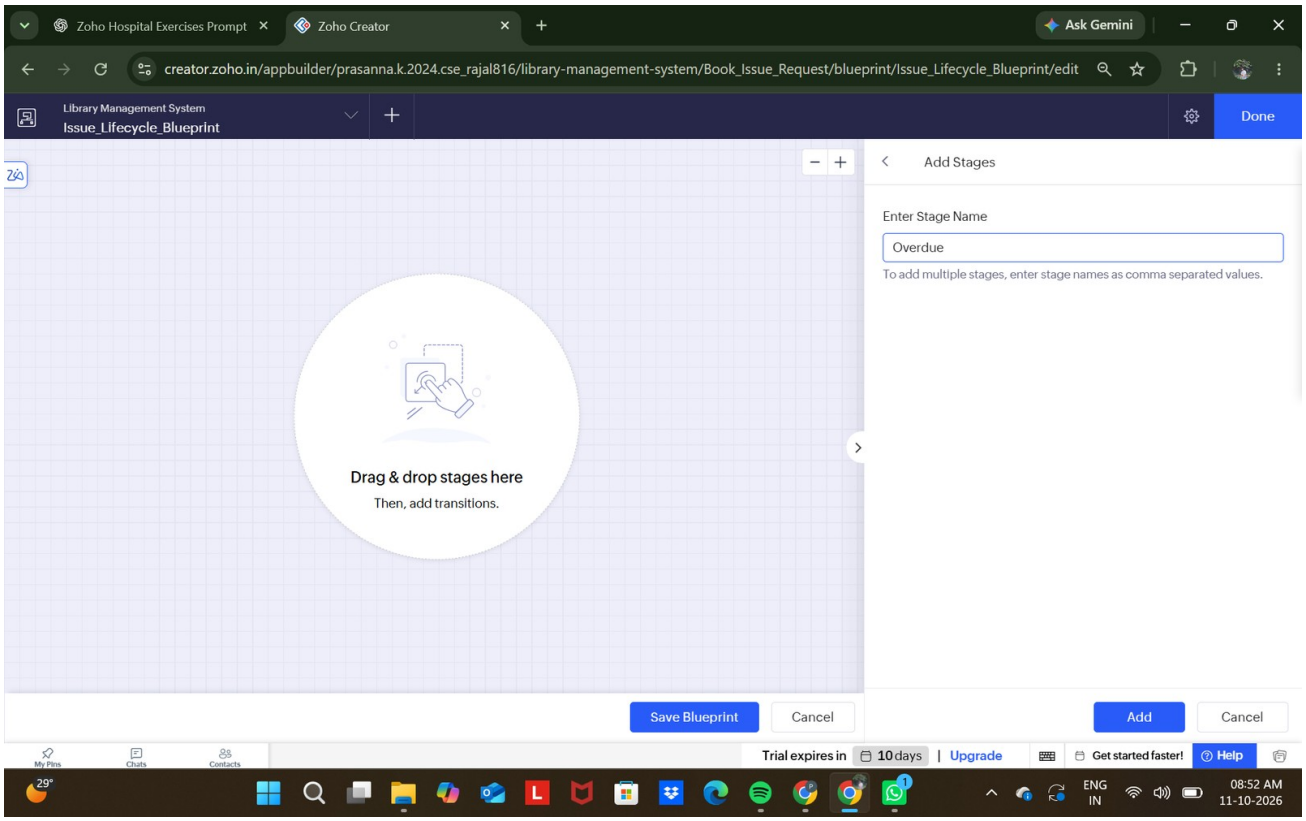


Figure 59: Reviewing the stage list as the blueprint is populated.

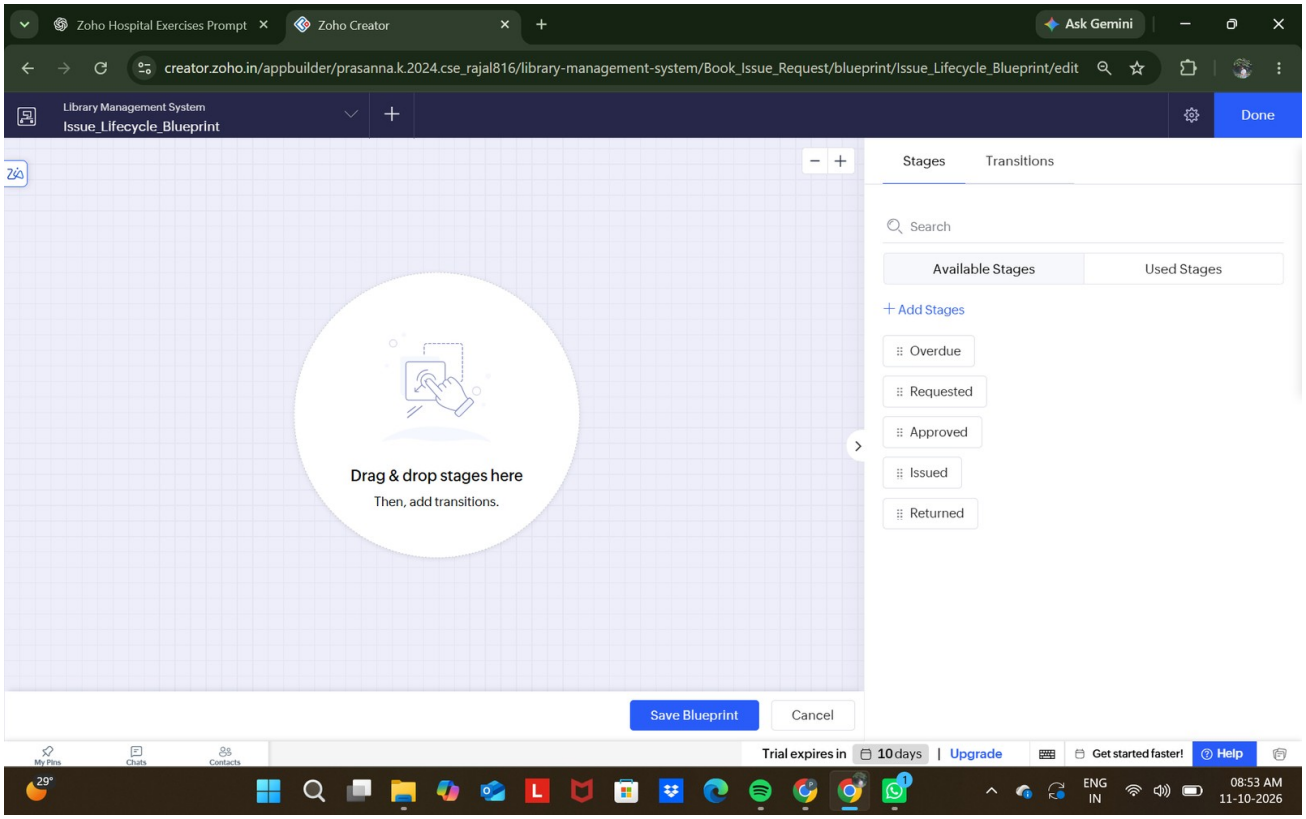


Figure 60: Placing the stages on the blueprint canvas.

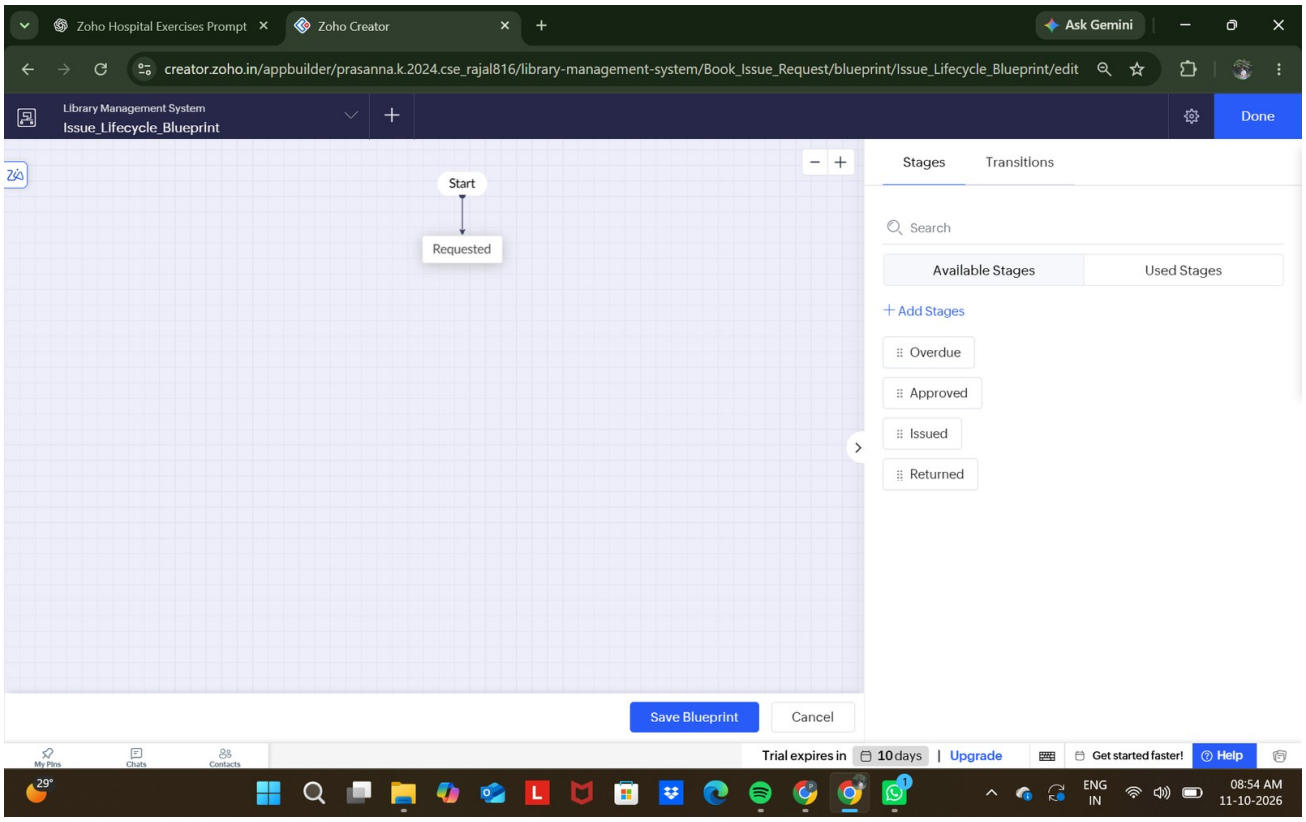


Figure 61: Beginning the main request-to-approval path.

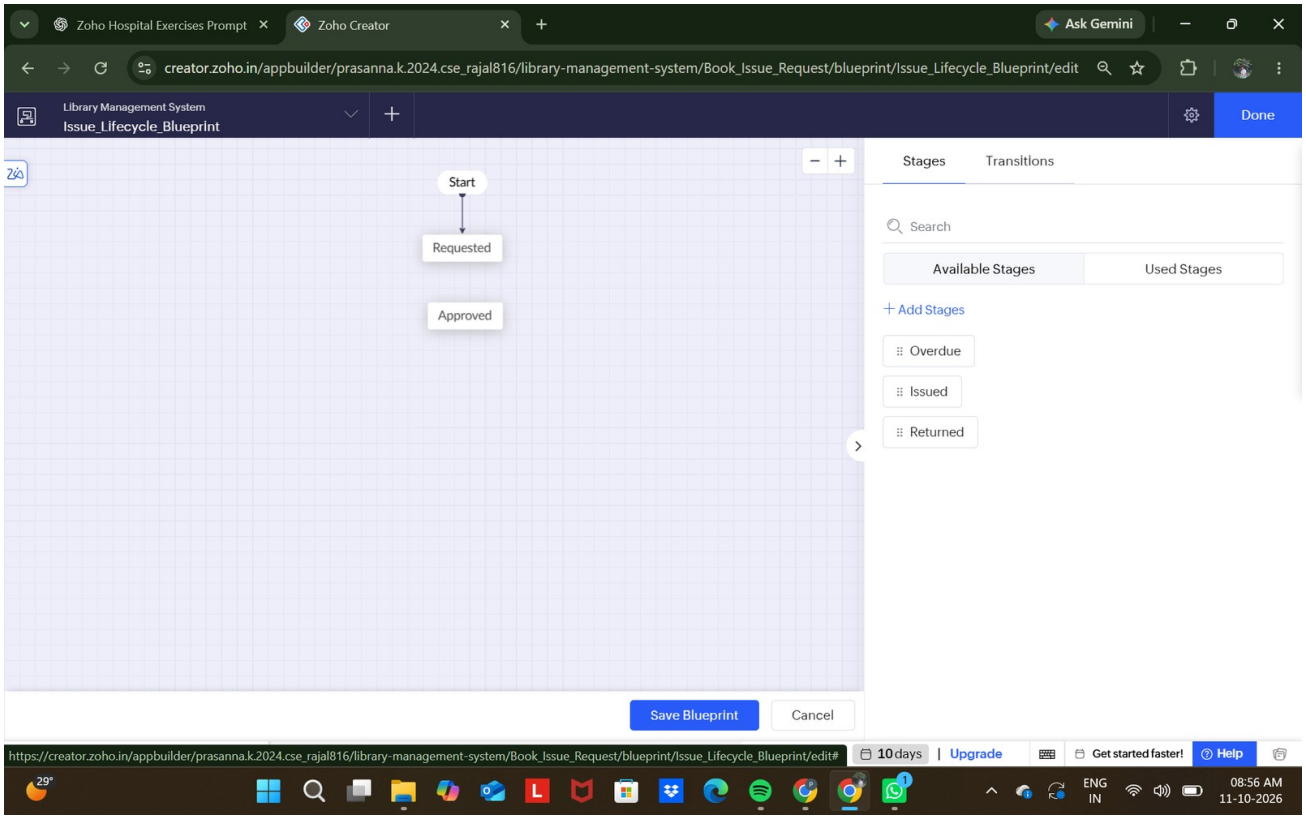


Figure 62: Connecting the first stages in the lifecycle.

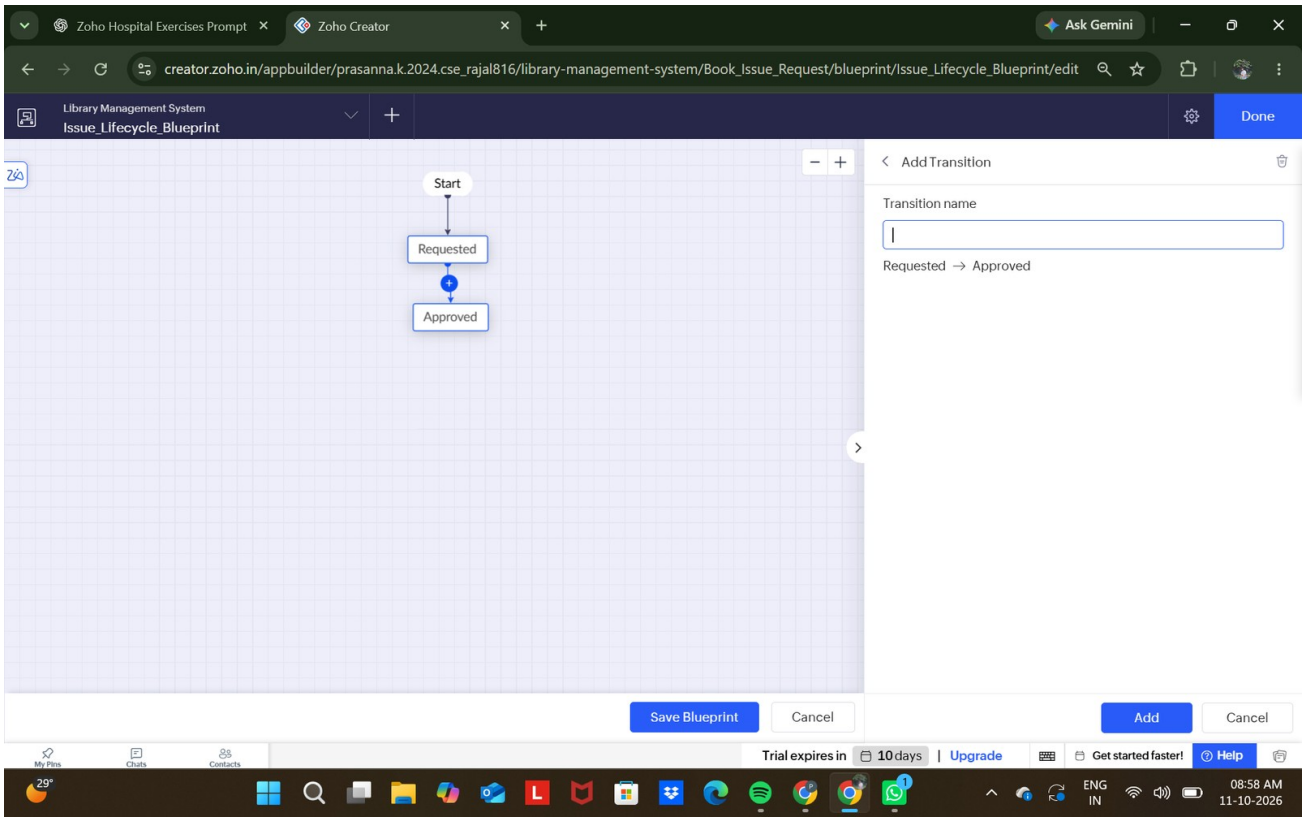


Figure 63: Adding the next stage after the request stage.

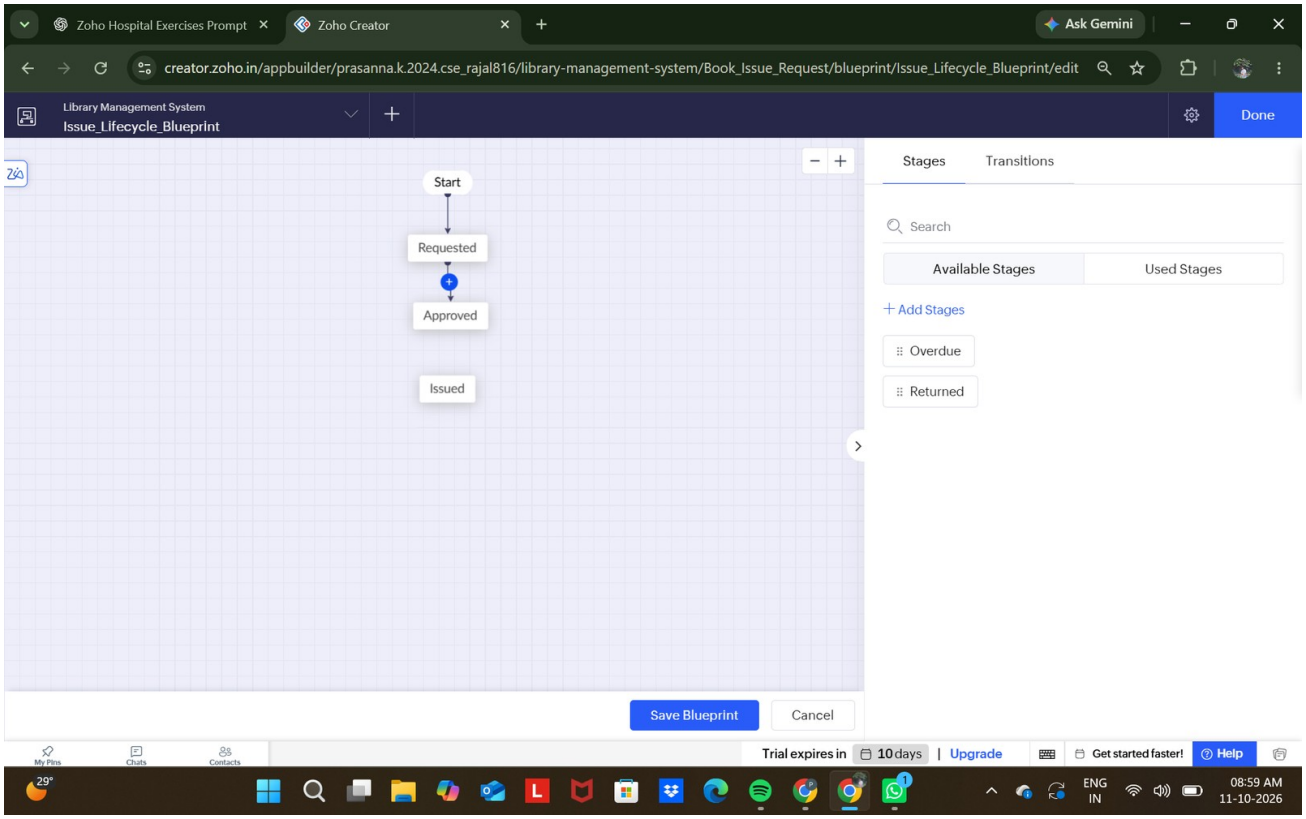


Figure 64: Reviewing the connected approval and issue stages.

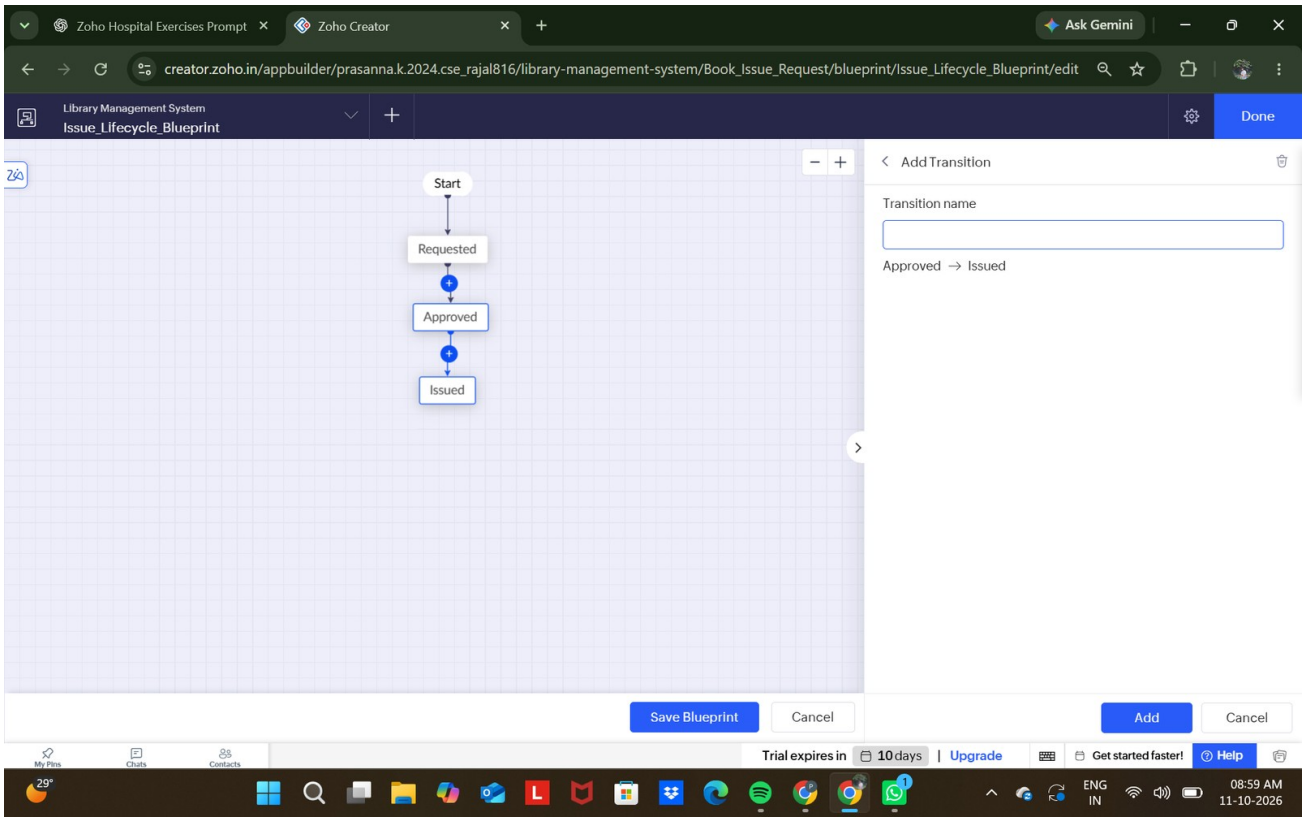


Figure 65: Continuing the main vertical stage sequence.

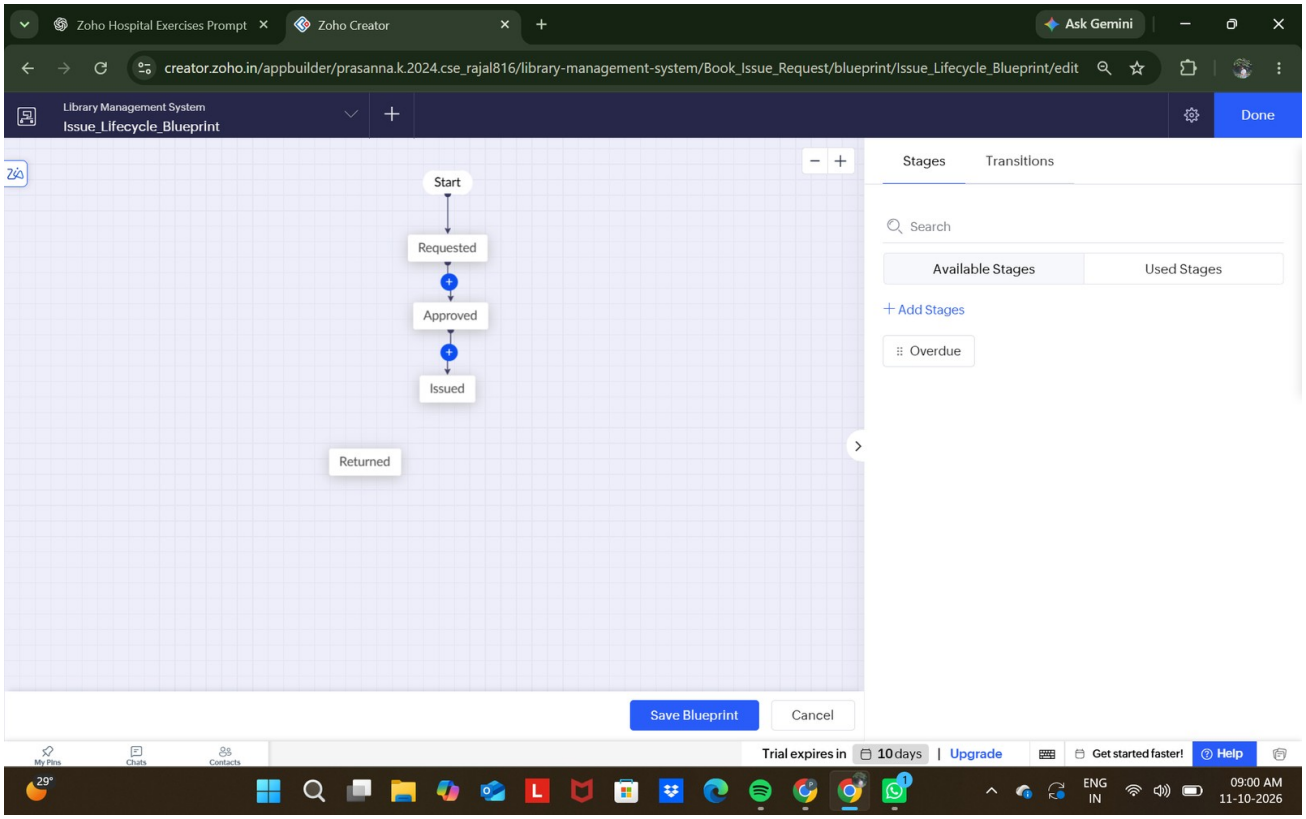


Figure 66: Adding another stage to the lifecycle diagram.

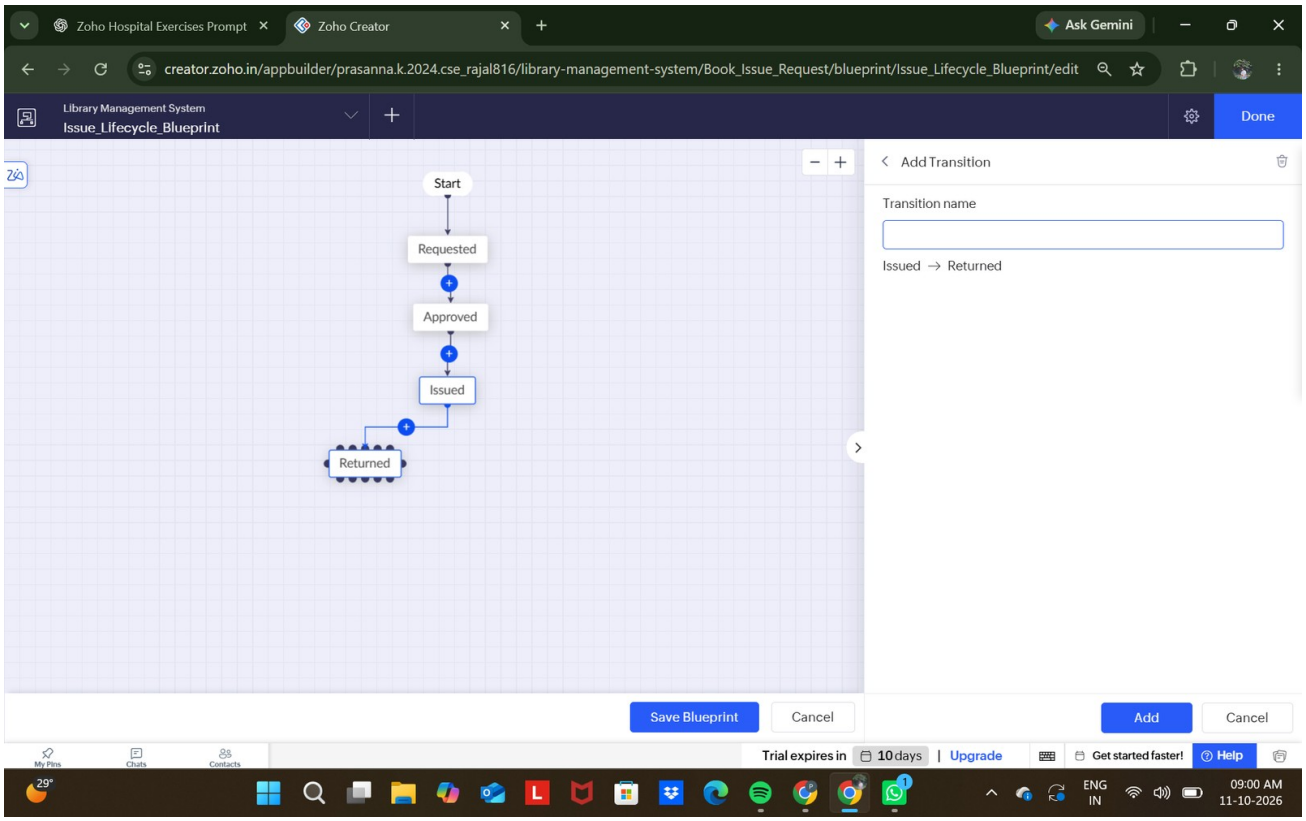


Figure 67: Opening a transition configuration between stages.

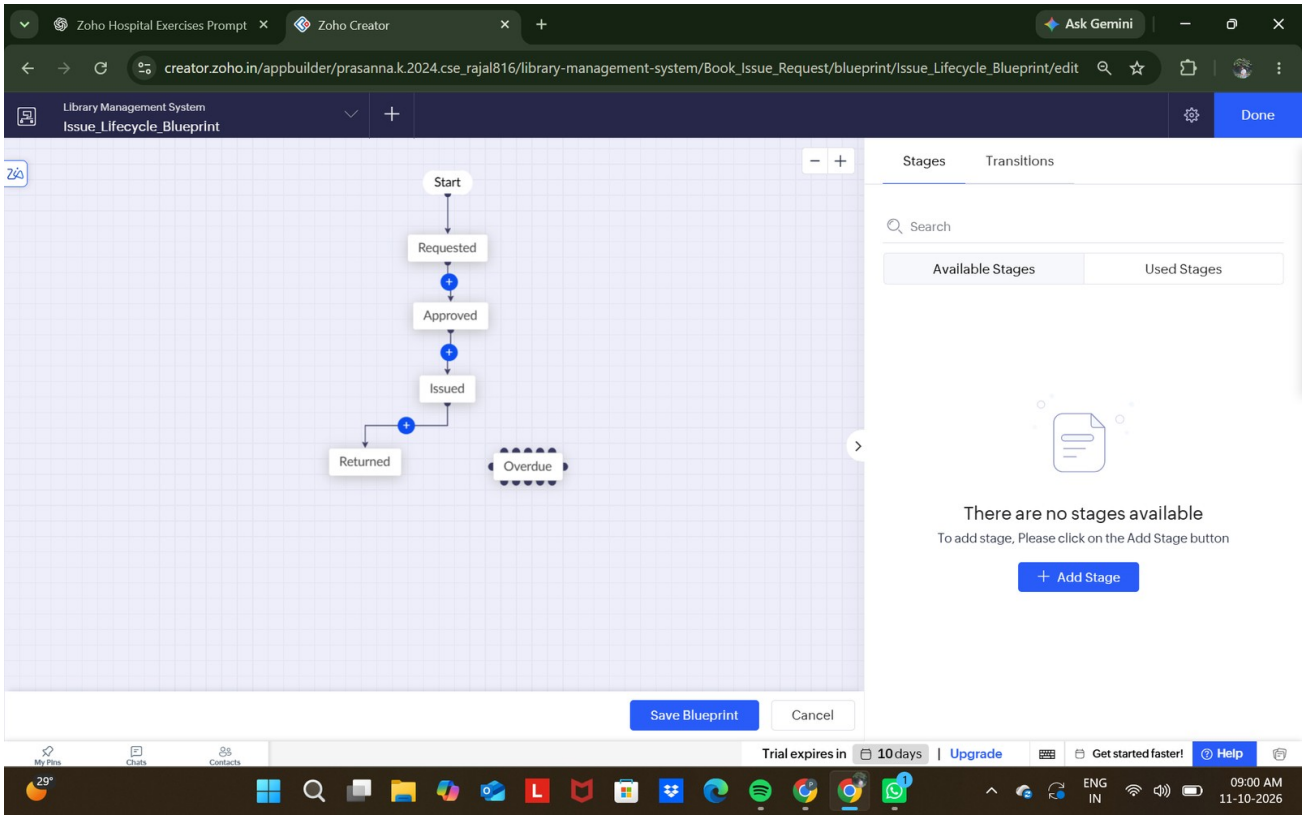


Figure 68: Defining a transition in the issue process.

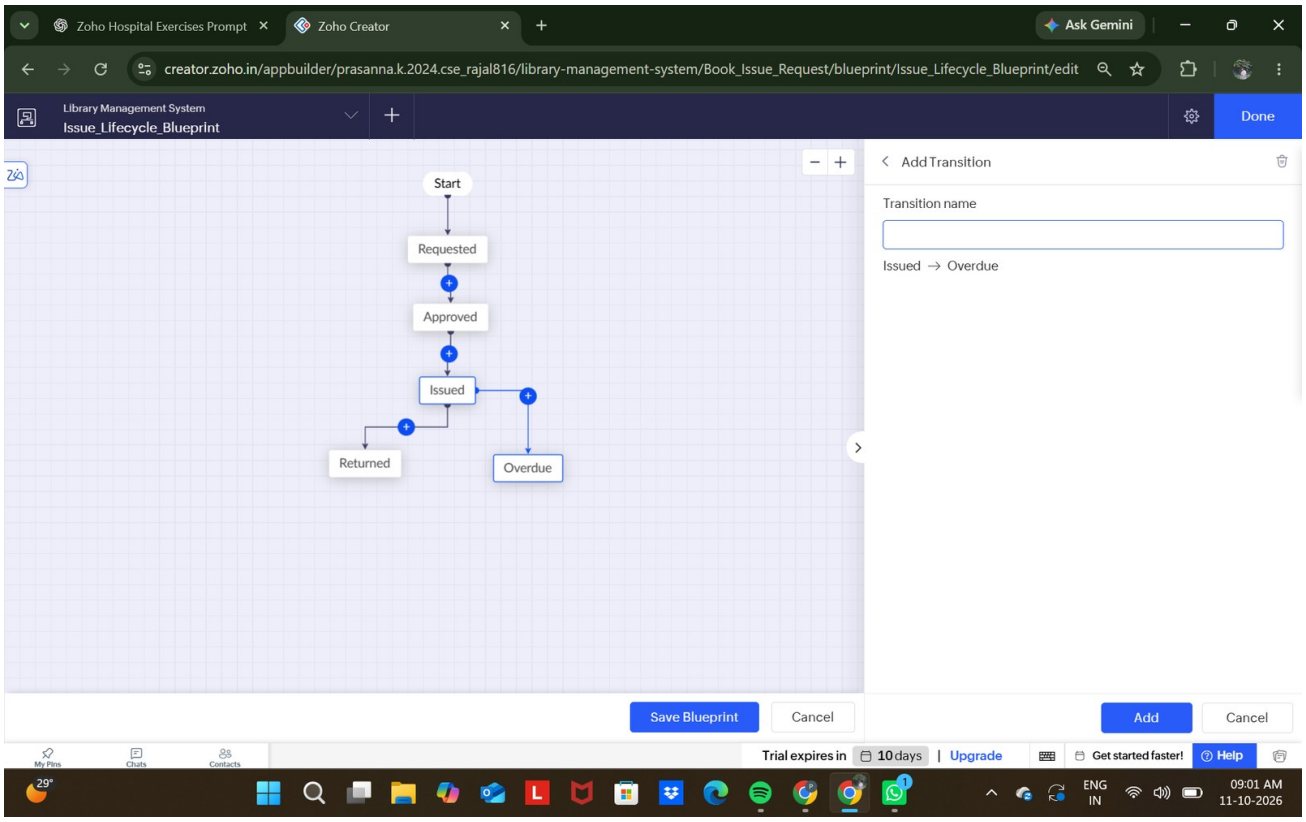


Figure 69: Reviewing the diagram with a connected side branch.

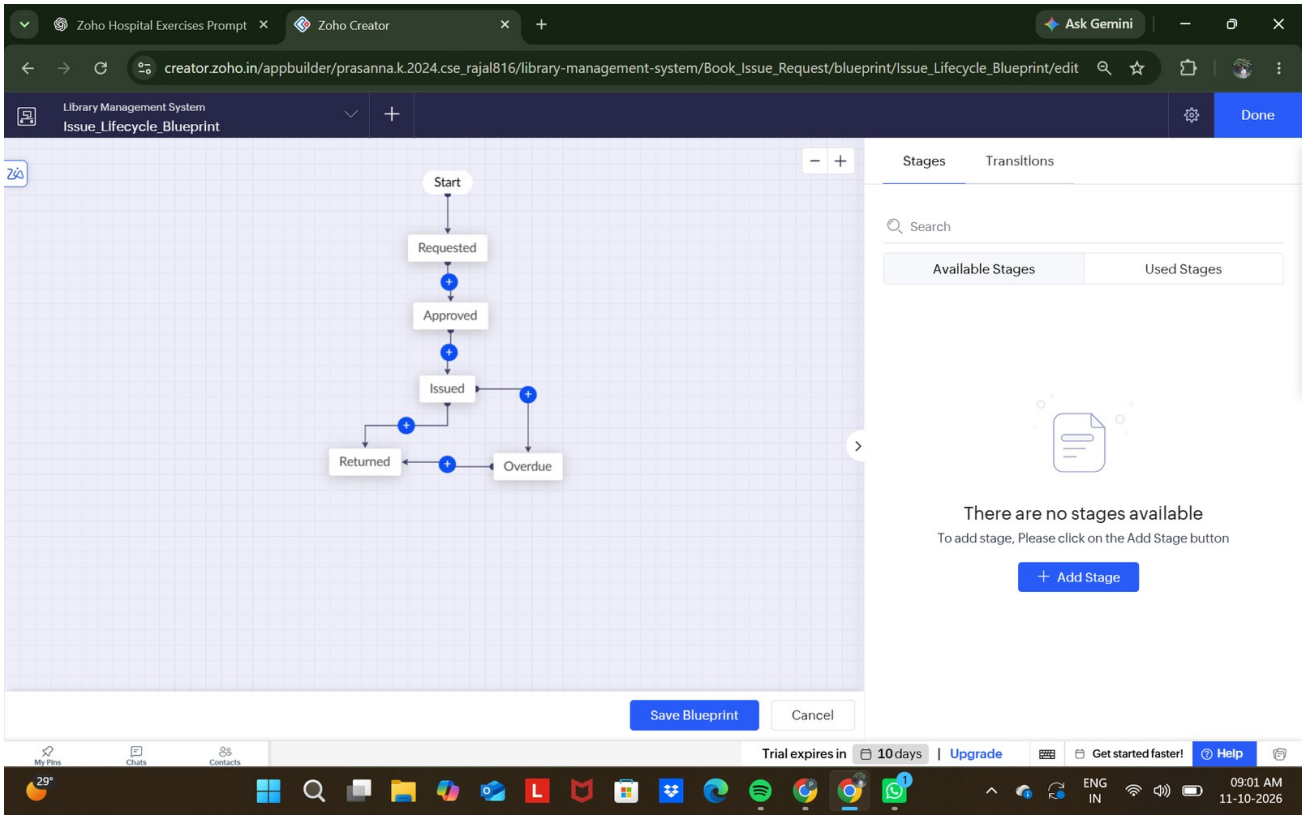


Figure 70: Reviewing the stage diagram after the transition change.

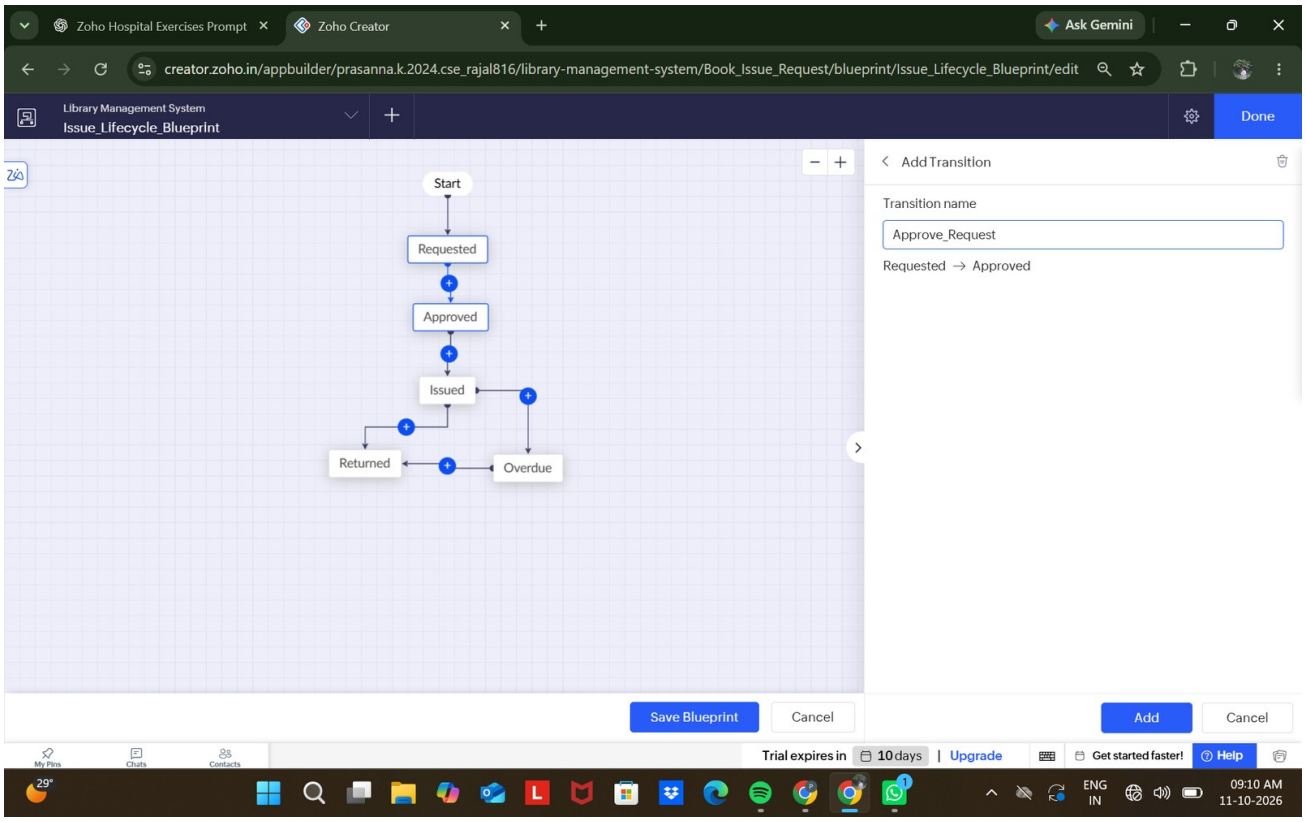


Figure 71: Continuing the configuration of the Return and Overdue branches.

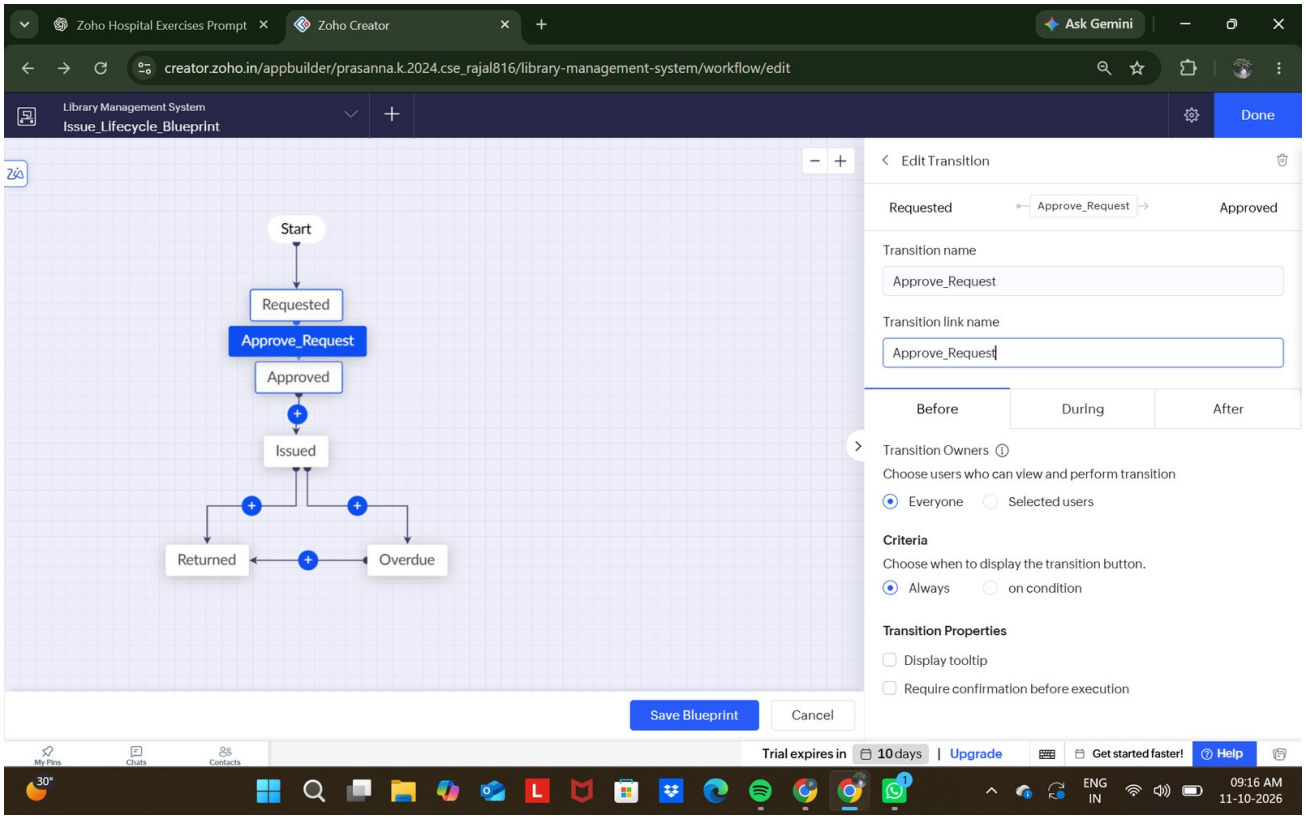


Figure 72: Selecting a transition on the blueprint canvas.

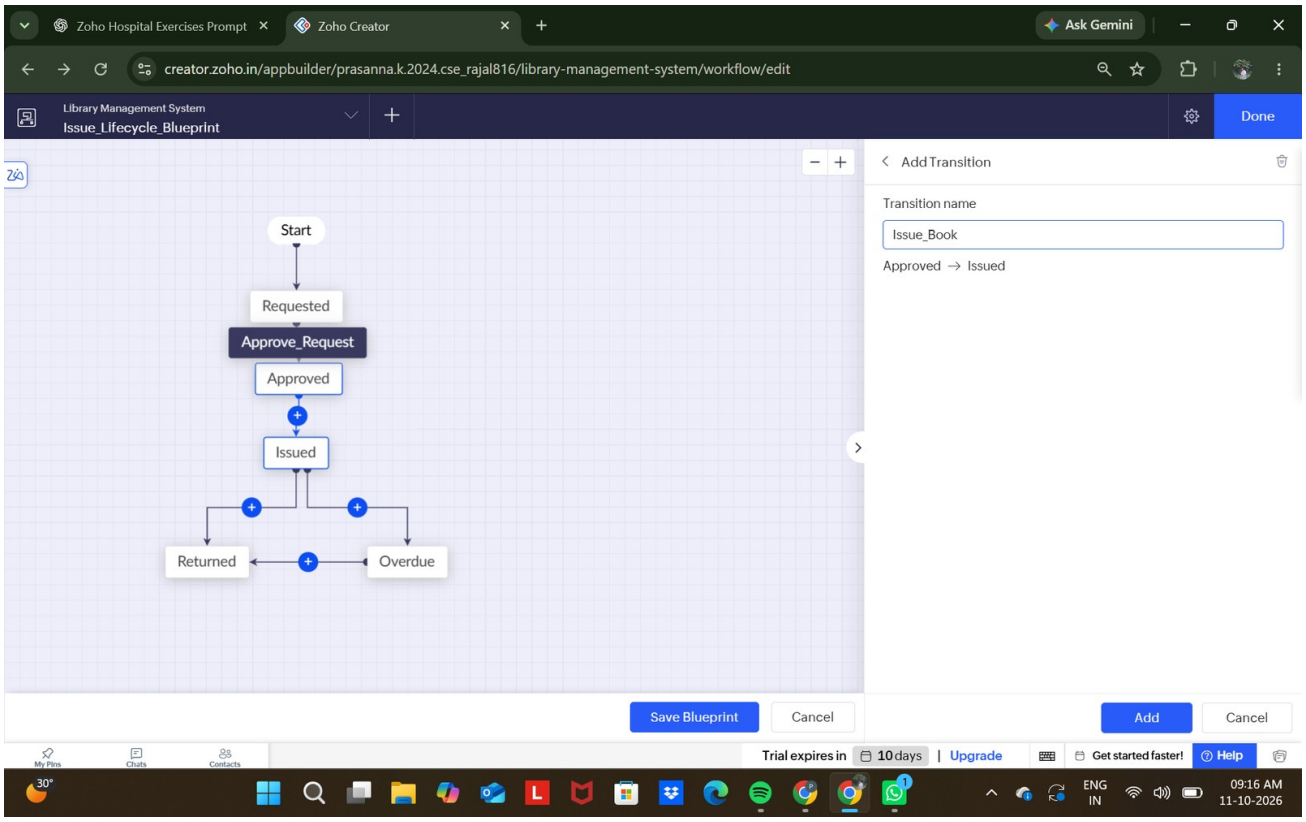


Figure 73: Naming a transition between lifecycle stages.

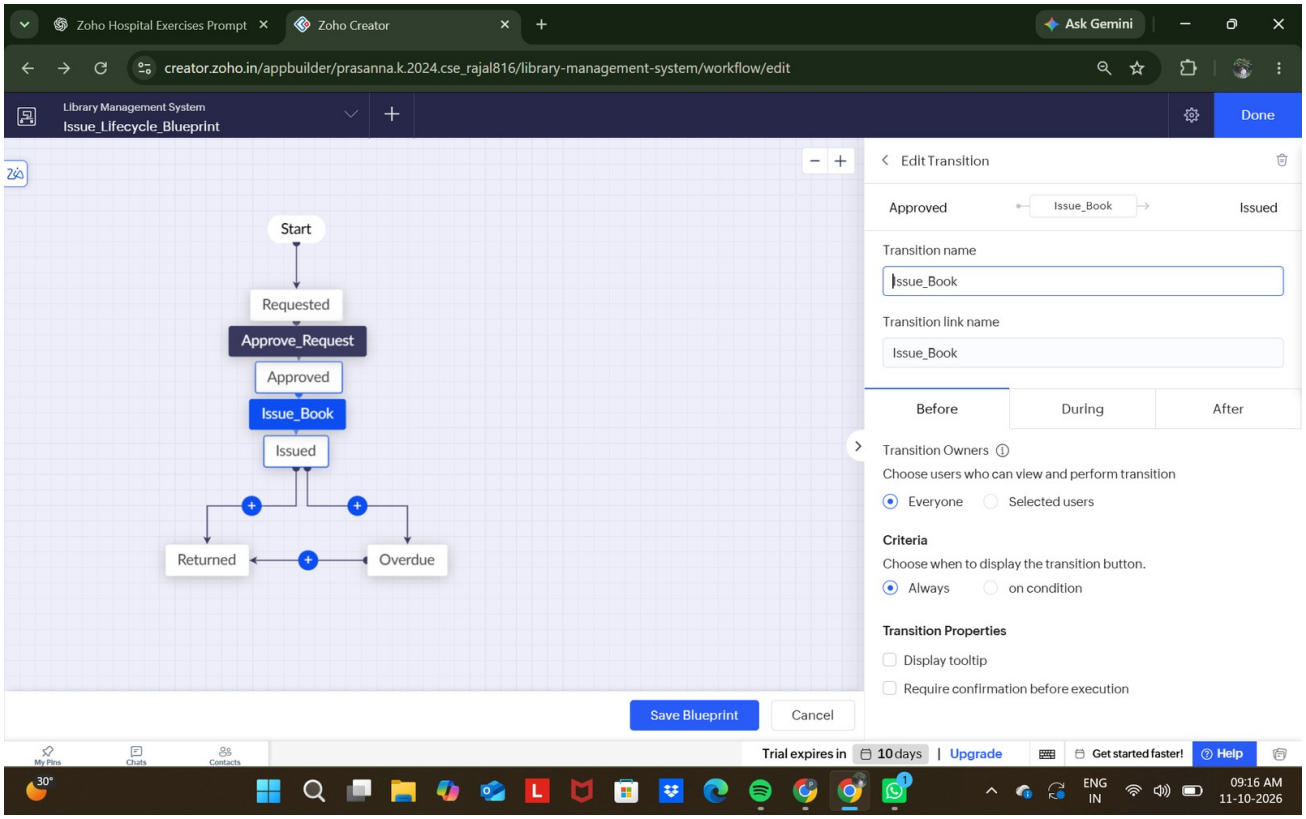


Figure 74: Reviewing the transition settings and connected stages.

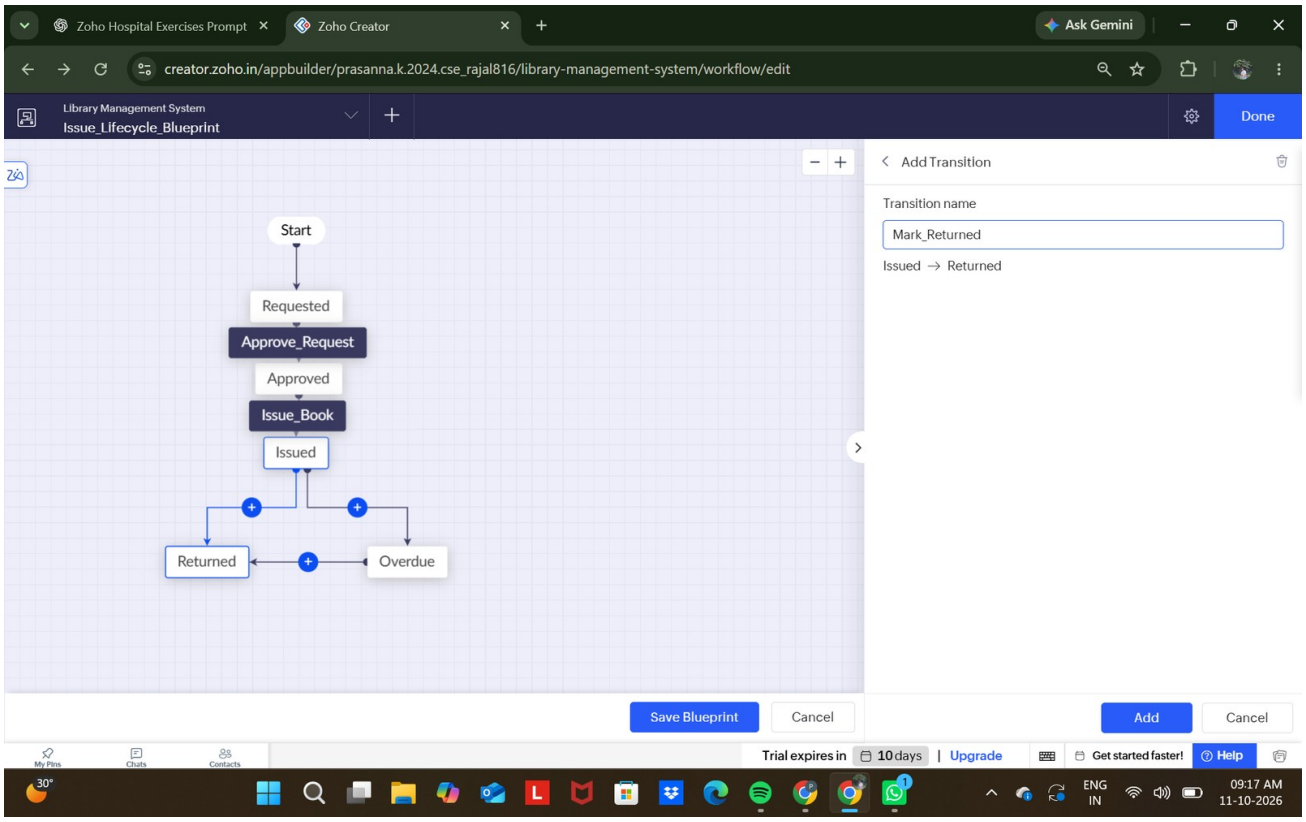


Figure 75: Configuring another transition in the issue lifecycle.

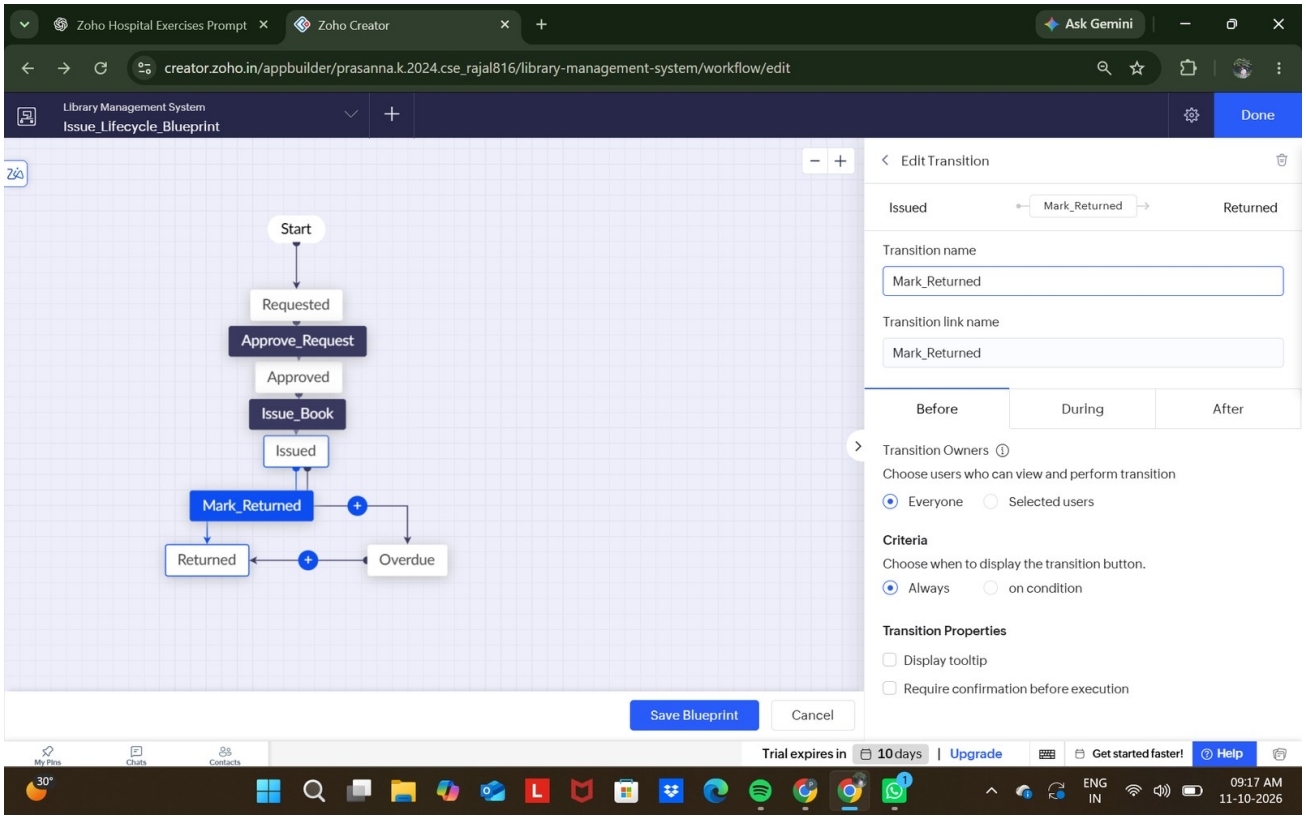


Figure 76: Reviewing transition details in the side panel.

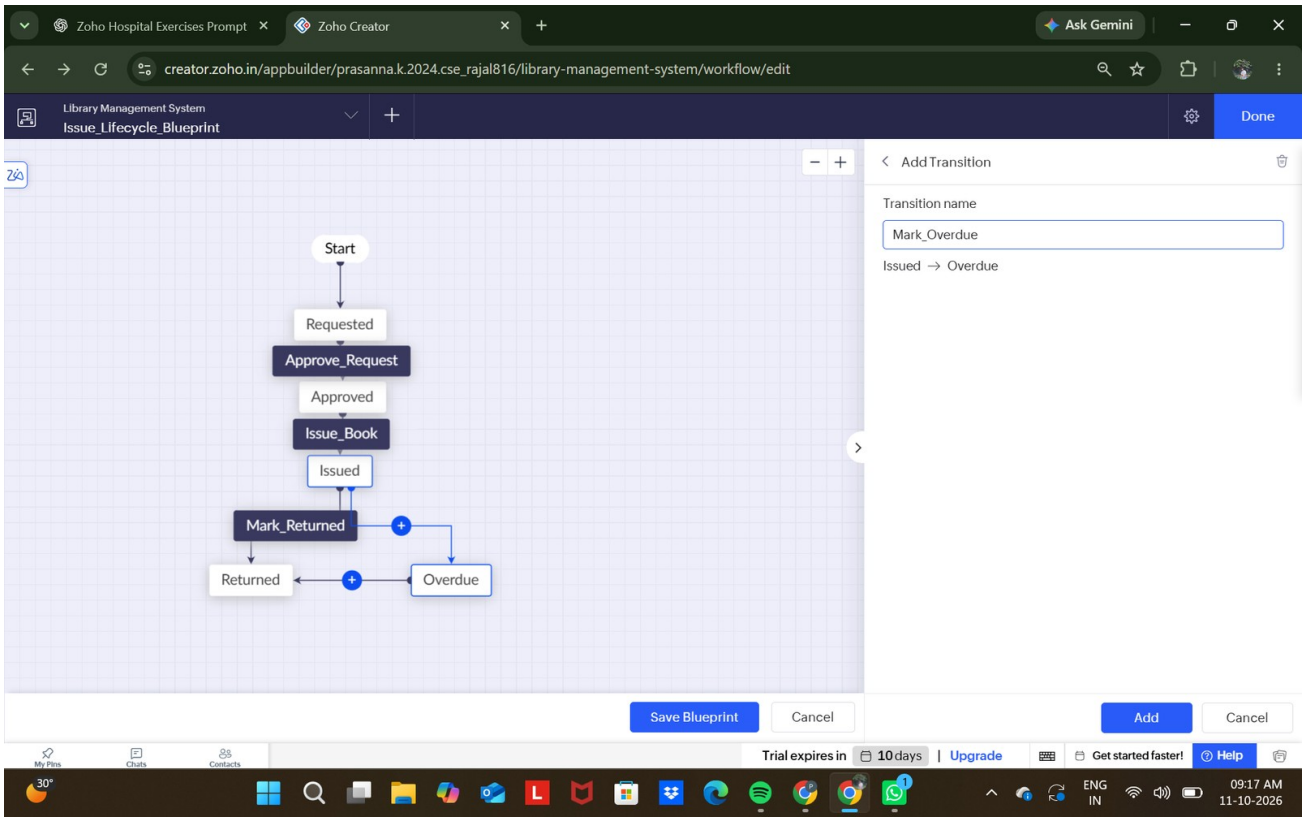


Figure 77: Connecting the Returned stage to the main process.

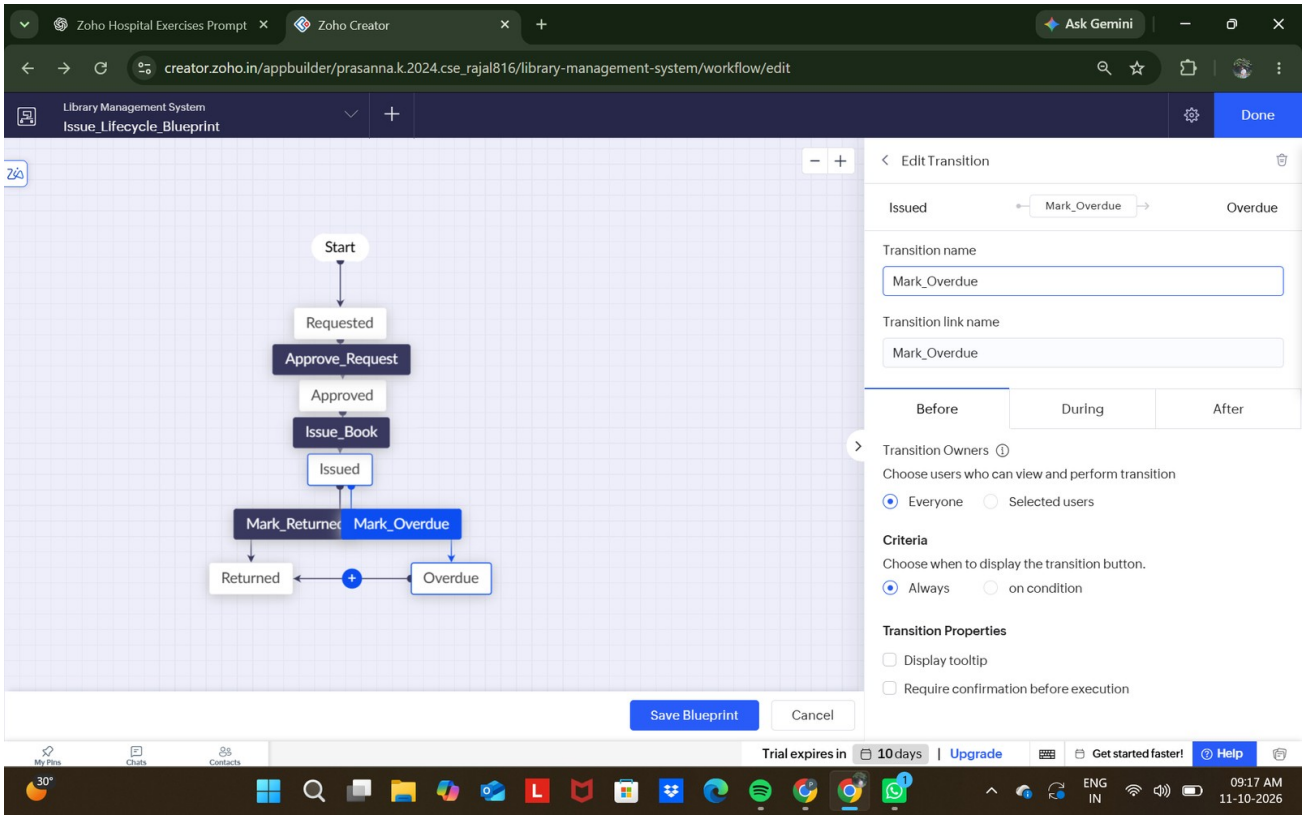


Figure 78: Connecting the Overdue branch from the Issued stage.

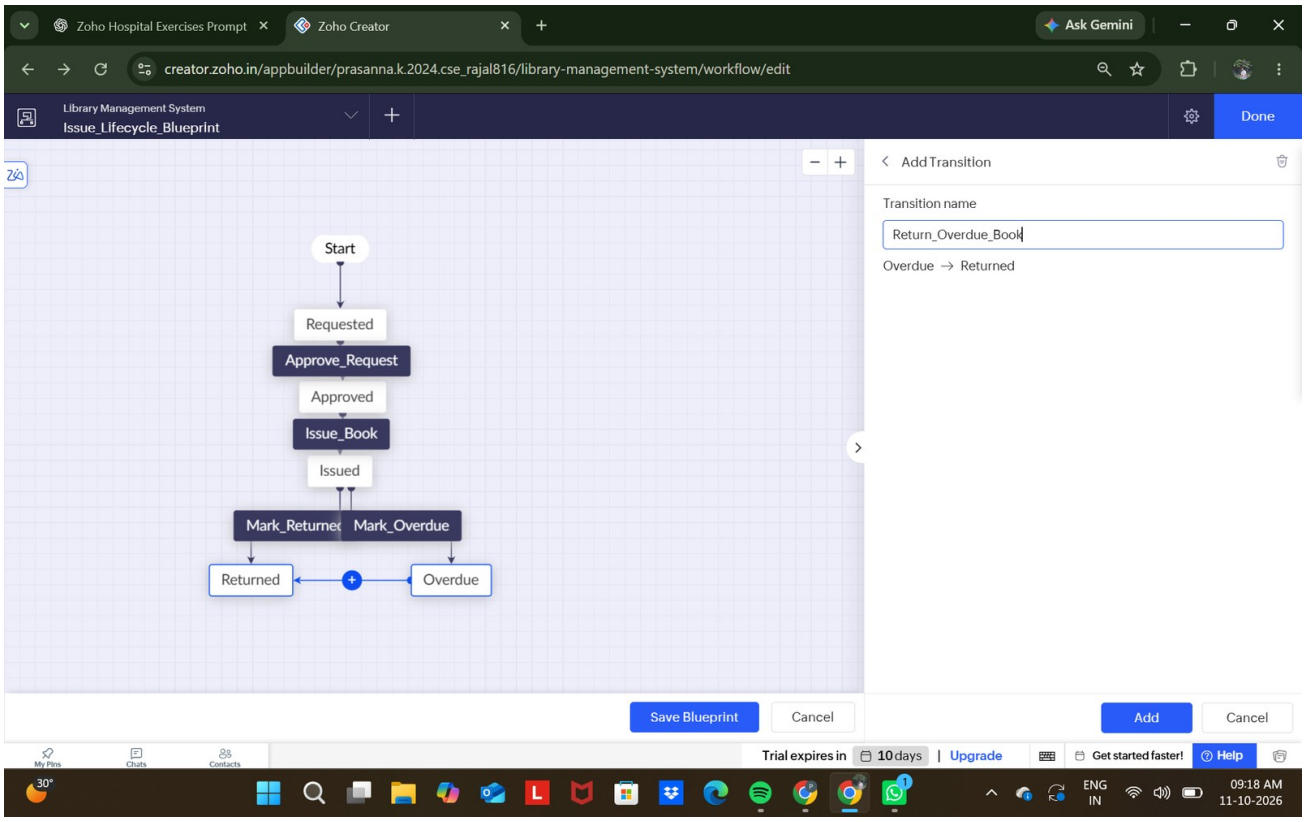


Figure 79: Reviewing transition links in the completed diagram.

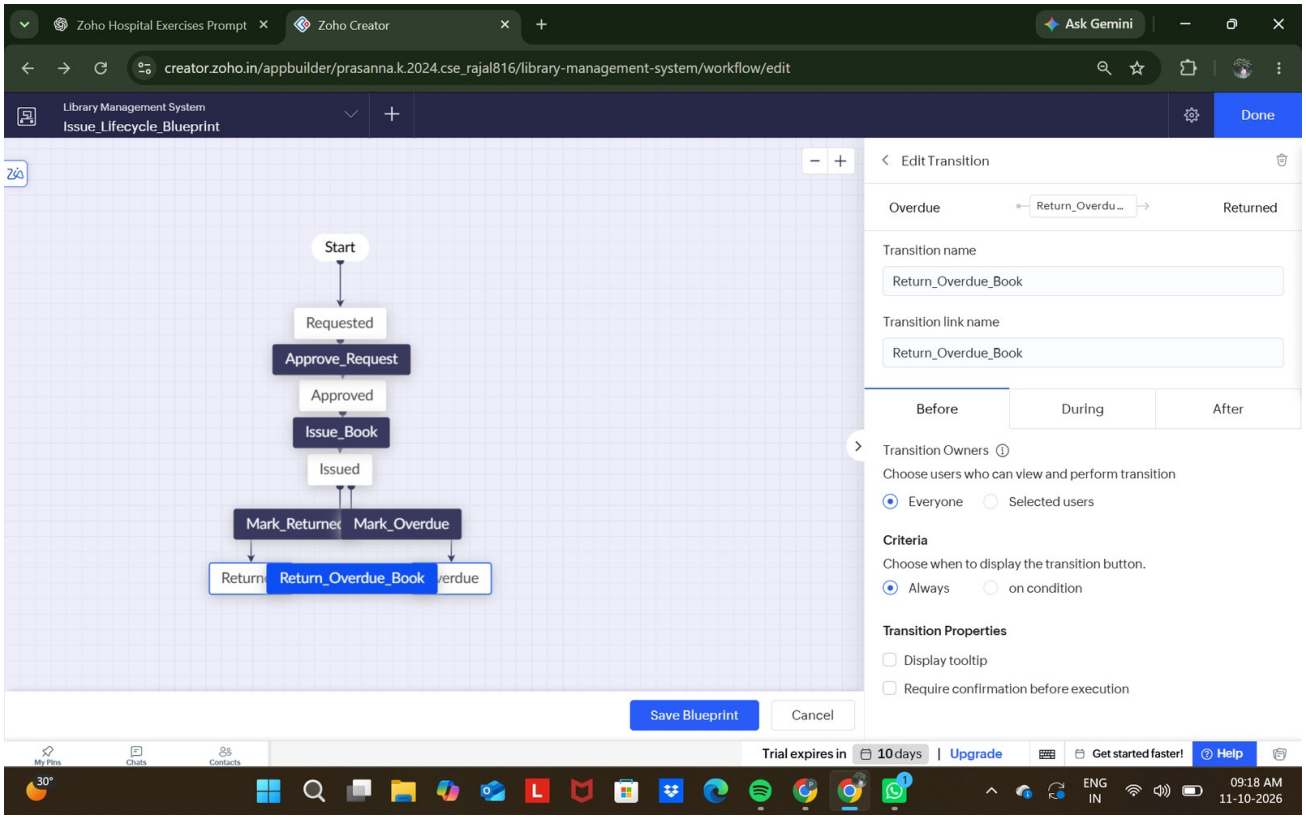


Figure 80: Checking the stage and transition layout.

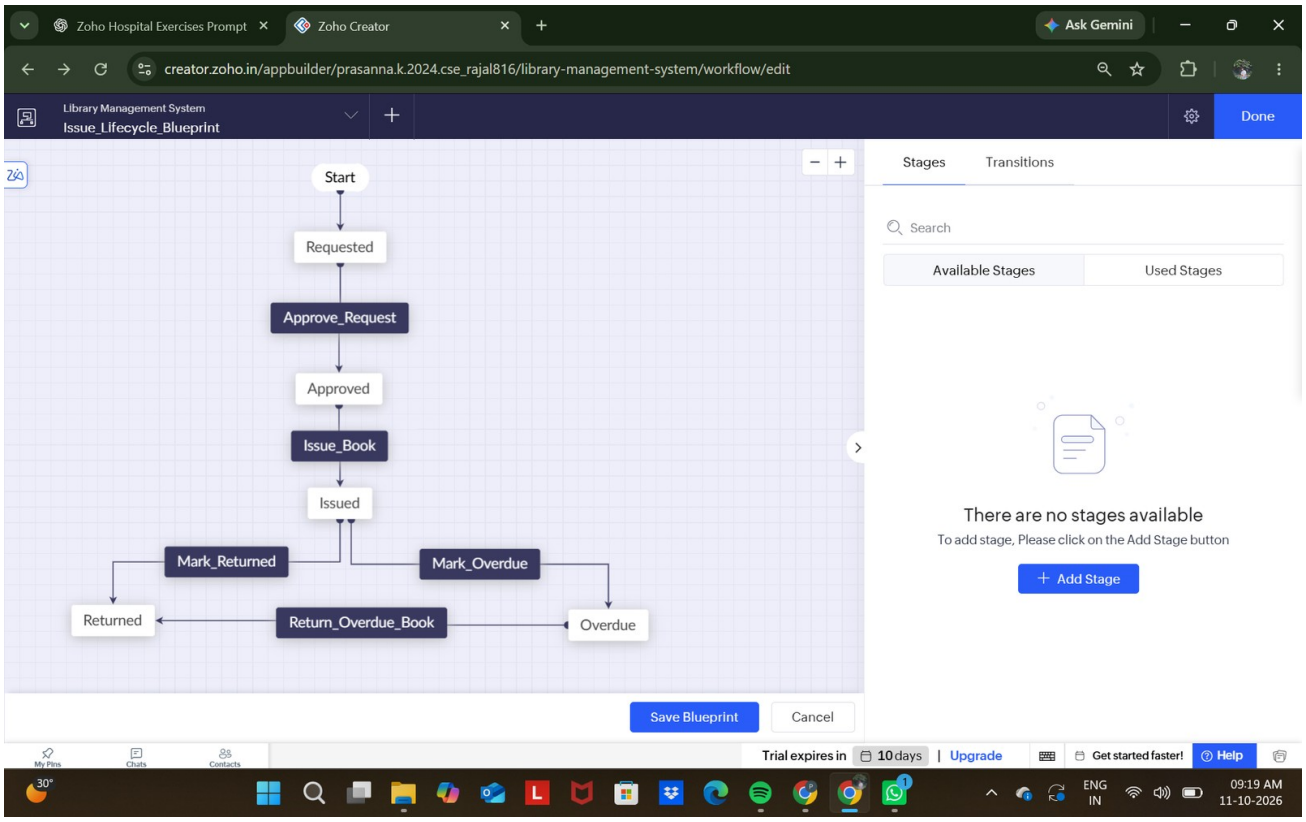


Figure 81: Reviewing the complete issue lifecycle blueprint.

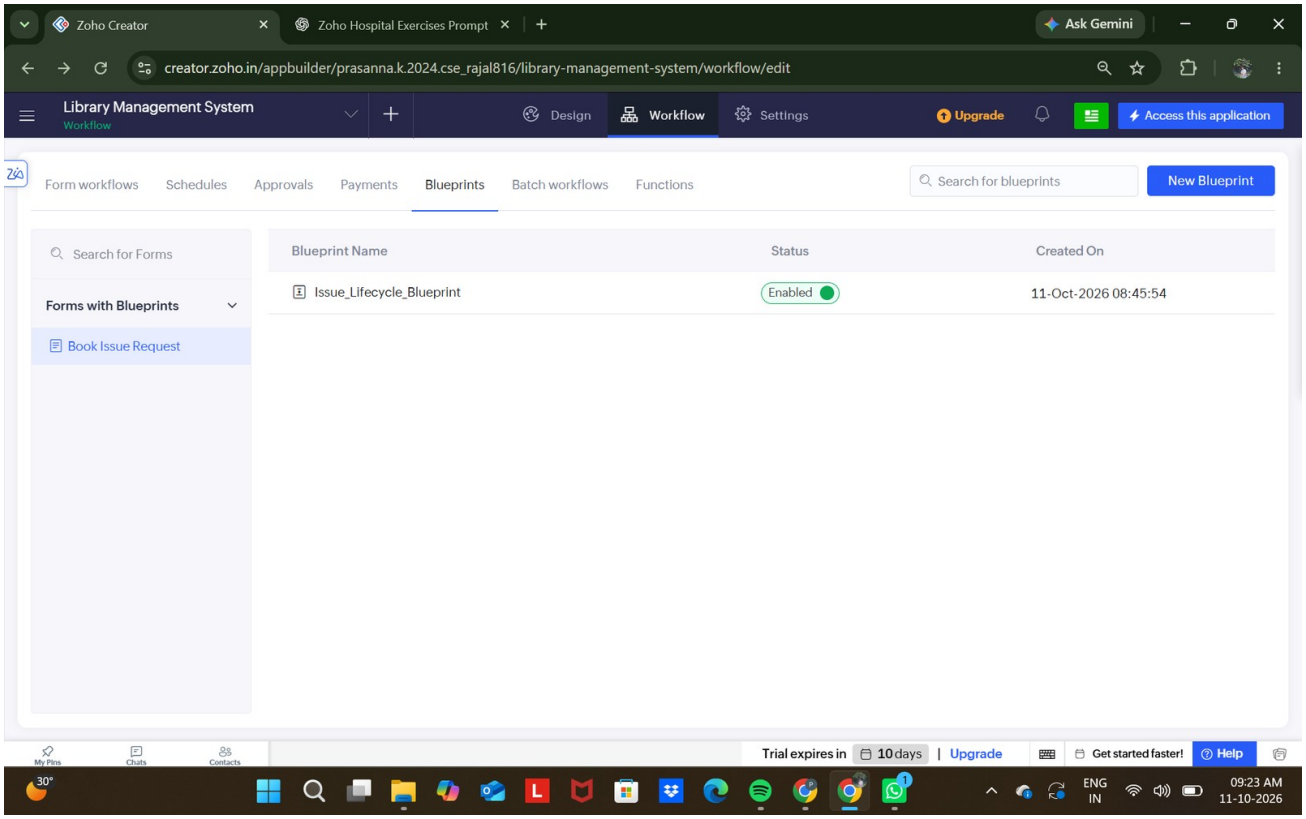


Figure 82: Verifying that Issue_Lifecycle_Blueprint is enabled.

6. Batch Workflows Configuration

Bulk_Overdue_Notice_Batch was configured for Book Issue Request records and processes records in blocks of 1,000. Its Deluge script checks for Status equal to Overdue and Overdue_Notice_Status equal to Not Sent, sends an overdue notice when the member email is present, and marks the notice status as Sent.

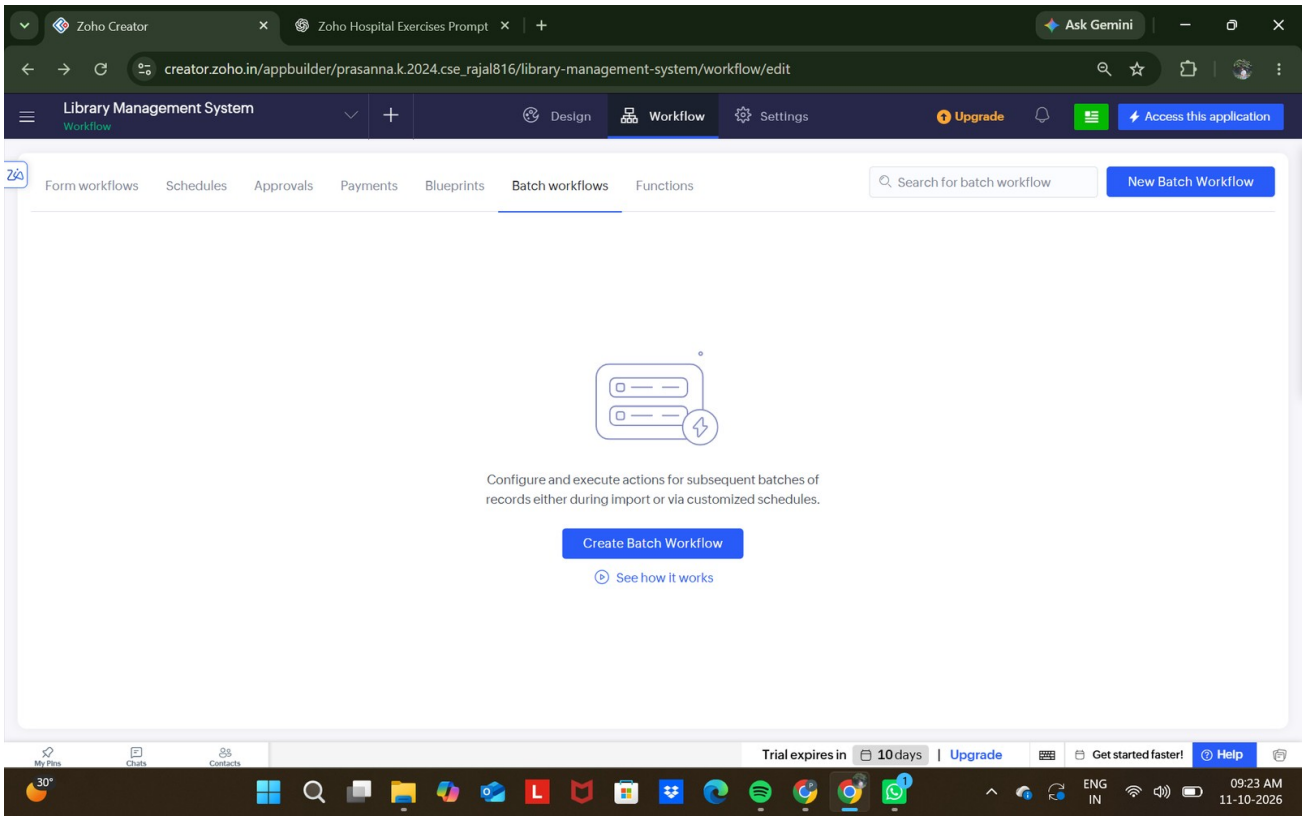


Figure 83: Opening the workflow option for processing records in bulk.

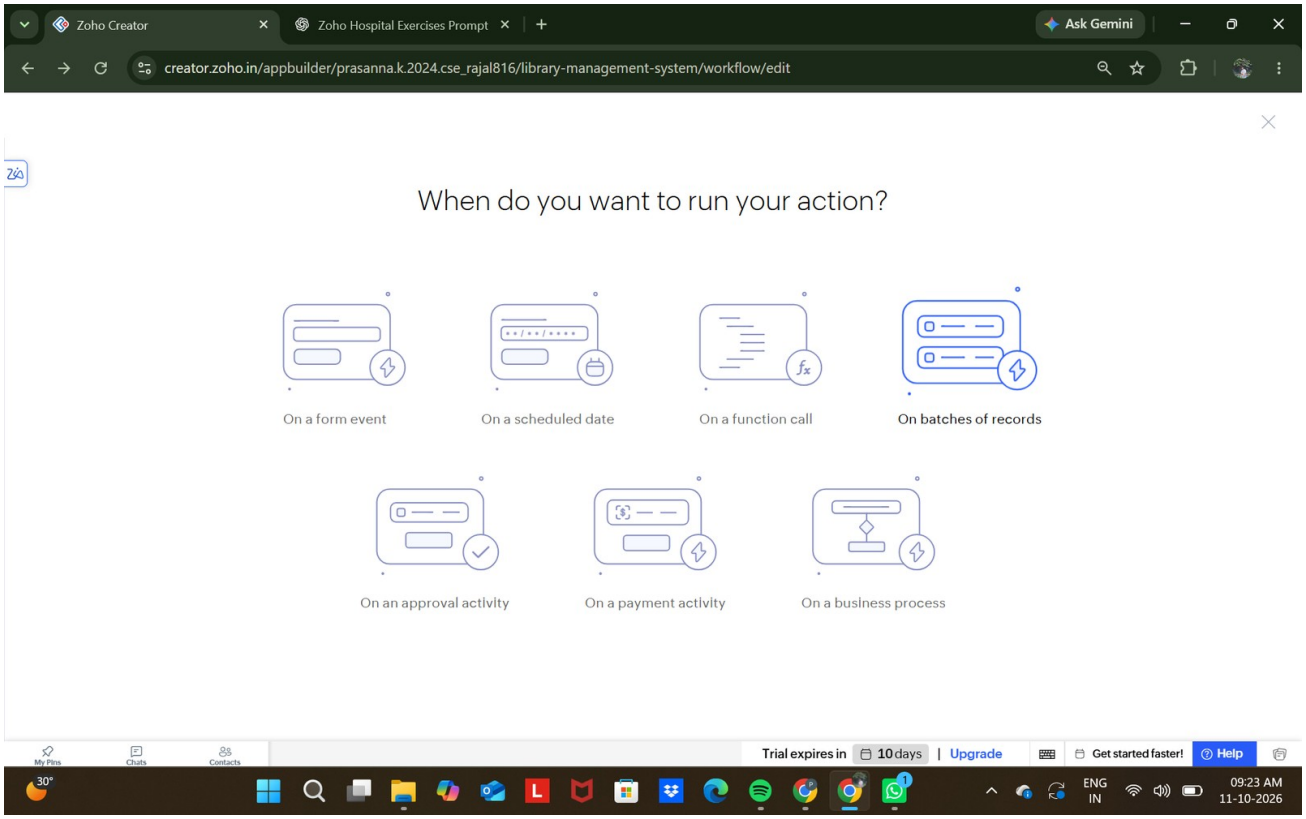


Figure 84: Selecting the batch-records trigger type.

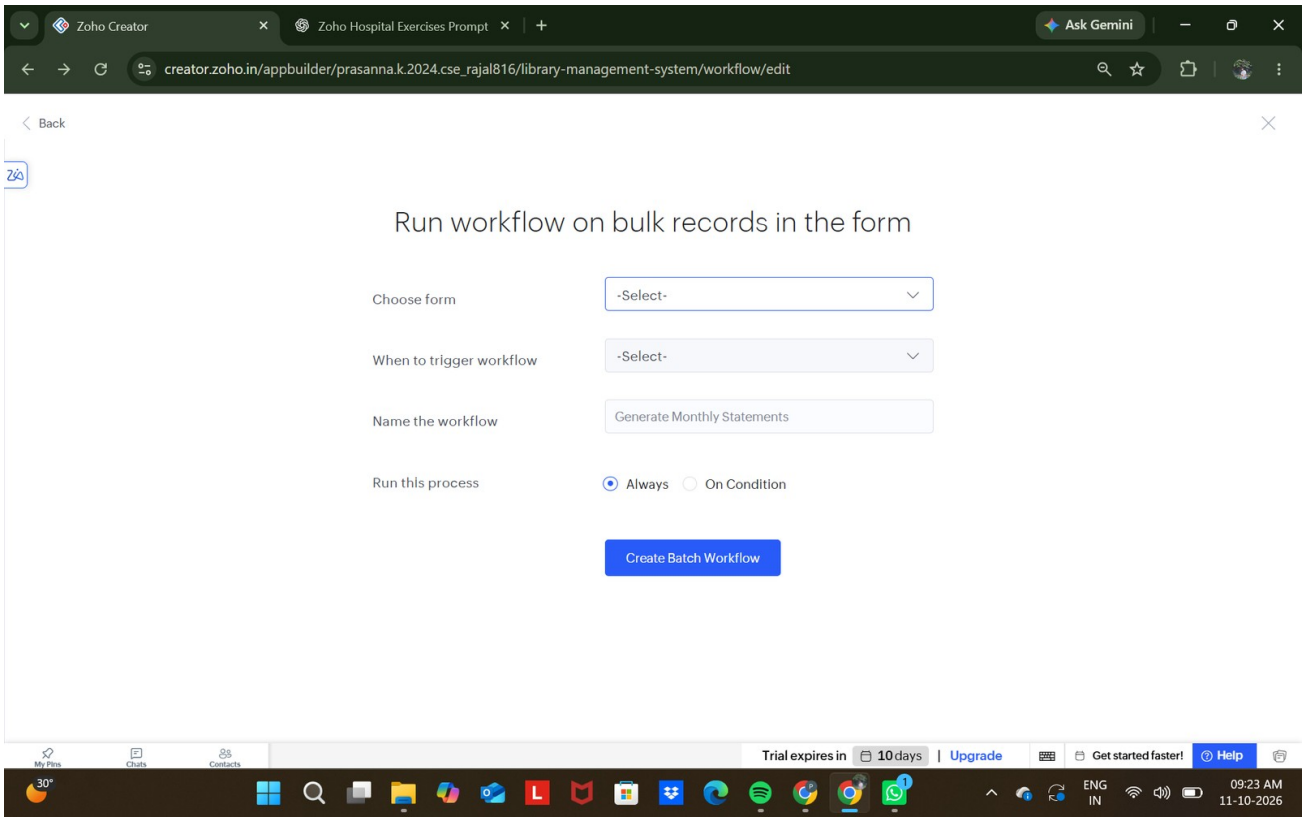


Figure 85: Configuring the batch workflow form, name and execution condition.

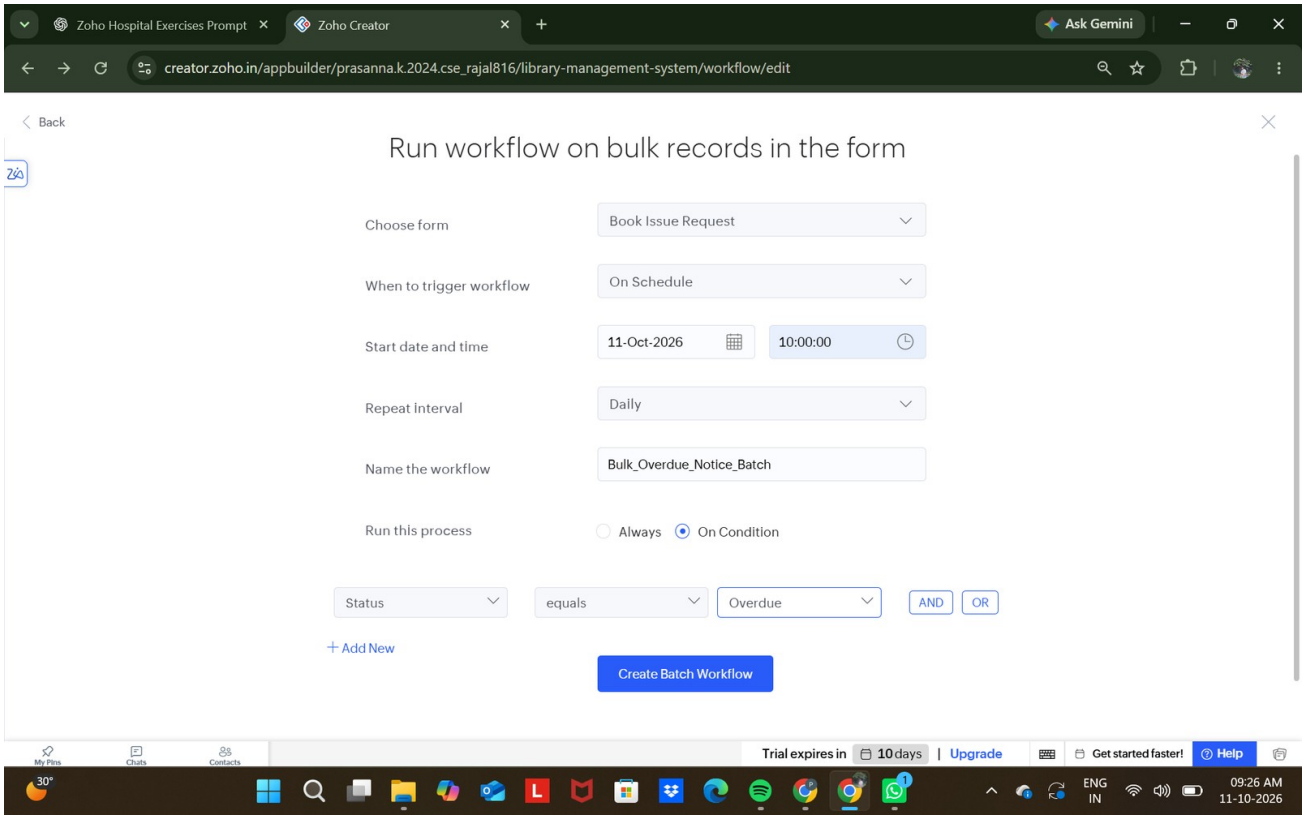


Figure 86: Specifying the form and trigger details for the batch workflow.

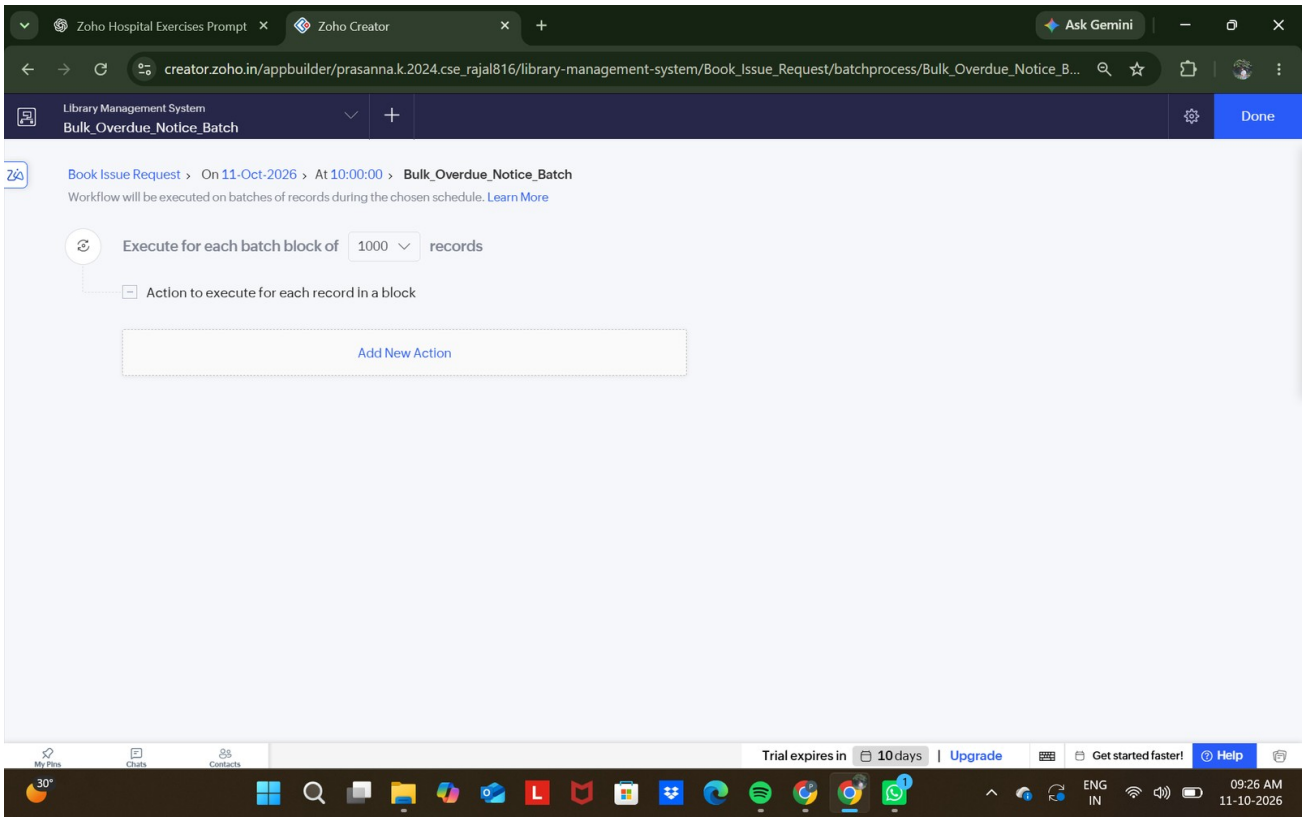


Figure 87: Opening the batch-workflow editor.

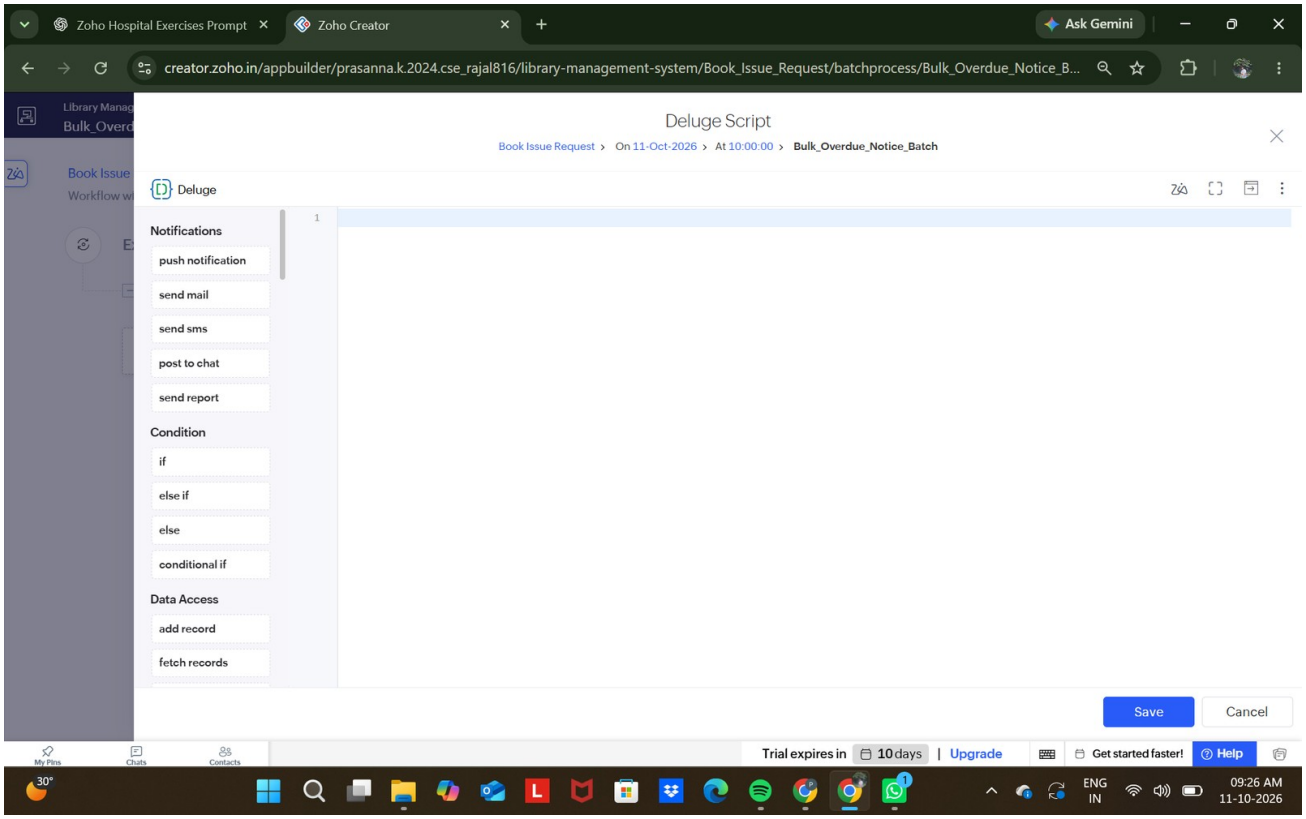


Figure 88: Opening the Deluge script editor for Bulk_Overdue_Notice_Batch.

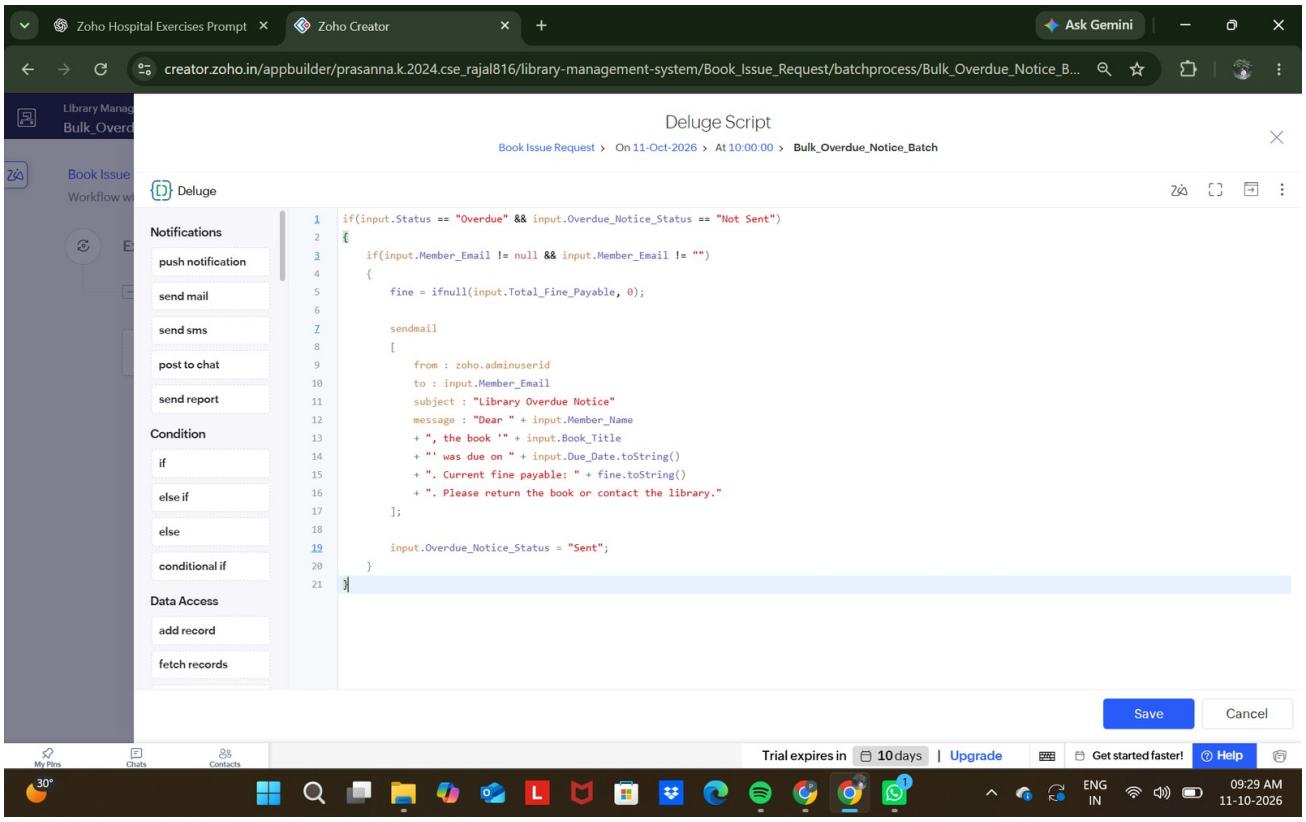


Figure 89: Writing conditions for overdue records and notice status.

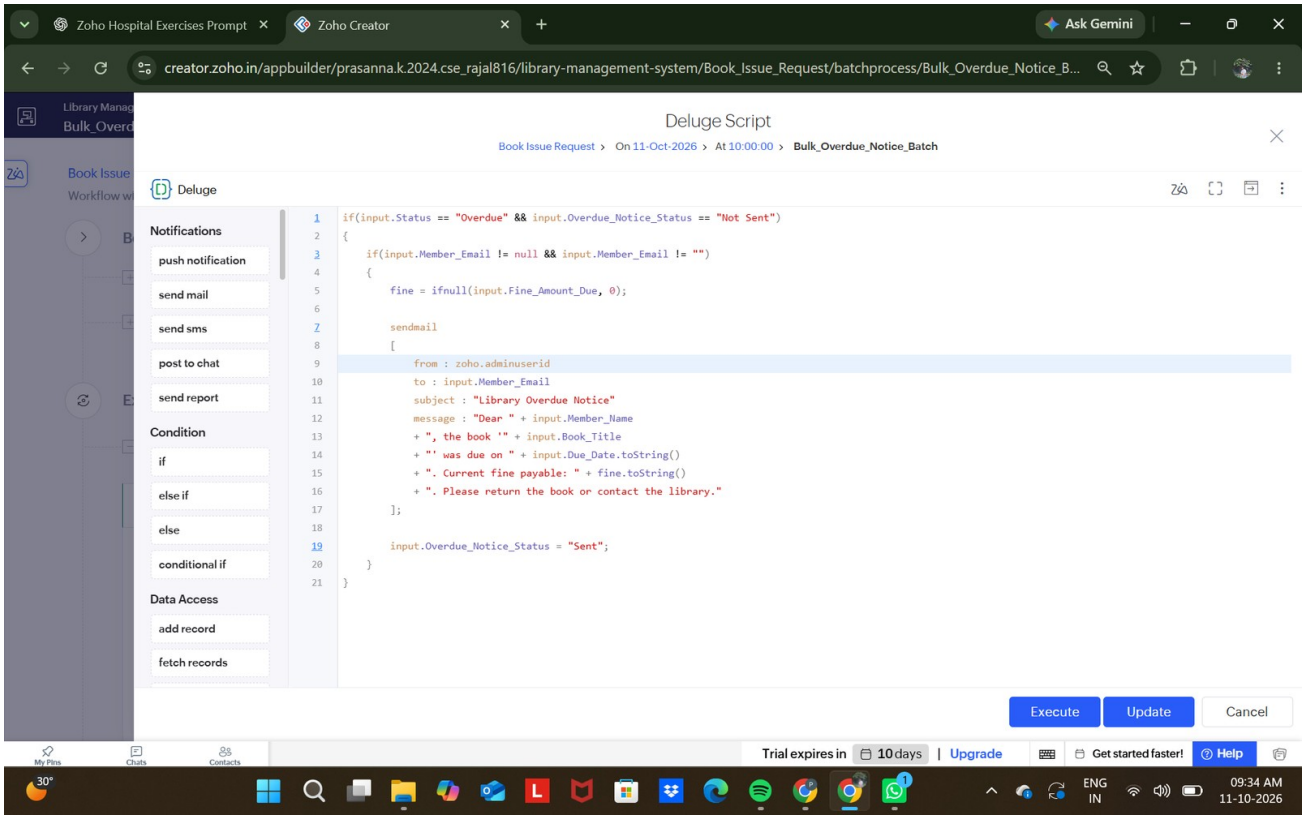


Figure 90: Adding the overdue email notification and status update logic.

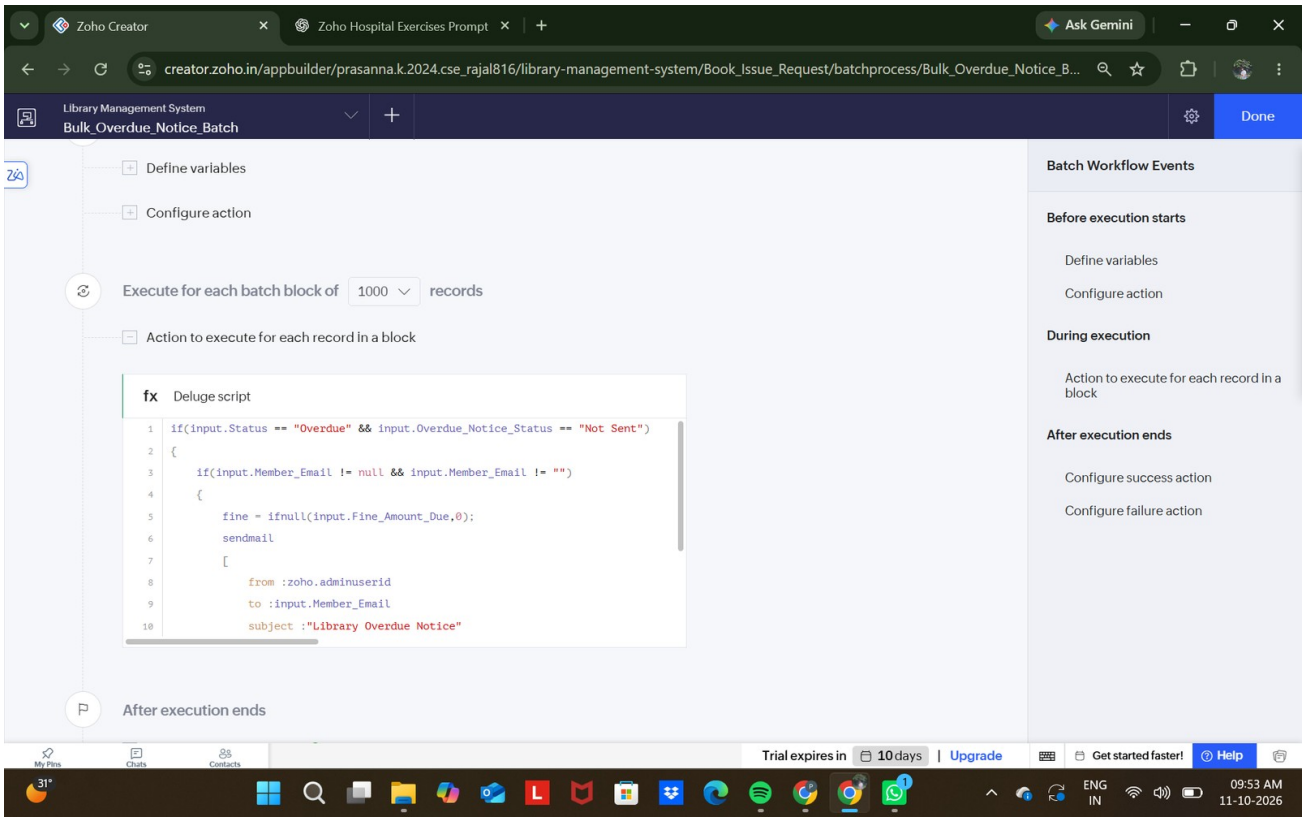


Figure 91: Reviewing the batch action code and the 1,000-record execution block.

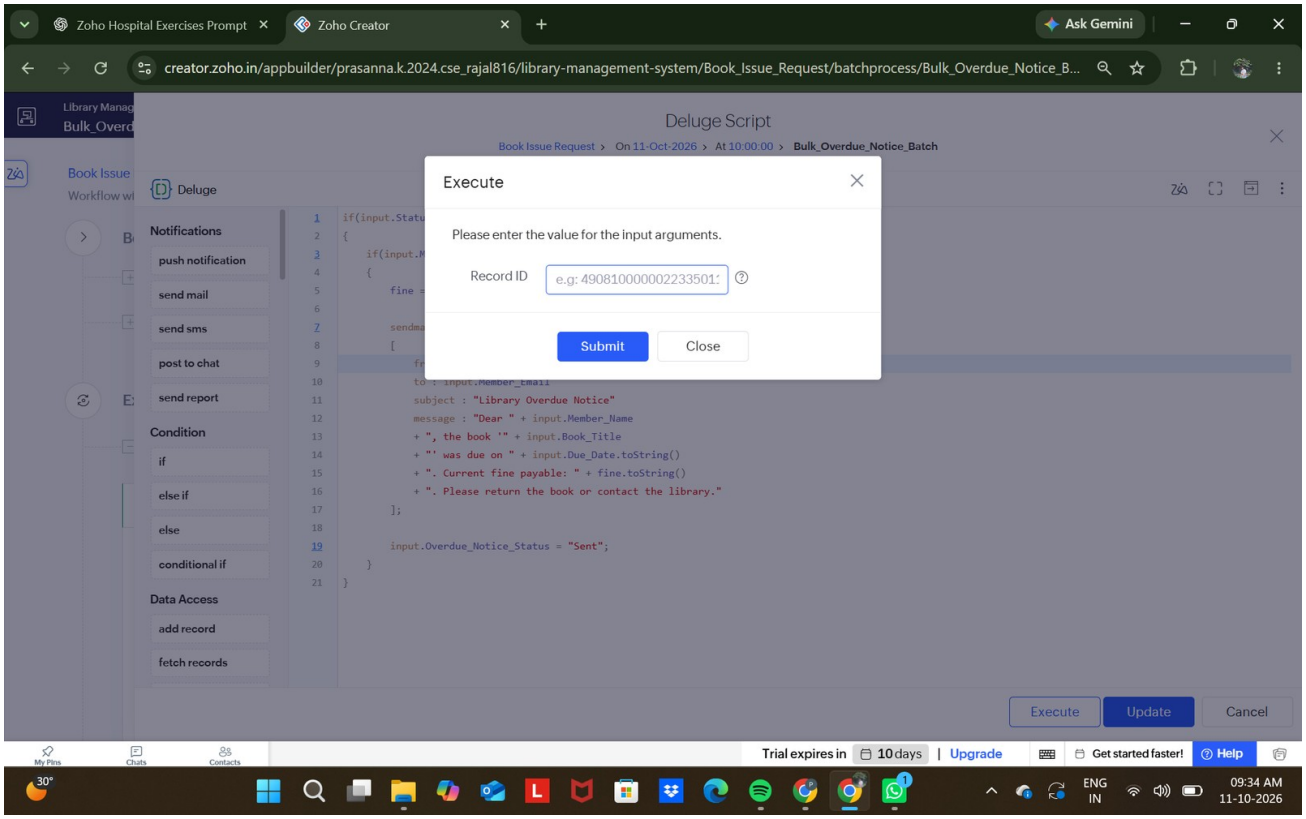


Figure 92: Opening the script execution dialog and reviewing its Record ID input.

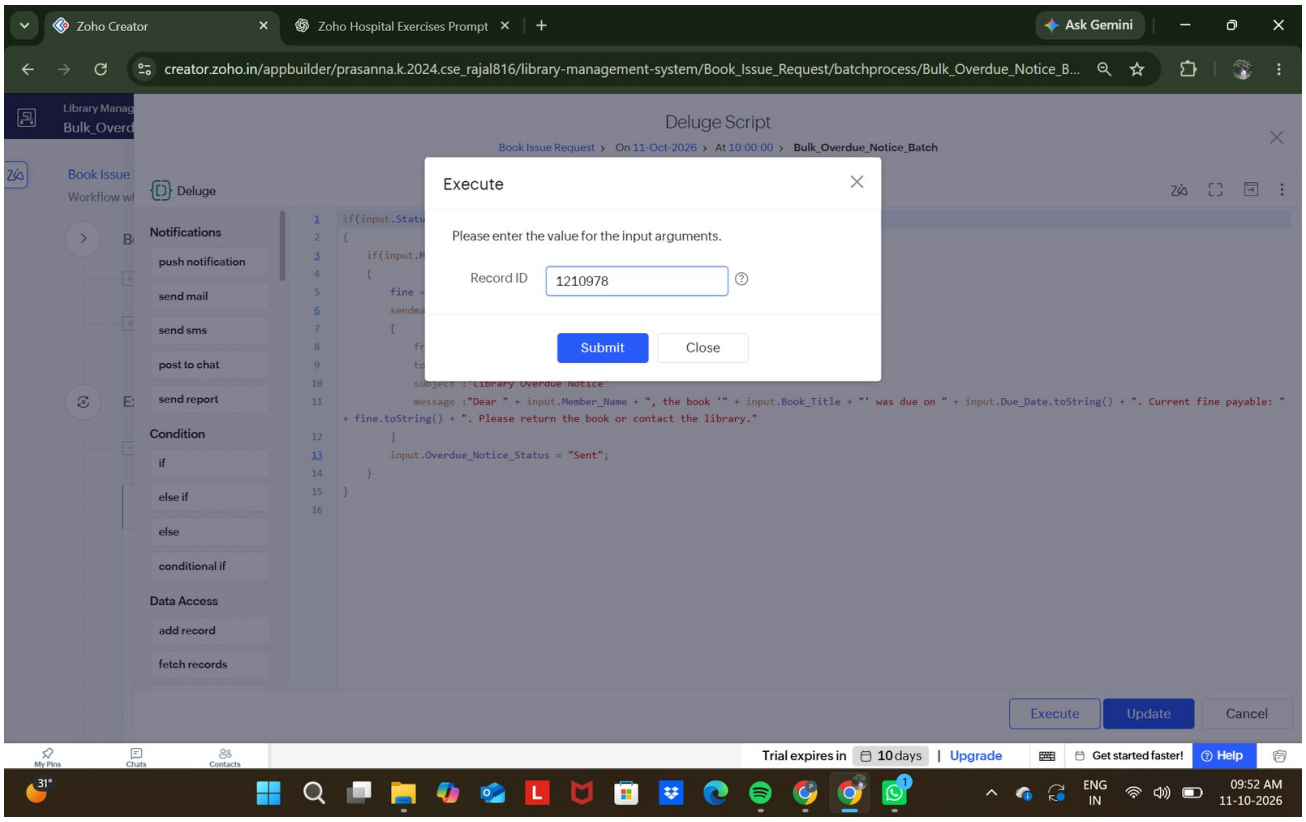


Figure 93: Providing a Book Issue Request Record ID to test the batch action.

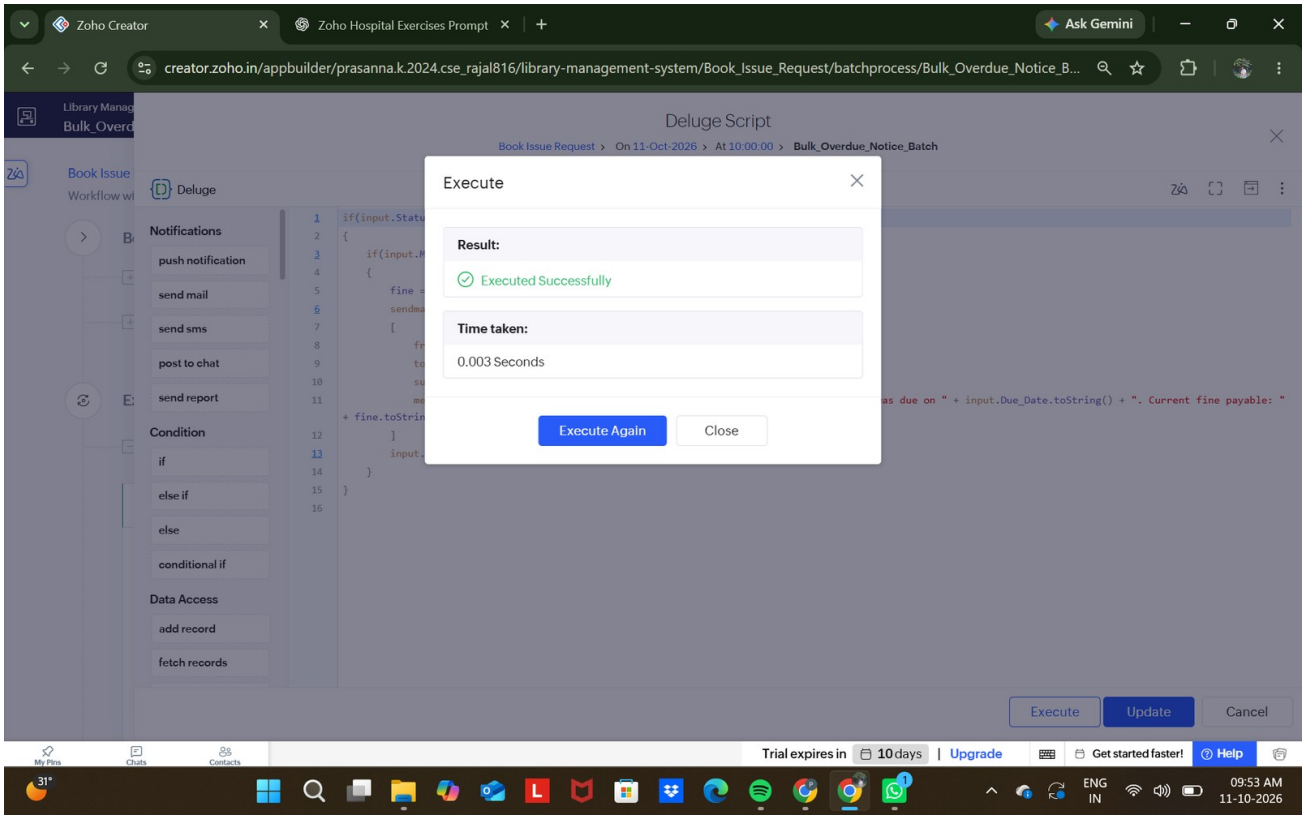


Figure 94: Confirming the batch overdue-notice script reports Executed Successfully.

7. Report Configuration

The Book Issue Request report was reviewed and its field labels, visible columns and report layout were configured. The supplied screens show the report field settings and the resulting structured record view.

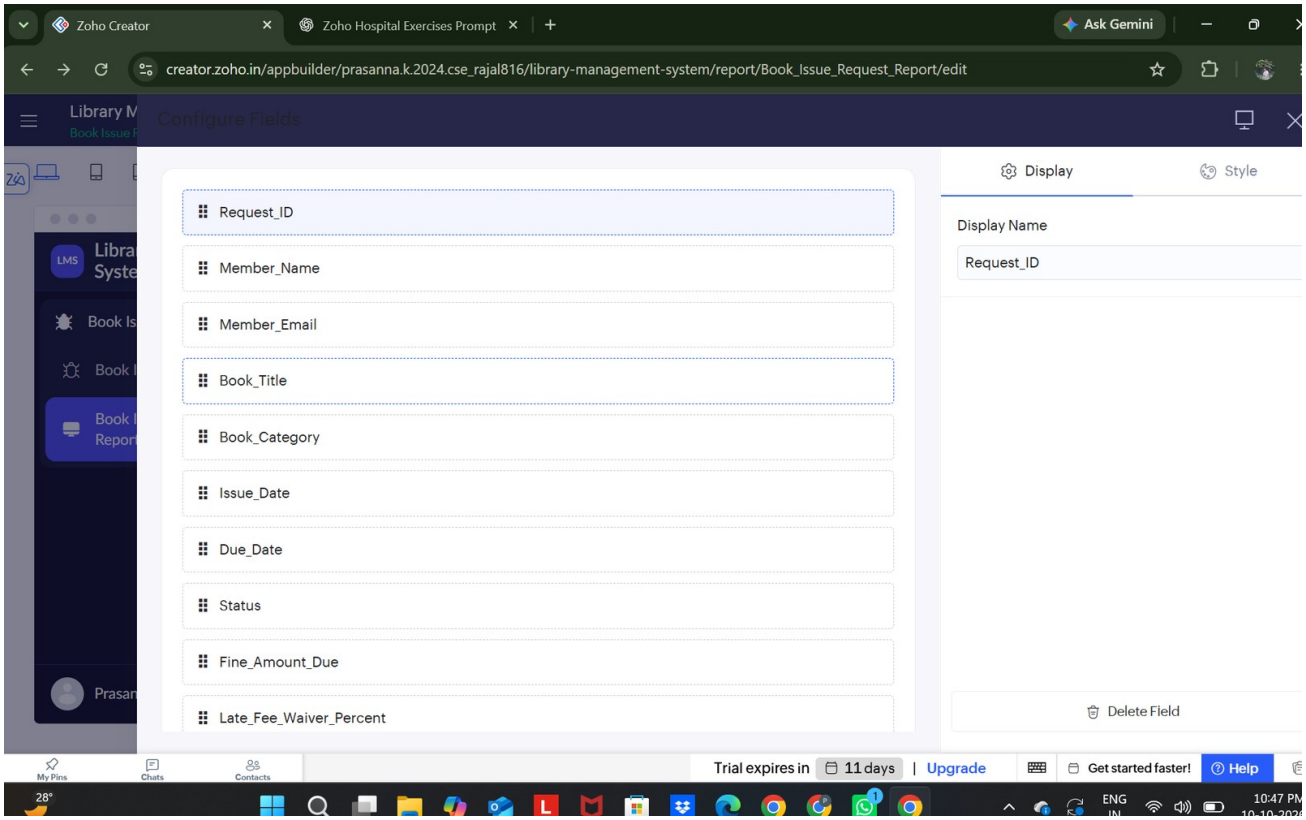


Figure 95: Opening Configure Fields for the Book Issue Request report.

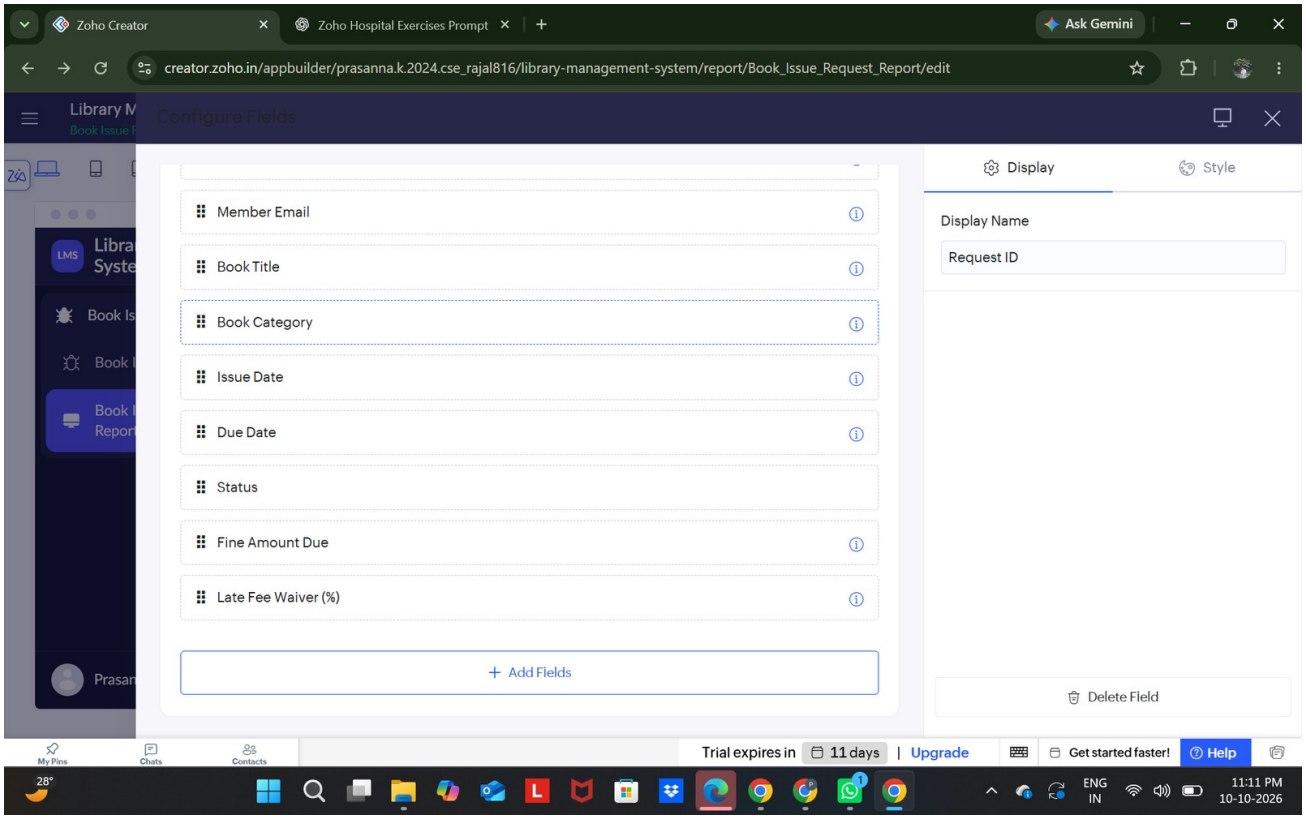


Figure 96: Updating display names for fields in the report.

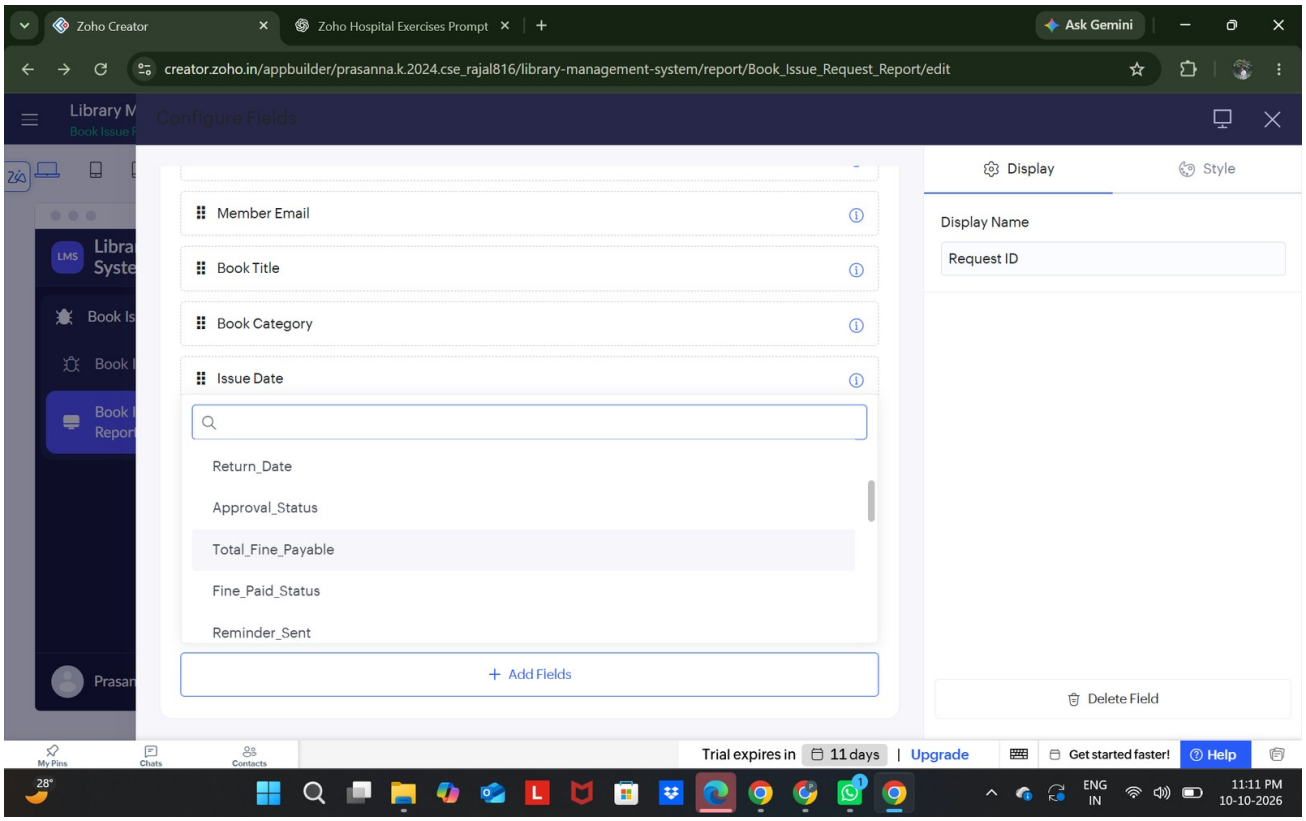


Figure 97: Reviewing selected columns and field labels.

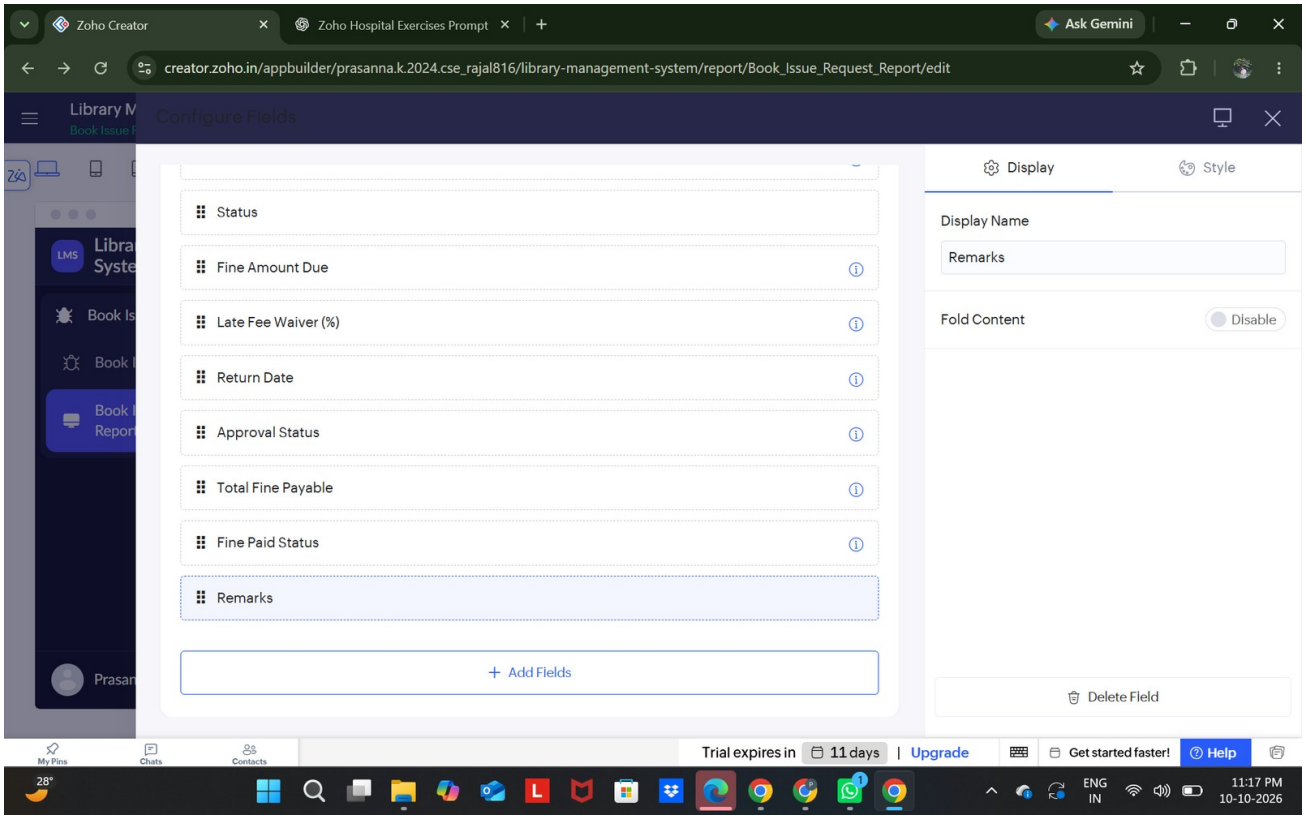


Figure 98: Configuring the Remarks field and other report columns.

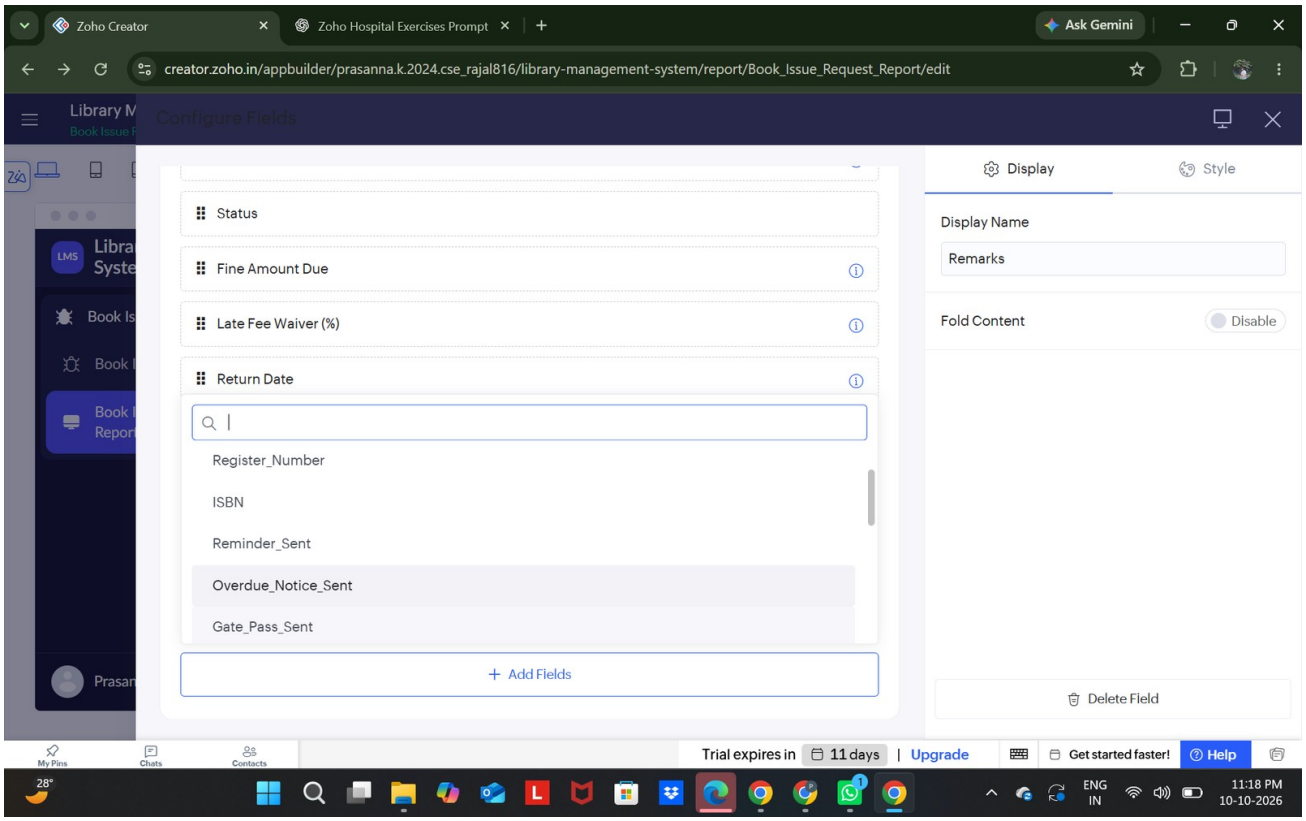


Figure 99: Reviewing the lower portion of the report field list.

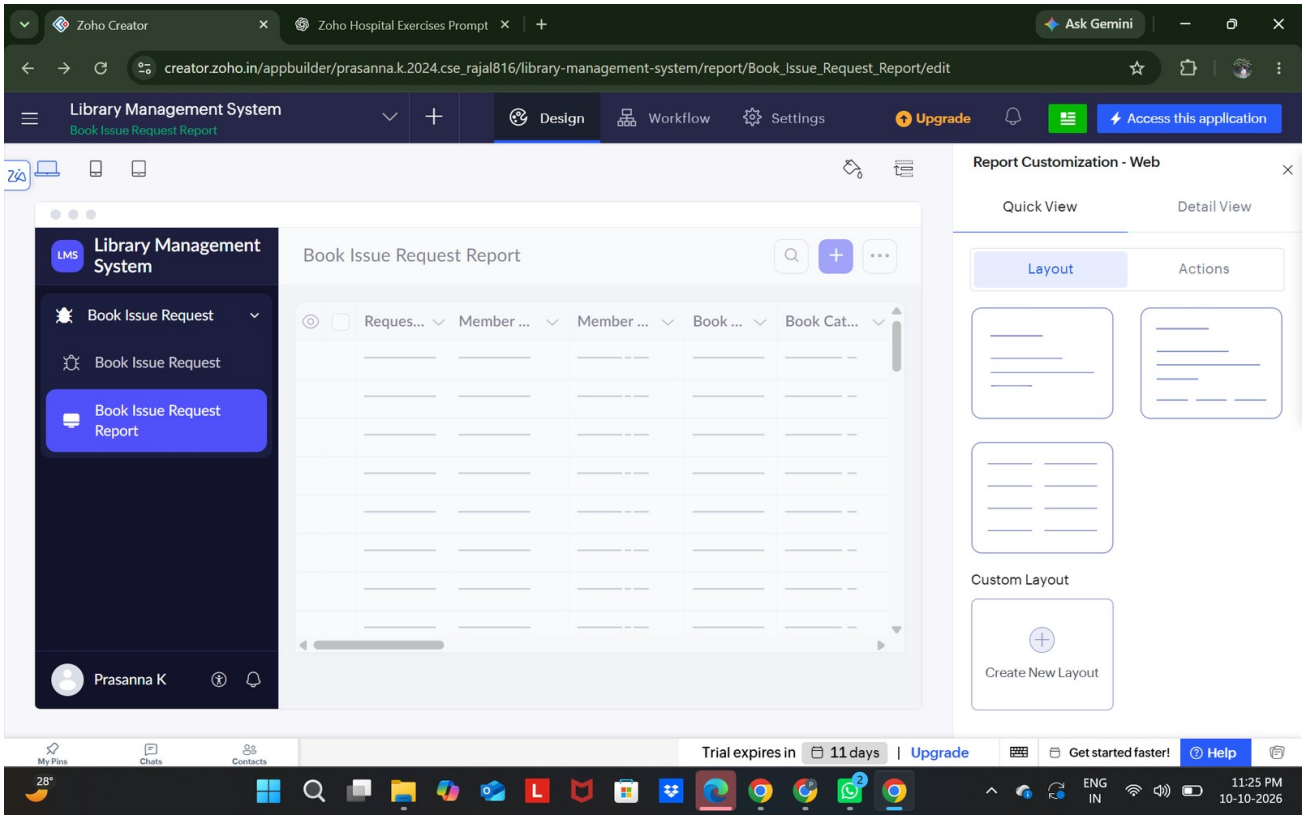


Figure 100: Reviewing the configured report within the application.

Functions Configuration

A reusable Deluge function named Generate_GatePass_Function was created in the Default namespace. It uses a void return type and an integer request_id argument, retrieves the associated Book Issue Request record, checks the issue status and member email, and prepares a gate-pass email with the due date.

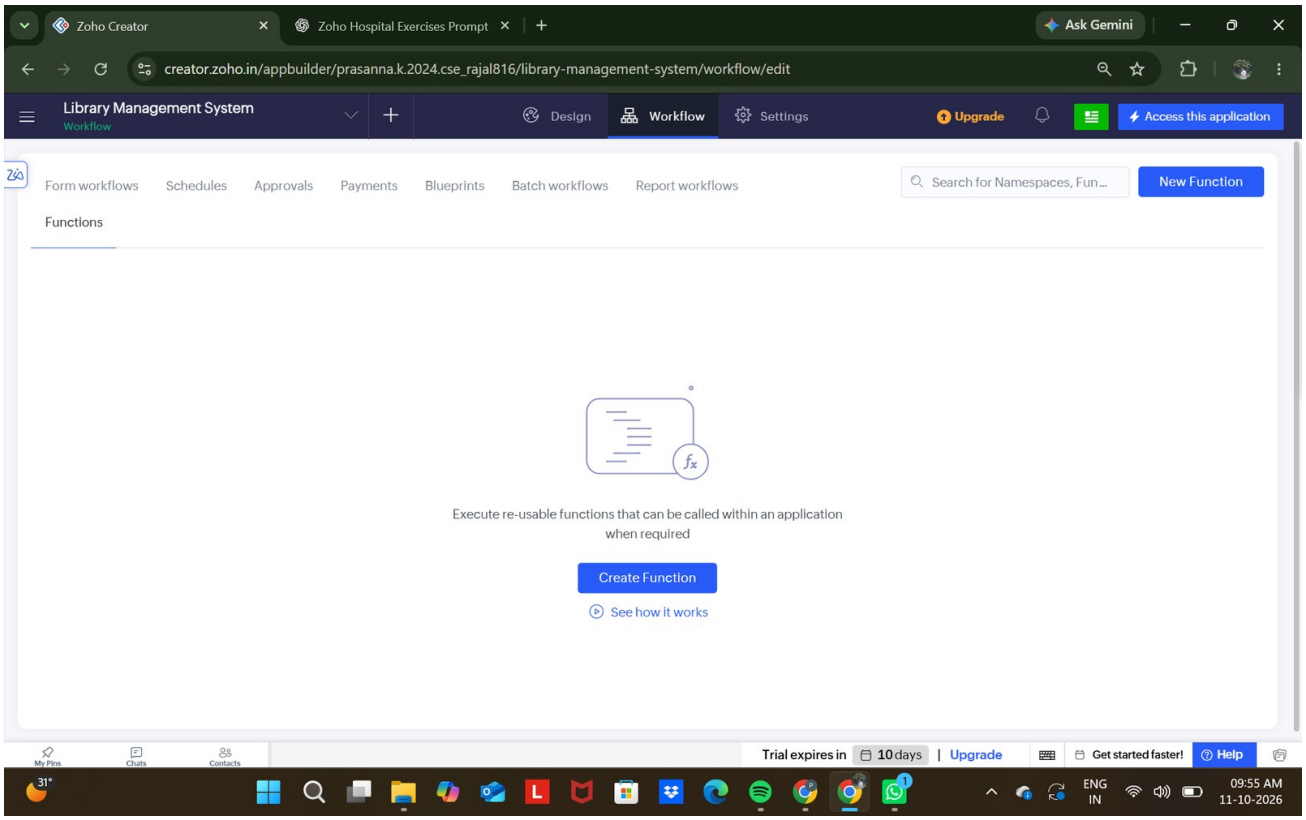


Figure 101: Opening the Functions area to create a reusable Deluge function.

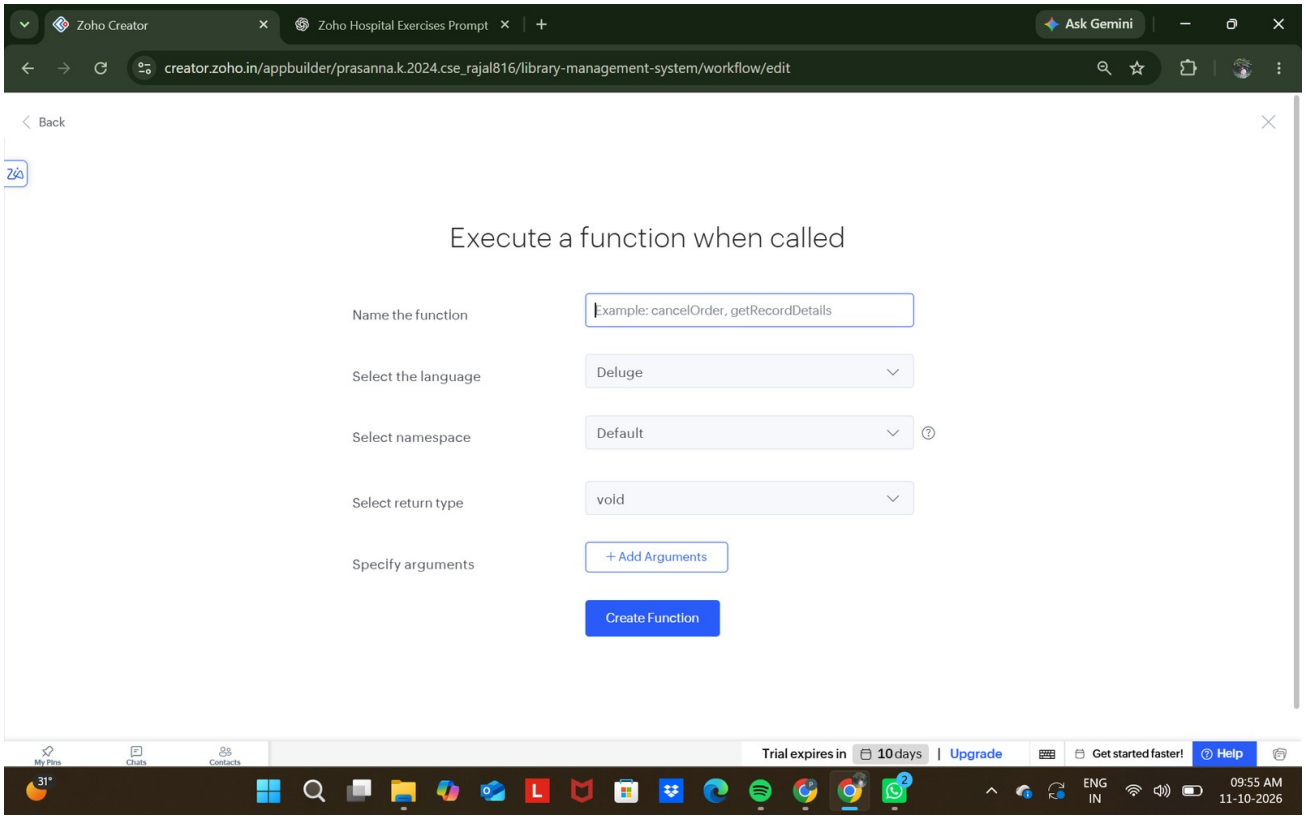


Figure 102: Reviewing the function language, namespace and return-type settings.

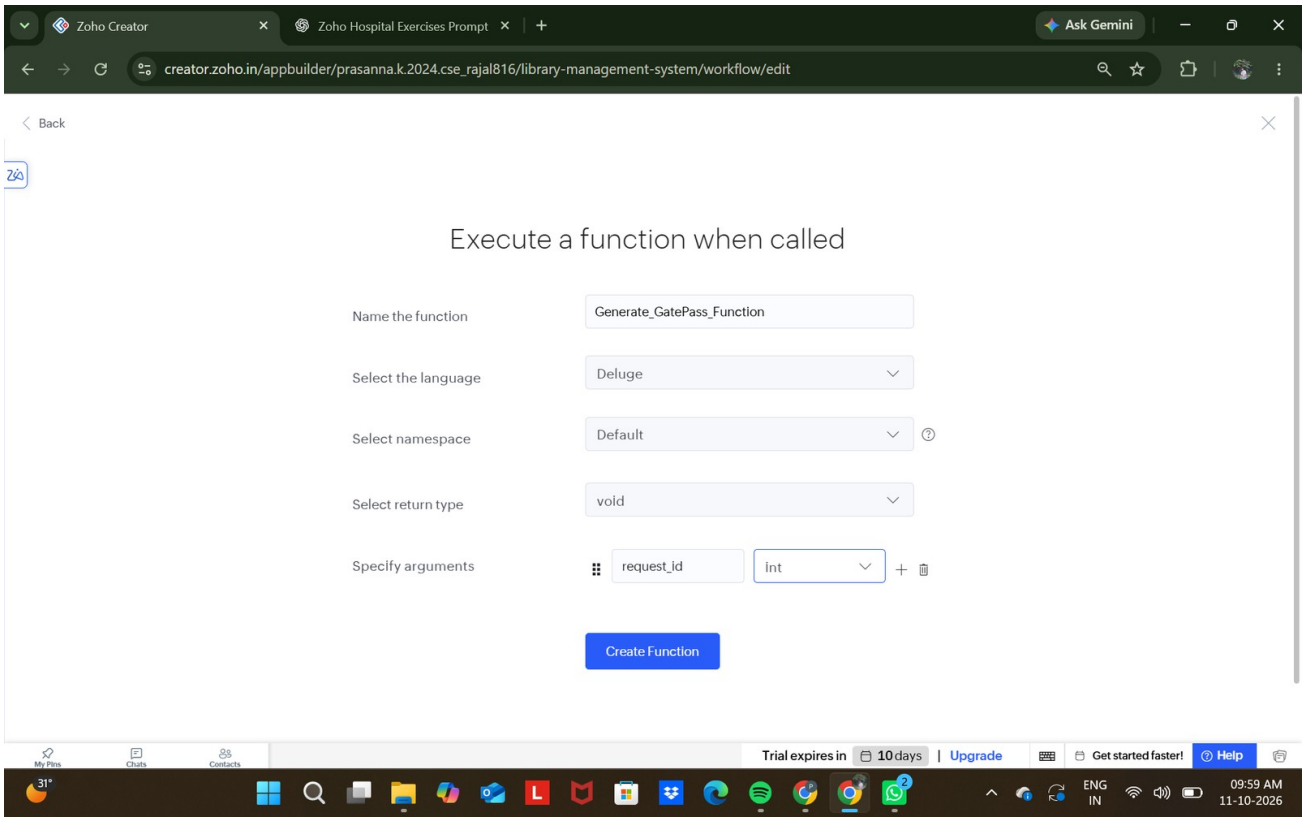


Figure 103: Naming Generate_GatePass_Function and adding the integer request_id argument.

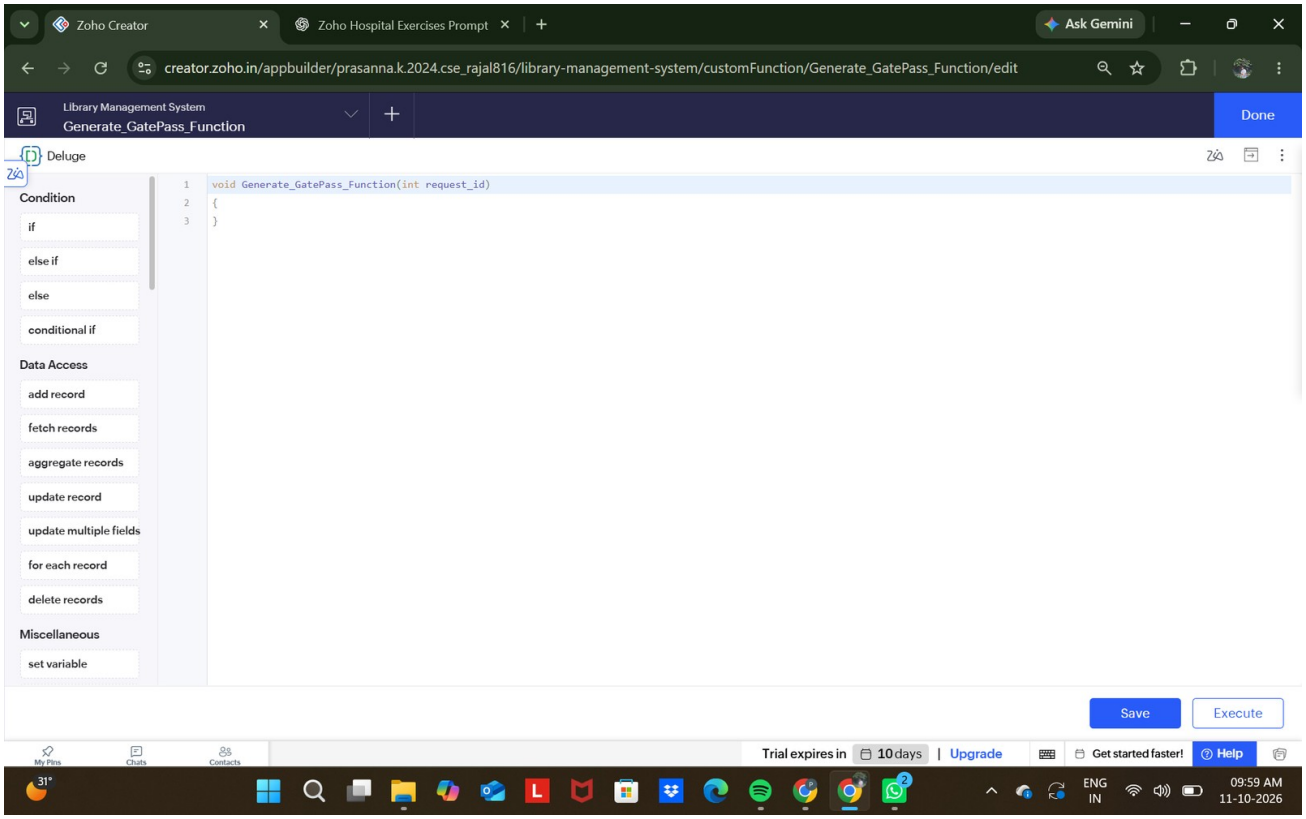


Figure 104: Opening the function script editor.

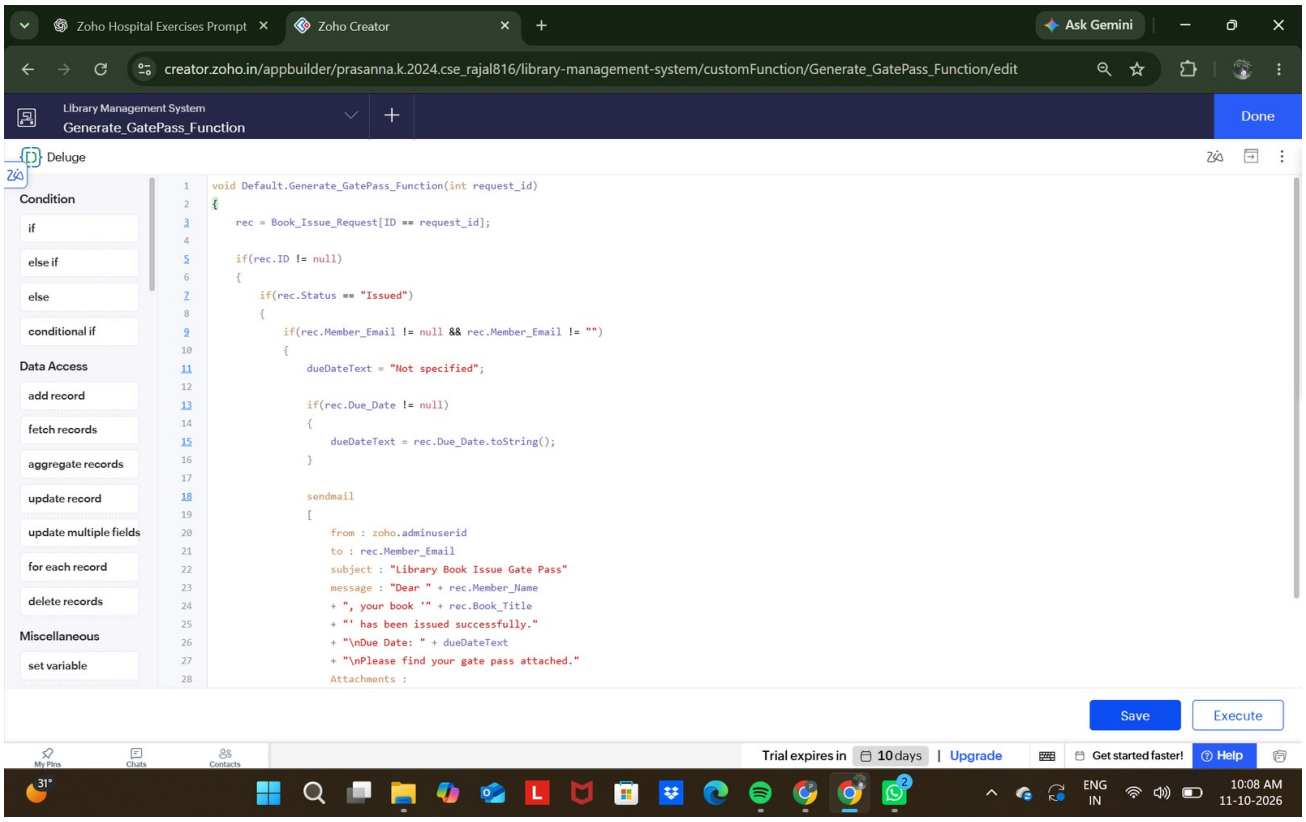


Figure 105: Writing the record lookup and issue-status checks.

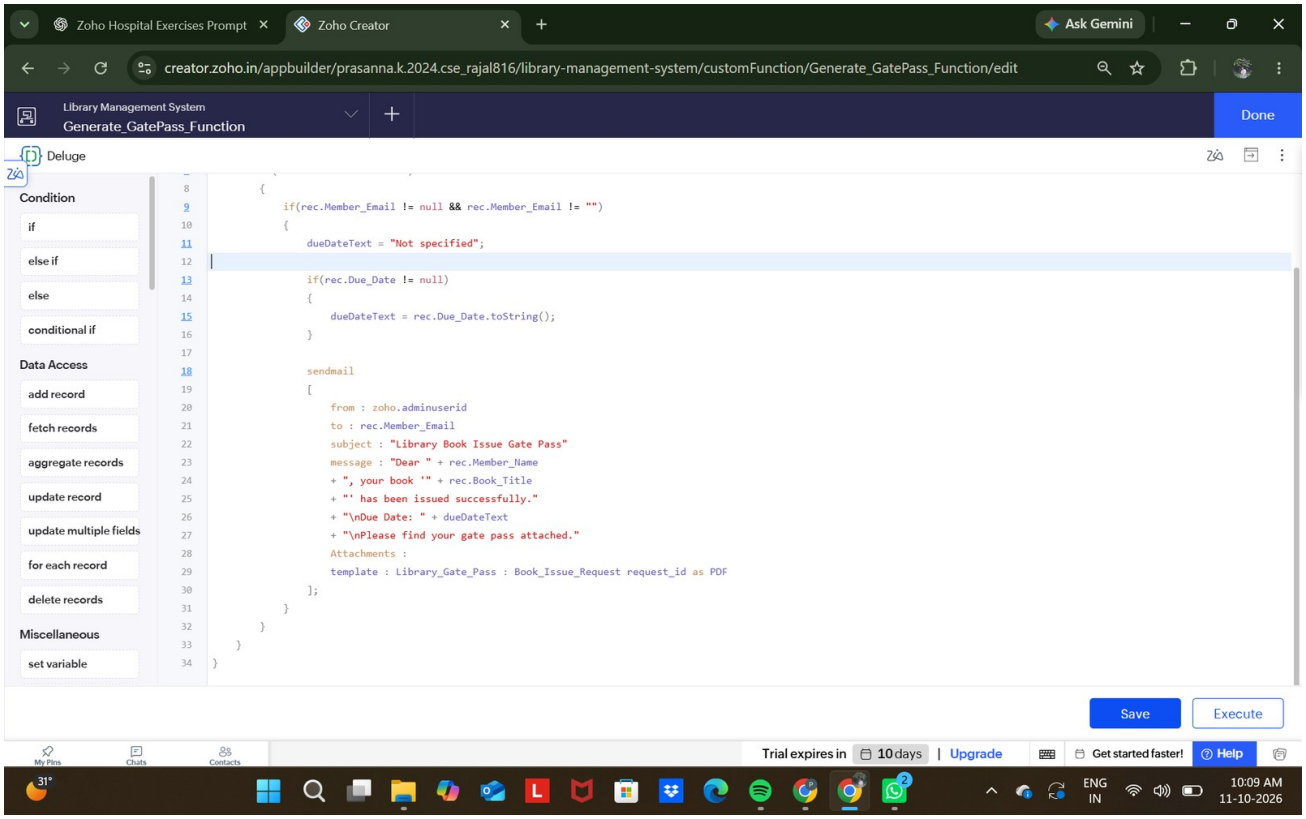


Figure 106: Finishing the gate-pass email logic and due-date message.

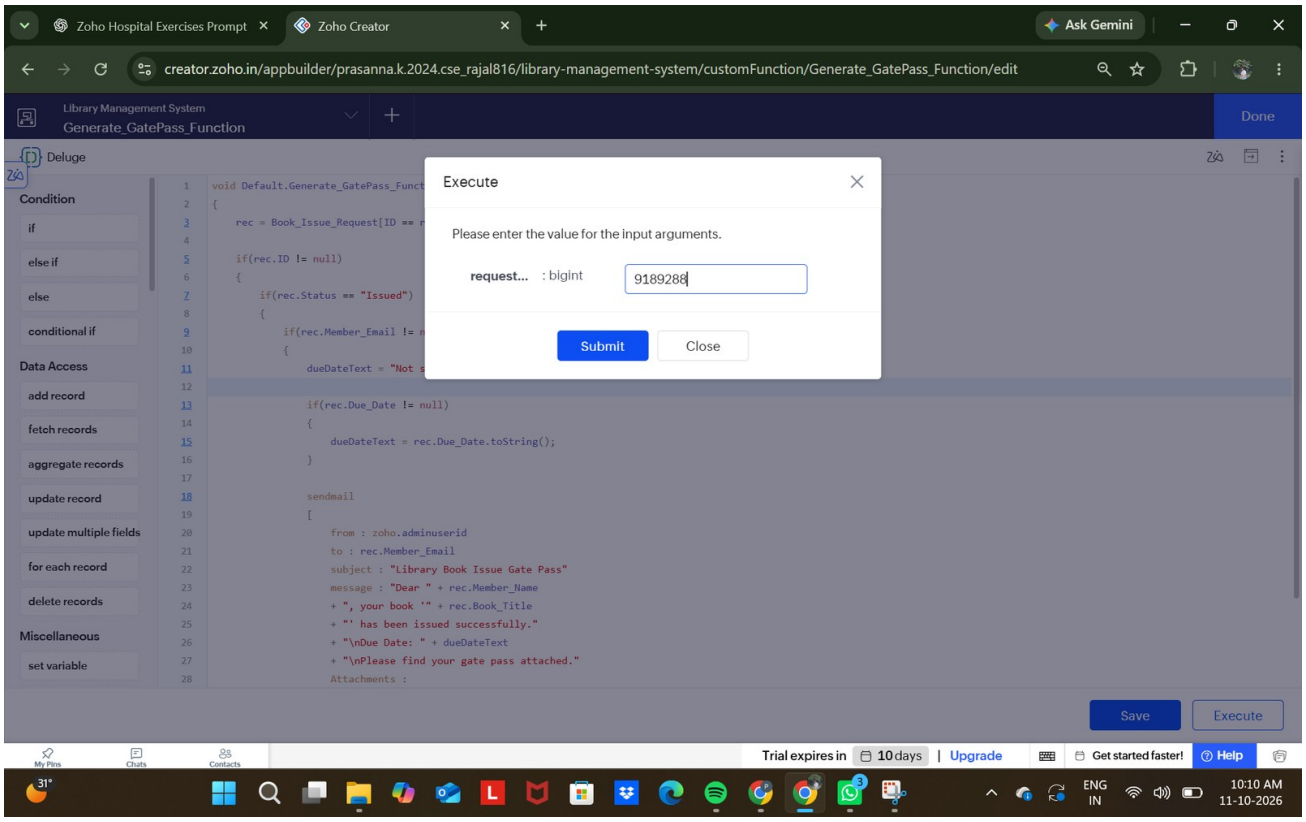


Figure 107: Providing request_id to test Generate_GatePass_Function.

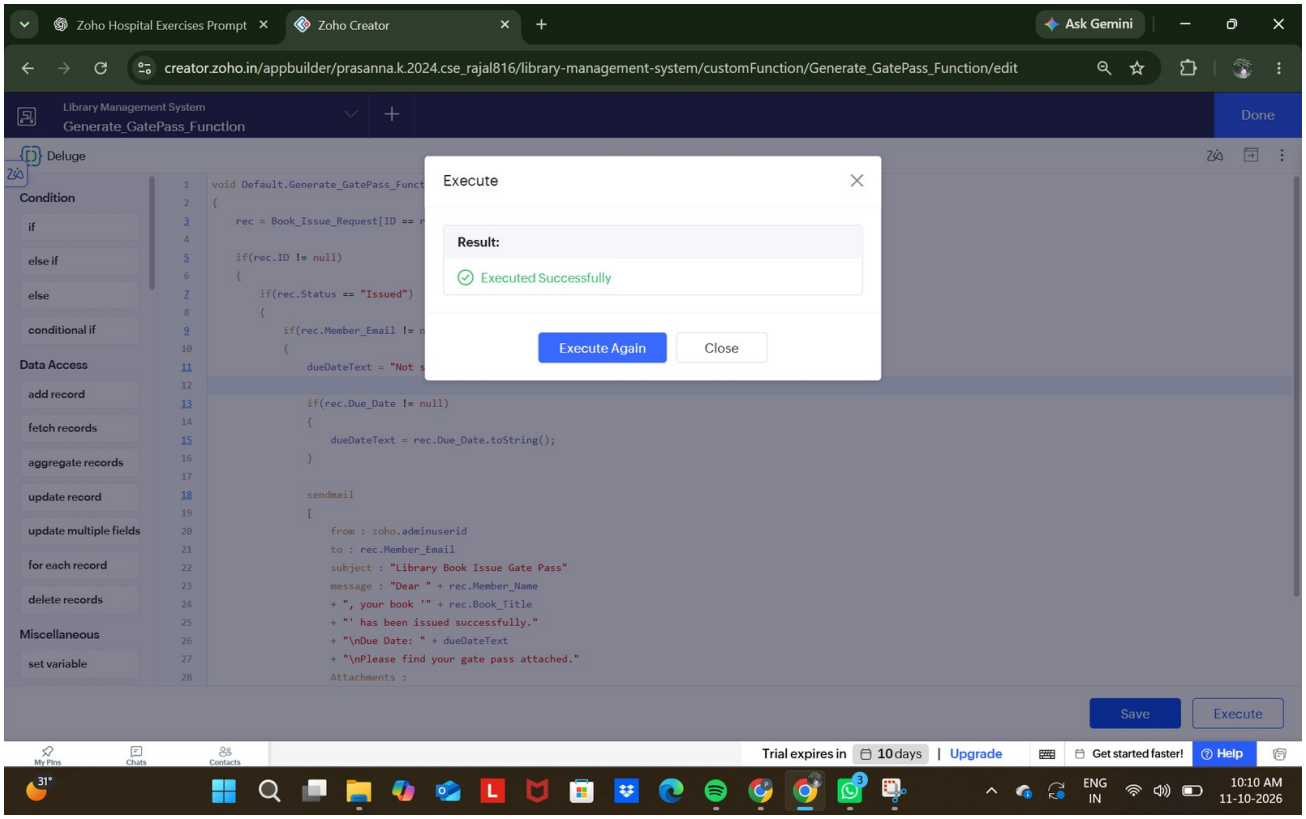


Figure 108: Confirming the gate-pass function reports Executed Successfully.

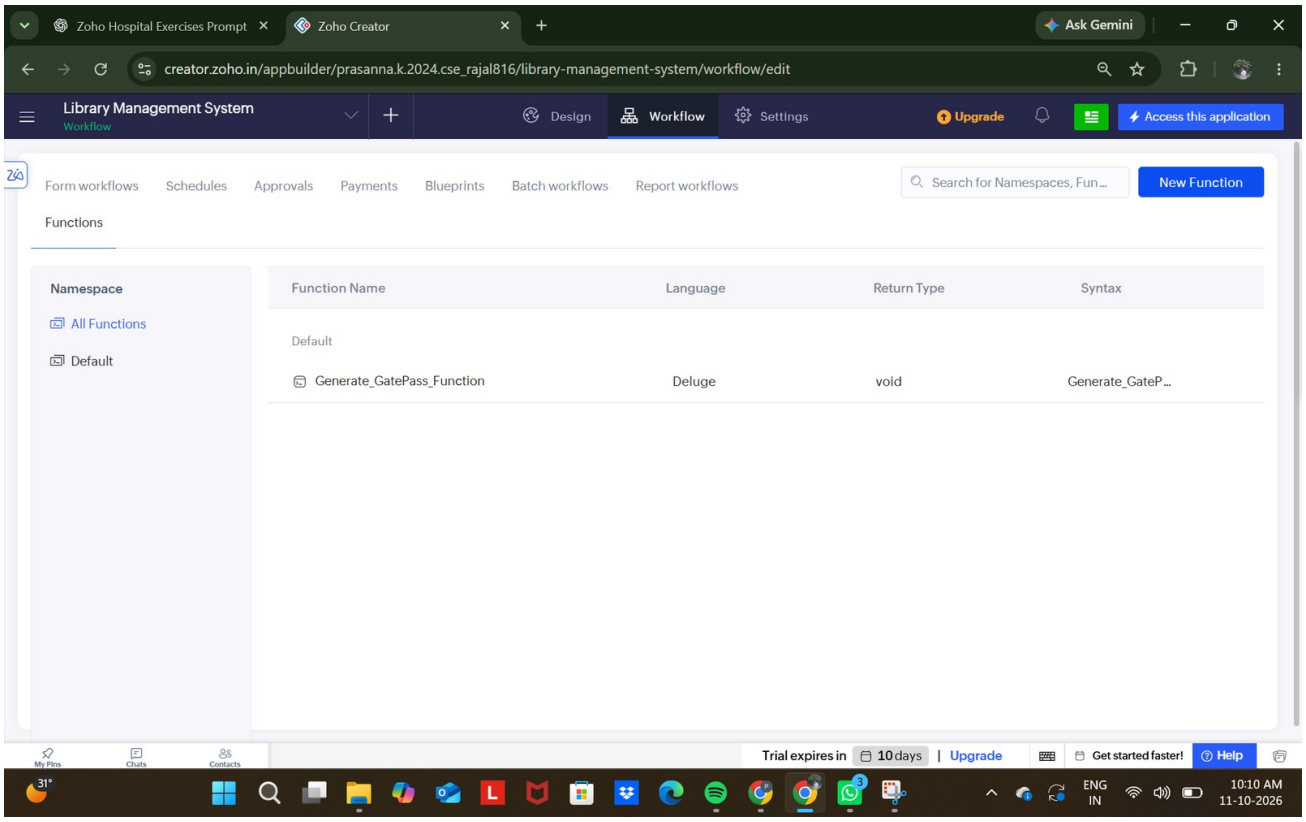


Figure 109: Verifying Generate_GatePass_Function appears in the Functions list.

8. Live Workflow Execution & Testing

The final application-level check opens the Book Issue Request form after the configuration steps. The related workflow status and execution screenshots are documented at the ends of the relevant subsections.

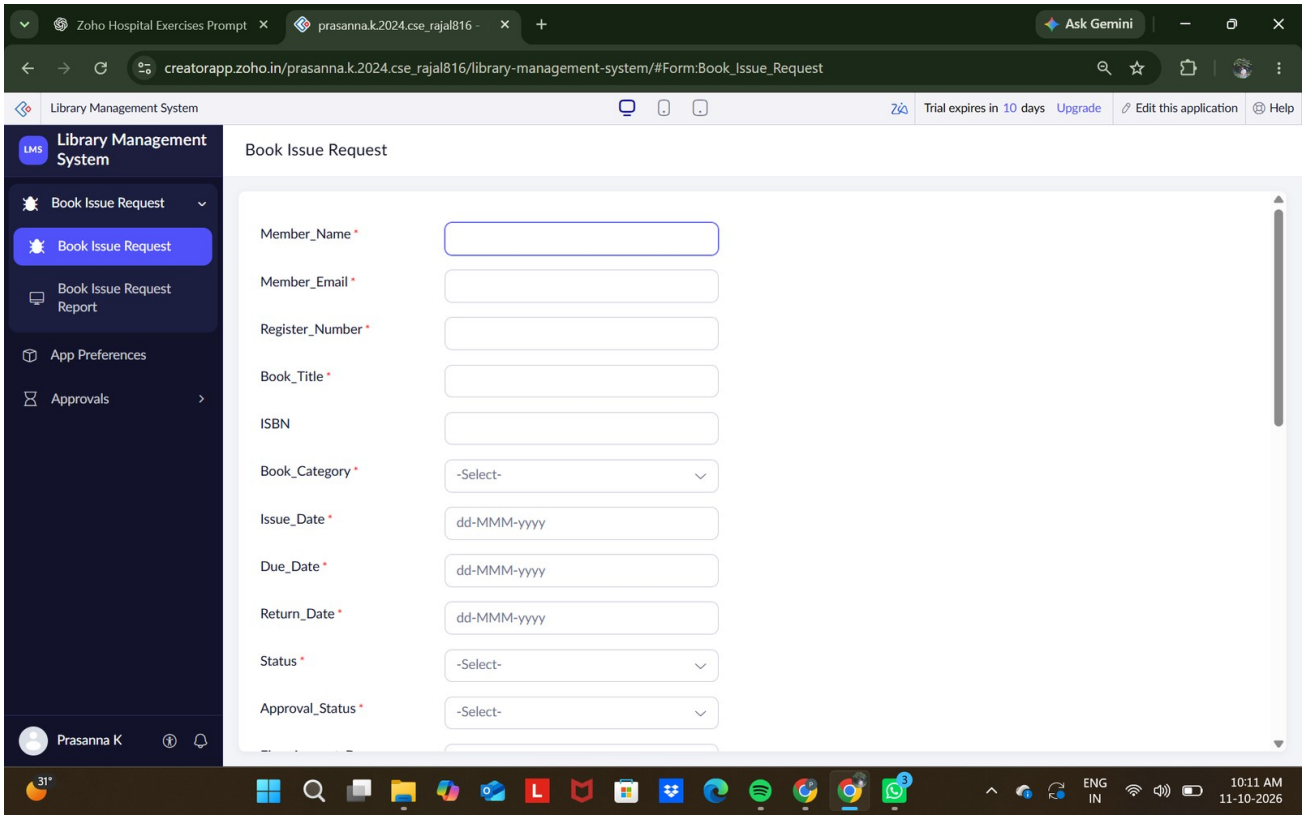


Figure 110: Opening the live Book Issue Request form for a final application check.

Result

The workflow options for the Library Management System were configured and documented in Zoho Creator. The screenshots show form-submission confirmation, scheduled reminders, conditional approval, fine-payment actions, issue-lifecycle stages and transitions, batch overdue processing, report field configuration, and the reusable gate-pass function. Workflow status and execution screenshots are placed at the ends of their relevant subsections. Actual online payment collection remains subject to confirming the payment gateway setup.